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On the Volume Conjectures for Dehn-filled 3-manifolds along fundamental shadow link complements with big volume

Yunpeng Meng, Baojun Wu, Tian Yang, and Yuxuan Yang

Abstract

We prove the Volume Conjecture and the Asymptotic Expansion Conjecture for closed oriented hyperbolic 3-manifolds obtained by integral Dehn-fillings along a fundamental shadow link complement, provided that their volumes are sufficiently close to that of the complement. More generally, we show that the conjectured asymptotic expansions hold for pairs obtained by change-of-pair operations on a fundamental shadow link complement whenever the volume of the corresponding hyperbolic cone manifold is sufficiently close to that of the complement. A key ingredient in the proof is the introduction of extended quantum $6j$-symbols and a Weak Reflection Symmetry satisfied by them. The Weak Reflection Symmetry is established by relating the extended quantum $6j$-symbols to the b-$6j$ symbols through a real Gevrey-1 estimate and a truncated Borel-Laplace transform.

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1 Introduction

A central theme in quantum topology is the principle that the asymptotic behavior of quantum invariants reflects the geometry of 3-manifolds. This idea first emerged in Kashaev's volume conjecture, which relates the exponential growth of a knot invariant constructed from the quantum dilogarithm to the hyperbolic volume of the knot complement [20]. Murakami and Murakami subsequently showed that Kashaev's invariant can be identified with a particular specialization of the colored Jones polynomial [29]. This perspective was later extended to hyperbolic 3-manifolds through analogous volume conjectures for Reshetikhin-Turaev and relative Reshetikhin-Turaev invariants [6, 33, 17, 41, 42].

Let M be a closed oriented hyperbolic 3-manifold. For any integer $r \geqslant 3$, let $\mathrm{RT}_r(M)$ be the r-th Reshetikhin-Turaev invariant of M evaluated at the root of unity $q = e^{\frac{2\pi\mathbf{i}}{r}}$.

Conjecture 1.1. *[6, Conjecture 1.2] As r varies over all positive odd integers,*

$$\lim_{r\to+\infty} \frac{4\pi}{r} \log\big(\mathrm{RT}_r(M)\big) = \mathrm{Vol}(M) + \mathbf{i}\mathrm{CS}(M) \mod \mathbf{i}\pi^2\mathbb{Z},$$

where $\mathrm{Vol}(M)$ and $\mathrm{CS}(M)$ are respectively the volume and the Chern-Simons invariant of M in the hyperbolic metric.

Conjecture 1.2. *[33, 17] As r varies over all positive odd integers,*

$$\mathrm{RT}_r(M) = \frac{C_r}{\sqrt{\pm\mathrm{Tor}(M)}} e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M)+\mathbf{i}\mathrm{CS}(M)\right)} \Big(1 + O\Big(\frac{1}{r}\Big)\Big),$$

where C_r is a complex number of norm 1 independent of the geometric structure on M, and $\mathrm{Tor}(M)$ is the adjoint twisted Reidemeister torsion of M.

Conjectures 1.1 and 1.2 are respectively proved in [33, 40, 7, 18] for hyperbolic 3-manifolds obtained by Dehn-fillings along the figure-8 knot, and by long Dehn-fillings along the family of twist knots.

Let M be a closed oriented 3-manifold and let L be a framed hyperbolic link in M with n components. For an odd integer $r \geqslant 3$, let $\boldsymbol{b} = (b_1, \dots, b_n) \in \{0, 2, \dots, r-3\}^n$ and let $\mathrm{RT}_r(M, L, \boldsymbol{b})$ be the r-th relative Reshetikhin-Turaev invariant of the pair (M, L) colored by $\boldsymbol{b}$ and evaluated at the root of unity $q = e^{\frac{2\pi\mathbf{i}}{r}}$. For a sequence $\boldsymbol{b}^{(r)} = (b_1^{(r)}, \dots, b_n^{(r)})$, let

$$\theta_k = \left| \lim_{r\to\infty} \frac{4\pi b_k^{(r)}}{r} - 2\pi \right|$$

for $k \in \{1, \dots, n\}$, and let $\boldsymbol{\theta} = (\theta_1, \dots, \theta_n)$.

Conjecture 1.3. *[41, Conjecture 1.1] If $M_{L_{\boldsymbol{\theta}}}$ is a hyperbolic cone manifold consisting of M and a hyperbolic cone metric on M with singular locus L and cone angles $\boldsymbol{\theta}$, then as r varies over all positive odd integers,*

$$\lim_{r\to\infty}\frac{4\pi}{r}\log \mathrm{RT}_r(M,L,\boldsymbol{b}^{(r)}) = \mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) + \mathbf{i}\mathrm{CS}(M_{L_{\boldsymbol{\theta}}}) \mod \mathbf{i}\pi^2\mathbb{Z},$$

where $\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}})$ and $\mathrm{CS}(M_{L_{\boldsymbol{\theta}}})$ are respectively the volume and the Chern-Simons invariant of $M_{L_{\boldsymbol{\theta}}}$.

Suppose further that for each $k\in\{1,\dots,n\}$, either $b_k^{(r)} > \frac{r}{2}$ for all r sufficiently large or $b_k^{(r)} < \frac{r}{2}$ for all r sufficiently large. In the former case we let $\mu_k = 1$ and in the latter case we let $\mu_k = -1$, and we let

$$\theta_k^{(r)} = \mu_k\left(\frac{4\pi b_k^{(r)}}{r} - 2\pi\right).$$

Let $\boldsymbol{\theta}^{(r)} = (\theta_1^{(r)},\dots,\theta_n^{(r)})$, and suppose that $\{\boldsymbol{\theta}^{(r)}\}$ converges as r tends to infinity. Also suppose that for all r sufficiently large, a hyperbolic cone metric on M with singular locus L and cone angles $\boldsymbol{\theta}^{(r)}$ exists. We denote M with such a hyperbolic cone metric by $M^{(r)}$, and let $\mathrm{Vol}(M^{(r)})$ and $\mathrm{CS}(M^{(r)})$ respectively be the volume and the Chern-Simons invariant of $M^{(r)}$. For the parallel copies $\{\gamma_k\}_{k\in\{1,\dots,n\}}$ of the core curves of L given by the framing, let $\mathrm{H}^{(r)}(\gamma_k)$ be the logarithmic holonomy in $M^{(r)}$. Let $\rho_{M^{(r)}}:\pi_1(M\smallsetminus L)\to \mathrm{PSL}(2;\mathbb{C})$ be the holonomy representation of the restriction of $M^{(r)}$ to $M\smallsetminus L$, and let $\mathbb{T}_{(M\smallsetminus L,\boldsymbol{m})}([\rho_{M^{(r)}}])$ be the Reidemeister torsion of $M\smallsetminus L$ twisted by the adjoint action of $\rho_{M^{(r)}}$ with respect to the system of meridians $\boldsymbol{m}$ of a tubular neighborhood of the core curves of L.

Conjecture 1.4. *[42, Conjecture 1.1] As r varies over all positive odd integers,*

$$\mathrm{RT}_r(M,L,\boldsymbol{b}^{(r)}) = C_r\frac{e^{\sum_{k=1}^n \mu_k\frac{\mathrm{H}^{(r)}(\gamma_k)}{2}}}{\sqrt{\pm\mathbb{T}_{(M\smallsetminus L,\boldsymbol{m})}([\rho_{M^{(r)}}])}}e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M^{(r)})+\mathbf{i}\mathrm{CS}(M^{(r)})\right)}\left(1+O\Big(\frac{1}{r}\Big)\right),$$

where C_r is a complex number of norm 1 independent of the geometric structure on (M,L).

Remark 1.5. If we let

$$\widetilde{\theta}_k^{(r)} = \mu_k\left(\frac{4\pi(b_k^{(r)}+1)}{r} - 2\pi\right)$$

for each $k\in\{1,\dots,n\}$, let $\widetilde{\boldsymbol{\theta}}^{(r)} = (\widetilde{\theta}_1^{(r)},\dots,\widetilde{\theta}_n^{(r)})$, and by abuse of notation let $M^{(r)} = M_{L_{\widetilde{\boldsymbol{\theta}}^{(r)}}}$ be the corresponding hyperbolic cone manifold, then Conjecture 1.4 can be rephrased as: As r varies over all positive odd integers,

$$\mathrm{RT}_r(M,L,\boldsymbol{b}^{(r)}) = \frac{C_r}{\sqrt{\pm\mathbb{T}_{(M\smallsetminus L,\boldsymbol{m})}([\rho_{M^{(r)}}])}}e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M^{(r)})+\mathbf{i}\mathrm{CS}(M^{(r)})\right)}\left(1+O\Big(\frac{1}{r}\Big)\right).$$

The reason is due to the Complexified Schläfli Formula of cone metrics (see, [8, Theorem 3.20] for the real part and [16, Theorem 3.5] for the imaginary part) that, if $\widetilde{\boldsymbol{\theta}}^{(r)} = \boldsymbol{\theta}^{(r)} + \frac{4\pi}{r}\boldsymbol{\mu}$, where $\boldsymbol{\mu} = (\mu_k)_{k\in\{1,\dots,n\}}\in\{-1,1\}^n$, then

$$\mathrm{Vol}(M_{L_{\widetilde{\boldsymbol{\theta}}^{(r)}}}) + \mathbf{i}\mathrm{CS}(M_{L_{\widetilde{\boldsymbol{\theta}}^{(r)}}}) = \mathrm{Vol}(M_{L_{\boldsymbol{\theta}^{(r)}}}) + \mathbf{i}\mathrm{CS}(M_{L_{\boldsymbol{\theta}^{(r)}}}) - \frac{1}{2}\sum_{k=1}^n \mu_k\mathrm{H}^{(r)}(\gamma_k)\cdot\frac{4\pi}{r} + O\Big(\frac{1}{r^2}\Big).$$

Conjectures 1.3 and 1.4 were proved in [41, 42, 34] for the pairs (M, L) such that $M \setminus L$ is homeomorphic to a fundamental shadow link complement (see Section 2.3) and all the cone angles are sufficiently small. They were also proved in [34] for pairs (M, L) such that M is obtained by doing a Dehn-filling along a fundamental shadow link complement with sufficiently large surgery coefficients, L consists of the core of the filled solid tori, and all the cone angles are strictly less than π. Topologically, (M, L) is obtained by doing a change-of-pair operation from the fundamental shadow link complement (see Section 2.4.)

As shown in [11], every oriented 3-manifold with toroidal or empty boundary can be obtained from a suitable fundamental shadow link complement by performing integral Dehn-fillings on some of its boundary components. Notice also that, on the one hand, the relative Reshetikhin-Turaev invariants of the triple $(M, L, \mathbf{0})$ coincide with the Reshetikhin-Turaev invariants of M, while, on the other hand, the coloring $\mathbf{0}$ corresponds to the cone metric on (M, L) for which all the cone angles are equal to 2π; this is precisely the hyperbolic metric on M.

Therefore, as proposed in [41], to prove Conjectures 1.1 and 1.2 for all the closed oriented hyperbolic 3-manifolds, it suffices to prove Conjectures 1.3 and 1.4 for pairs (M, L) such that M is obtained by doing an integral Dehn-filling along a fundamental shadow link complement, L consists of the cores of the filled solid tori, and the coloring is $\mathbf{0}$. In view of the result of [34] mentioned above, it therefore suffices to

(1) extend the range of the cone angles from $[0, \pi)$ to $[0, 2\pi]$, and

(2) remove the assumption that the Dehn-filling coefficients are sufficiently large.

The main results of this article accomplish step (1). More precisely, we have

Theorem 1.6. *For any fundamental shadow link complement M_{FSL}, there exists a $\delta > 0$ such that if the pair (M, L) obtained from M_{FSL} by doing a change-of-pair operation and the cone angles $\boldsymbol{\theta}$ satisfy*

$$\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) > \mathrm{Vol}(M_{FSL}) - \delta,$$

then Conjecture 1.3 and Conjecture 1.4 hold for (M, L).

The change-of-pair operation will be recalled in Section 2.4. As part of the construction, the closed 3-manifold M in Theorem 1.6 is obtained by doing an integral Dehn-filling along a fundamental shadow link complement M_{FSL}, and as a consequence, we have the following

Theorem 1.7. *For any fundamental shadow link complement M_{FSL}, there exists a $\delta > 0$ such that if the closed oriented hyperbolic 3-manifold M obtained by doing an integral Dehn-filling along M_{FSL} satisfies*

$$\mathrm{Vol}(M) > \mathrm{Vol}(M_{FSL}) - \delta,$$

then Conjecture 1.1 and Conjecture 1.2 hold for M.

Remark 1.8. With techniques in [40, 34, 18], one can generalize Theorem 1.7 to any closed or cusped hyperbolic 3-manifold M obtained by doing a rational Dehn-filling along some components of M_{FSL} that satisfies $\mathrm{Vol}(M) > \mathrm{Vol}(M_{\mathrm{FSL}}) - \delta$.

The main technical difficulty we overcome in this paper is the following. In a direct application of Ohtsuki's method that writes the summation in the definition of the Reshetikhin-Turaev invariant into the summation of Fourier coefficients (each of which is a contour integral) then estimates each of these contour integrals using the saddle point approximation, there are certain problematic Fourier coefficients

that cannot be estimated by the usual method. That is essentially why the assumption that the cone angles are less than π is required in [34].

To resolve this issue, we introduce the *extended quantum* $6j$*-symbols* (see Definition 3.5) and establish a *Weak Reflection Symmetry* for them (see Proposition 3.7 and Theorem 6.2). We then carefully add suitable extra terms arising from the extended $6j$-symbols into the summation defining the Reshetikhin-Turaev invariant (see Formula (4.1)). These additional terms, on the one hand, do not change the asymptotic behavior of the invariant (see Propositions 4.2 and 4.4) and, on the other hand, annihilate those problematic Fourier coefficients (see Propositions 4.3 and 4.5).

To prove the Weak Reflection Symmetry of the extended quantum $6j$-symbols, we relate them to the b-$6j$ symbols, objects that have recently been intensively studied by the second and third authors and their collaborators in [23, 24] and satisfy a reflection symmetry. This relation is established using a *real Gevrey*-1 *estimate* and a truncated Borel-Laplace transform developed in this paper (see Section 6 and Appendix A).

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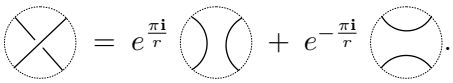
2 Preliminaries

Most of the material in this section, except Proposition 2.15 (2), is taken from [41, 23, 24] and is included here for the reader's convenience.

2.1 Relative Reshetikhin-Turaev invariants

We recall the relative Reshetikhin-Turaev invariants [3, 22] following the skein theoretical approach and focusing on the $\mathrm{SO}(3)$-theory. In particular, we consider the values at the root of unity $q = e^{\frac{2\pi\mathbf{i}}{r}}$ for odd integers $r \geqslant 3$. To be more precise, we will follow the normalization of the invariants as described in [22].

A framed link in an oriented 3-manifold M is a smooth embedding L of a disjoint union of finitely many thickened circles $\mathrm{S}^1 \times [0,1]$ into M. The Kauffman bracket skein module $\mathrm{K}_r(M)$ of M is the $\mathbb{C}$-module generated by the isotopic classes of framed links in M modulo the following two relations:

(1) *Kauffman Bracket Skein Relation:*  .

(2) *Framing Relation:* $L \cup$  $= \left(-e^{\frac{2\pi\mathbf{i}}{r}} - e^{-\frac{2\pi\mathbf{i}}{r}} \right) L.$

There is a canonical isomorphism

$$\langle\ \rangle : \mathrm{K}_r(\mathrm{S}^3) \to \mathbb{C}$$

defined by sending the empty link to 1. The image $\langle L \rangle$ of the framed link L is called the Kauffman bracket of L.

Let $\mathrm{K}_r(A\times[0,1])$ be the Kauffman bracket skein module of the product of an annulus A with a closed interval. For any link diagram D in $\mathbb{R}^2$ with k ordered components and $b_1,\dots,b_k\in \mathrm{K}_r(A\times[0,1])$, let

$$\langle b_1,\dots,b_k\rangle_D$$

be the complex number obtained by cabling $b_1,\dots,b_k$ along the components of D considered as an element of $K_r(\mathrm{S}^3)$ then taking the Kauffman bracket $\langle\ \rangle$.

On $\mathrm{K}_r(A\times[0,1])$ there is a commutative multiplication induced by the juxtaposition of annuli, making it a $\mathbb{C}$-algebra; and as a $\mathbb{C}$-algebra $\mathrm{K}_r(A\times[0,1])\cong\mathbb{C}[z]$, where z is the core curve of A. For an integer $n\geqslant 0$, let $e_n(z)$ be the n-th Chebyshev polynomial defined recursively by $e_0(z)=1$, $e_1(z)=z$ and $e_n(z)=ze_{n-1}(z)-e_{n-2}(z)$. Let $\mathrm{I}_r=\{0,2,\dots,r-3\}$ be the set of even integers in between 0 and $r-2$. Then the Kirby coloring $\Omega_r\in\mathrm{K}_r(A\times[0,1])$ is defined by

$$\Omega_r=\mu_r\sum_{n\in\mathrm{I}_r}[n+1]e_n,$$

where

$$\mu_r=\frac{2\sin\frac{2\pi}{r}}{\sqrt{r}}$$

and $[n]$ is the quantum integer defined by

$$[n]=\frac{q^n-q^{-n}}{q-q^{-1}}=\frac{\sin\frac{2n\pi}{r}}{\sin\frac{2\pi}{r}}.$$

Let M be a closed oriented 3-manifold and let L be a framed link in M with n components. Suppose M is obtained from S^3 by doing a surgery along a framed link L', $D(L')$ is a standard diagram of L' (i.e., the blackboard framing of $D(L')$ coincides with the framing of L'). Then L adds extra components to $D(L')$ forming a linking diagram $D(L\cup L')$ with $D(L)$ and $D(L')$ linking in possibly a complicated way. Let U_+ be the diagram of the unknot with framing 1, $\sigma(L')$ be the signature of the linking matrix of L' and $\boldsymbol{b}=(b_1,\dots,b_n)$ be a multi-element of I_r. Then the r-th *relative Reshetikhin-Turaev invariant of M with L colored by $\boldsymbol{b}$* is defined as

$$\mathrm{RT}_r(M,L,\boldsymbol{b})=\mu_r\langle e_{b_1},\dots,e_{b_n},\Omega_r,\dots,\Omega_r\rangle_{D(L\cup L')}\langle\Omega_r\rangle_{U_+}^{-\sigma(L')}. \tag{2.1}$$

Note that if $L=\emptyset$ or $b_1=\cdots=b_n=0$, then $\mathrm{RT}_r(M,L,\boldsymbol{b})=\mathrm{RT}_r(M)$, the r-th Reshetikhin-Turaev invariant of M.

2.2 Hyperbolic cone manifolds

According to [8], a 3-dimensional *hyperbolic cone-manifold* is a 3-manifold M, which can be triangulated so that the link of each simplex is piecewise linear homeomorphic to a standard sphere and M is equipped with a complete path metric such that the restriction of the metric to each simplex is isometric to a hyperbolic geodesic simplex. The *singular locus* L of a cone-manifold M consists of the points with no neighborhood isometric to a ball in a Riemannian manifold. It follows that

(1) L is a link in M such that each component is a closed geodesic.

(2) At each point of L there is a *cone angle* θ which is the sum of dihedral angles of 3-simplices containing the point.

(3) The restriction of the metric on $M\smallsetminus L$ is a smooth hyperbolic metric, but is incomplete if $L\neq\emptyset$.

Hodgson-Kerckhoff [19] proved that hyperbolic cone metrics on M with singular locus L are locally parametrized by the cone angles provided all the cone angles are less than or equal to 2π, and Kojima [21] proved that hyperbolic cone manifolds (M, L) are globally rigid provided all the cone angles are less than or equal to π. It is expected to be globally rigid if all the cone angles are less than or equal to 2π.

For a 3-manifold N with boundary consisting of tori $T_1, \dots, T_n$, with a choice of generators (u_i, v_i) for $\pi_1(T_i)$ and an integer p_i for each $i \in \{1, \dots, n\}$, the $(p_1, \dots, p_n)$-Dehn filling on N attaches a solid torus to each T_i so that $p_i u_i + v_i$ bounds a disk. If $\mathrm{H}(u_i)$ and $\mathrm{H}(v_i)$ are respectively the logarithmic holonomy for u_i and v_i, then a solution to

$$p_i \mathrm{H}(u_i) + \mathrm{H}(v_i) = \mathbf{i}\theta_i \tag{2.2}$$

near the complete structure gives a cone-manifold structure on the resulting manifold M with the cone angle θ_i along the core curve L_i of the solid torus attached to T_i; it is a smooth structure if $\theta_1 = \cdots = \theta_n = 2\pi$.

In this setting, the Chern-Simons invariant for a hyperbolic cone manifold (M, L) can be defined by using the Neumann-Zagier potential function [31]. To do this, we need a framing on each component, namely, a choice of a curve γ_i on T_i that is isotopic to the core curve L_i of the solid torus attached to T_i. If we choose the orientation of γ_i so that $(p_i u_i + v_i) \cdot \gamma_i = 1$, then we can let $\gamma_i = -u_i$. We consider the following function

$$\frac{\Phi(\mathrm{H}(u_1), \dots, \mathrm{H}(u_n))}{\mathbf{i}} - \sum_{i=1}^{n} \frac{\mathrm{H}(u_i)\mathrm{H}(v_i)}{4\mathbf{i}} + \sum_{i=1}^{n} \frac{\theta_i \mathrm{H}(\gamma_i)}{4},$$

where Φ is the Neumann-Zagier potential function (see [31]) defined on the deformation space of hyperbolic structures on $M \smallsetminus L$ parametrized by the holonomy of the meridians $\{\mathrm{H}(u_i)\}$, characterized by

$$\begin{cases} \frac{\partial \Phi(\mathrm{H}(u_1), \dots, \mathrm{H}(u_n))}{\partial \mathrm{H}(u_i)} = \frac{\mathrm{H}(v_i)}{2}, \\ \\ \Phi(0, \dots, 0) = \mathbf{i}\Big(\mathrm{Vol}(M \smallsetminus L) + \mathbf{i}\mathrm{CS}(M \smallsetminus L)\Big) \qquad \mod \pi^2\mathbb{Z}, \end{cases} \tag{2.3}$$

where $M \smallsetminus L$ is with the complete hyperbolic metric. Another important feature of Φ is that it is even in each of its variables $\mathrm{H}(u_i)$.

Following the argument in [31, Sections 4 & 5], one can prove that if the cone angles of components of L are $\theta_1, \dots, \theta_n$, then

$$\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) = \mathrm{Re}\bigg(\frac{\Phi(\mathrm{H}(u_1), \dots, \mathrm{H}(u_n))}{\mathbf{i}} - \sum_{i=1}^{n} \frac{\mathrm{H}(u_i)\mathrm{H}(v_i)}{4\mathbf{i}} + \sum_{i=1}^{n} \frac{\theta_i \mathrm{H}(\gamma_i)}{4} \bigg); \tag{2.4}$$

and in [43], Yoshida proved that when $\theta_1 = \cdots = \theta_n = 2\pi$,

$$\mathrm{Vol}(M) + \mathbf{i}\mathrm{CS}(M) = \frac{\Phi(\mathrm{H}(u_1), \dots, \mathrm{H}(u_n))}{\mathbf{i}} - \sum_{i=1}^{n} \frac{\mathrm{H}(u_i)\mathrm{H}(v_i)}{4\mathbf{i}} + \sum_{i=1}^{n} \frac{\theta_i \mathrm{H}(\gamma_i)}{4} \qquad \mod \mathbf{i}\pi^2\mathbb{Z}. \tag{2.5}$$

Definition 2.1. *[41, Definition 2.1] The* Chern-Simons invariant *of a hyperbolic cone manifold $M_{L_{\boldsymbol{\theta}}}$ with the framing $(\gamma_1, \dots, \gamma_n)$ is defined as*

$$\mathrm{CS}(M_{L_{\boldsymbol{\theta}}}) = \mathrm{Im}\bigg(\frac{\Phi(\mathrm{H}(u_1), \dots, \mathrm{H}(u_n))}{\mathbf{i}} - \sum_{i=1}^{n} \frac{\mathrm{H}(u_i)\mathrm{H}(v_i)}{4\mathbf{i}} + \sum_{i=1}^{n} \frac{\theta_i \mathrm{H}(\gamma_i)}{4} \bigg) \mod \pi^2\mathbb{Z}.$$

Then together with (2.4), we have

$$\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) + \mathbf{i}\mathrm{CS}(M_{L_{\boldsymbol{\theta}}}) = \frac{\Phi(\mathrm{H}(u_1), \ldots, \mathrm{H}(u_n))}{\mathbf{i}} - \sum_{i=1}^{n} \frac{\mathrm{H}(u_i)\mathrm{H}(v_i)}{4\mathbf{i}} + \sum_{i=1}^{n} \frac{\theta_i \mathrm{H}(\gamma_i)}{4} \qquad \mathrm{mod}\ \mathbf{i}\pi^2\mathbb{Z}. \tag{2.6}$$

2.3 Fundamental shadow link complements

If we take c truncated tetrahedra $\Delta_1, \ldots, \Delta_c$ and glue them together along the triangles of truncation, we obtain a (possibly non-orientable) handlebody of genus $c+1$ with a link in its boundary consisting of the edges of the truncated tetrahedra. By taking the orientable double (the orientable double covering with the boundary quotient out by the deck involution) of this handlebody, we obtain a link L_{FSL} inside $M_c = \#^{c+1}(S^2 \times S^1)$. We call a link obtained this way a *fundamental shadow link*, and its complement in M_c a *fundamental shadow link complement*. Alternatively, to construct a fundamental shadow link complement, we can first take the double of each tetrahedron along the hexagonal faces and then glue the resulting pieces together along homeomorphisms between the 3-punctured spheres coming from the double of the triangles of truncation. See Figure 1.

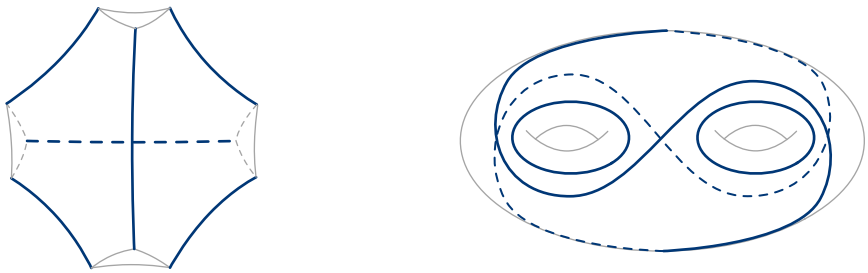


Figure 1: The handlebody on the right is obtained from the truncated tetrahedron on the left by identifying the triangles on the top and the bottom by a horizontal reflection and the triangles on the left and the right by a vertical reflection.

The fundamental importance of the family of the fundamental shadow link complements is the following

Theorem 2.2. *[11] Every compact oriented 3-manifold with toroidal or empty boundary can be obtained from a suitable fundamental shadow link complement by doing an integral Dehn-filling to some of the boundary components.*

A hyperbolic cone metric on M_c with singular locus L_{FSL} and with cone angles $\theta_1, \ldots, \theta_n$ can be constructed as follows. For each $s \in \{1, \ldots, c\}$, let $e_{s_1}, \ldots, e_{s_6}$ be the edges of the building block Δ_s, and let θ_{s_j} be half of the cone angle of the component of L_{FSL} containing e_{s_j}. If $\{\theta_{s_1}, \ldots, \theta_{s_6}\}$ form the set of dihedral angles of a truncated hyperideal tetrahedron, by abuse of notation still denoted by Δ_s, then the hyperbolic cone manifold M_c with singular locus L_{FSL} and cone angles $\theta_1, \ldots, \theta_n$, where n is the number of components of L_{FSL}, is obtained by gluing Δ_s's together along isometries of the triangles of truncation, and taking the double. In this metric, the logarithmic holonomy of the meridian u_i of a tubular neighborhood $N(L_i)$ of L_i satisfies

$$\mathrm{H}(u_i) = \mathbf{i}\theta_i. \tag{2.7}$$

A preferred longitude v_i on the boundary of $N(L_i)$ can be chosen as follows. Recall that a fundamental shadow link is obtained from the double of a set of truncated tetrahedra (along the hexagonal faces)

glued together by orientation preserving homeomorphisms between the three-punctured spheres coming from the double of the triangles of truncation, and recall also that the mapping class group of the three-punctured sphere is generated by mutations, which could be represented by the four 3-braids in Figure 2. For each mutation, we assign an integer ± 1 to each component of the braid as in Figure 2; and for a composition of a sequence of mutations, we assign the sum of the ± 1 assigned by the mutations to each component of the 3-braid. In this way, each orientation-preserving homeomorphism between the



Figure 2

three-punctured spheres assigns three integers to three of the components of L_{FSL}, one for each. For each $i \in \{1, \dots, n\}$, let ι_i be the sum of all the integers on L_i assigned by the homeomorphisms between the three-punctured spheres. Then we can choose a preferred longitude v_i on $N(L_i)$ such that $u_i \cdot v_i = 1$ and the logarithmic holonomy satisfies

$$\mathrm{H}(v_i) = -l_i + \mathbf{i}\frac{\iota_i \theta_i}{2}, \tag{2.8}$$

where l_i is the length of the closed geodesic L_i. In this way, a framing on L_i gives an integer p_i in the way that the parallel copy of L_i on $N(L_i)$ is isotopic to the curve representing $p_i u_i + v_i$.

In the complete hyperbolic metric, since $M_c \setminus L_{\text{FSL}}$ is the union of $2c$ regular ideal octahedra, we have the volume

$$\mathrm{Vol}(M_c \setminus L_{\text{FSL}}) = 2c v_8, \tag{2.9}$$

where v_8 is the volume of the regular ideal octahedron; and by [28, Corollary 2.5] and [27, Theorem 2.4], the Chern-Simons invariant

$$\mathrm{CS}(M_c \setminus L_{\text{FSL}}) = \Big(\sum_{i=1}^{n} \frac{\iota_i}{2}\Big)\pi^2 \qquad \text{mod } \pi^2\mathbb{Z}. \tag{2.10}$$

Proposition 2.3 ([9, 10]). *Let $L_{FSL} = L_1 \cup \cdots \cup L_n$ be a framed fundamental shadow link in $M_c = \#^{c+1}(S^2 \times S^1)$ with framing p_i on the component L_i, and with a coloring $\boldsymbol{a} = (a_1, \dots, a_n)$ of even integers in $\{0, 2, \dots, r-3\}$. Then the r-th relative Reshetikhin-Turaev invariant*

$$\mathrm{RT}_{\mathrm{r}}(M_c, L_{FSL}, \boldsymbol{a}) = \left(\frac{2\sin\frac{2\pi}{r}}{\sqrt{r}}\right)^{-c} \prod_{i=1}^{n} (-1)^{\frac{\iota_i a_i}{2}} q^{(p_i + \frac{\iota_i}{2})\frac{a_i(a_i+2)}{2}} \prod_{s=1}^{c} \begin{vmatrix} a_{s_1} & a_{s_2} & a_{s_3} \\ a_{s_4} & a_{s_5} & a_{s_6} \end{vmatrix},$$

where $a_{s_1}, \dots, a_{s_6}$ are the colors of the edges of the building block Δ_s inherited from the color $\boldsymbol{a}$ on L_{FSL}, and $\begin{vmatrix} a_{s_1} & a_{s_2} & a_{s_3} \\ a_{s_4} & a_{s_5} & a_{s_6} \end{vmatrix}$ is the quantum $6j$-symbol of $(a_{s_1}, \dots, a_{s_6})$ to be recalled in Section 2.5.

2.4 The change-of-pair operations

Let M be a closed oriented 3-manifold and let L be a framed link in M. The change-of-pair operation is a topological operation that changes the pair (M, L) without changing the complement $M \setminus L$, and these operations connect all such pairs that have homeomorphic complements.

Precisely, suppose $L = L_1 \cup \cdots \cup L_n$. For each component $L_i : [0,1] \times S^1 \to M$ of L, recall that $L_i(\{0\} \times S^1) \subset M$ is the core curve of L_i and $L_i(\{1\} \times S^1) \subset M$ is the parallel copy. For each i, let L_i^* be the framed unknot in $M \setminus L$ whose core curve is isotopic to the meridian of the tubular neighborhood of L_i, and let $L^* = L_1^* \cup \cdots \cup L_n^*$. Let M^* be the closed 3-manifold obtained from M by doing the surgery along L and consider L^* as a framed link in $M^* = M_L$ via the inclusions $L^* \subset M \setminus L \subset M_L$. The *change-of-pair operation* $T_{(L;L^*)}$ is defined by sending (M, L) to (M^*, L^*). See Figure 3.

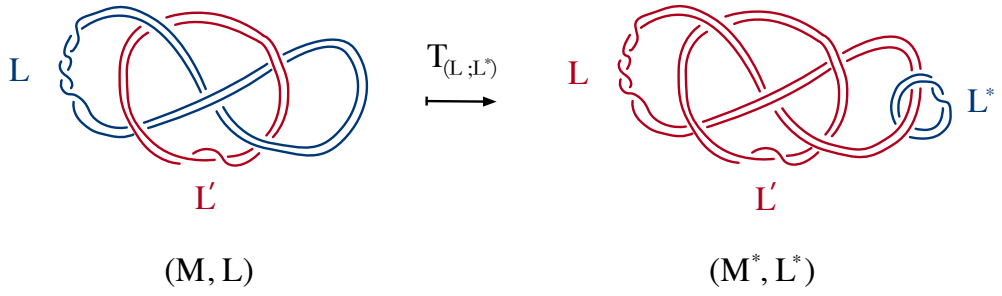


Figure 3: M is obtained from S^3 by the surgery along L'. M^* is obtained from M by doing the surgery along L, hence is obtained from S^3 by doing the surgery along $L' \cup L$.

Proposition 2.4. *[41, Propositions 1.3 and 1.4]*

(1) If (M^*, L^*) is obtained from (M, L) by doing a change-of-pair operation, then $M^* \setminus L^*$ is homeomorphic to $M \setminus L$.

(2) Every two pairs (M, L) and (M^*, L^*) such that $M \setminus L$ is homeomorphic to $M^* \setminus L^*$ are related by a sequence of the change-of-pair operations.

For the change-of-pair operations in full generality and the relationship with the discrete Fourier transforms, see [41, Sections 1.2 and 1.3].

2.5 Quantum $6j$-symbols

For $r \geqslant 3$ odd, let $q = e^{\frac{2\pi \mathbf{i}}{r}}$. For $n \in \mathbb{Z}_{\geqslant 0}$, let

$$[n] = \frac{q^n - q^{-n}}{q - q^{-1}} = \frac{\sin \frac{2n\pi}{r}}{\sin \frac{2\pi}{r}},$$

and let

$$\{n\}! = \prod_{k=1}^{n} \{k\} \quad \text{and} \quad \{0\}! = 0.$$

A triple (a_1, a_2, a_3) of even integers in $\{0, 2, \ldots, r-3\}$ is *r-admissible* if

(1) $a_i + a_j - a_k \geqslant 0$ for $\{i, j, k\} = \{1, 2, 3\}$, and

(2) $a_1 + a_2 + a_3 \leqslant 2(r-2)$.

A six-tuple $(a_1, \ldots, a_6)$ is *r-admissible* if the triples (a_1, a_2, a_3), (a_1, a_5, a_6), (a_2, a_4, a_6) and (a_3, a_4, a_5) are r-admissible. In the rest of this article, we let

$$T_1 = \frac{a_1 + a_2 + a_3}{2}, \quad T_2 = \frac{a_1 + a_5 + a_6}{2}, \quad T_3 = \frac{a_2 + a_4 + a_6}{2}, \quad T_4 = \frac{a_3 + a_4 + a_5}{2},$$
$$Q_1 = \frac{a_1 + a_2 + a_4 + a_5}{2}, \quad Q_2 = \frac{a_1 + a_3 + a_4 + a_6}{2} \quad \text{and} \quad Q_3 = \frac{a_2 + a_3 + a_5 + a_6}{2}.$$

Definition 2.5. *The* quantum $6j$-symbol *of an r-admissible 6-tuple $(a_1,\dots,a_6)$ is*

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{vmatrix} = \mathbf{i}^{-\sum_{k=1}^{6} a_k} \frac{\prod_{i=1}^{4}\prod_{j=1}^{3}\sqrt{[Q_j - T_i]!}}{\prod_{i=1}^{4}\sqrt{[T_i+1]!}} \sum_{k=\max\{T_1,T_2,T_3,T_4\}}^{\min\{Q_1,Q_2,Q_3\}} \frac{(-1)^k[k+1]!}{\prod_{i=1}^{4}[k-T_i]!\prod_{j=1}^{3}[Q_j-k]!},$$

where, as a convention, $\sqrt{-x} \doteq \mathbf{i}\sqrt{x}$ for $x > 0$.

The quantum $6j$-symbols satisfy the following Tetrahedral Symmetry and Reflection Symmetry, where Proposition 2.6 follows directly from the definition.

Proposition 2.6.

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{vmatrix} = \begin{vmatrix} a_2 & a_1 & a_3 \\ a_5 & a_4 & a_6 \end{vmatrix} = \begin{vmatrix} a_1 & a_3 & a_2 \\ a_4 & a_6 & a_5 \end{vmatrix} = \begin{vmatrix} a_1 & a_5 & a_6 \\ a_4 & a_2 & a_3 \end{vmatrix}.$$

Proposition 2.7. *[12, Lemma A.3 (c)] If the six-tuple $(a_1,a_2,a_3,a_4,a_5,a_6)$ is r-admissible, then both of the six-tuples*

$$(a_1,a_2,a_3,r-2-a_4,r-2-a_5,r-2-a_6) \quad \textit{and} \quad (r-2-a_1,r-2-a_2,a_3,r-2-a_4,r-2-a_5,a_6)$$

are r-admissible, and

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ r-2-a_4 & r-2-a_5 & r-2-a_6 \end{vmatrix}$$

and

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{vmatrix} = \begin{vmatrix} r-2-a_1 & r-2-a_2 & a_3 \\ r-2-a_4 & r-2-a_5 & a_6 \end{vmatrix}.$$

Proposition 2.8. *[1, Theorem 1.2] For any r-admissible six-tuple $(a_1,\dots,a_6)$,*

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{vmatrix} \leqslant O\Big(re^{\frac{r}{2\pi}v_8}\Big),$$

where v_8 is the volume of the regular ideal octahedron.

Definition 2.9. *An r-admissible six-tuple $(a_1,\dots,a_6)$ is called* r-hyperideal *if for $\{i,j,k\} = \{1,2,3\}$, $\{1,5,6\}$, $\{2,4,6\}$ and $\{3,4,5\}$,*

(1) $0 \leqslant a_i + a_j - a_k \leqslant r-2$, *and*

(2) $r-2 \leqslant a_i + a_j + a_k \leqslant 2(r-2)$.

Using the same argument as in the proof of [1, Theorem 1.2], one can prove the following

Proposition 2.10. *There exists a $\delta_1 > 0$ such that for any r-admissible six-tuple $(a_1,\dots,a_6)$ that is not r-hyperideal,*

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{vmatrix} < O\Big(e^{\frac{r}{2\pi}(v_8-\delta_1)}\Big).$$

2.6 b-6j symbols

In [24], b-6j symbols are defined as a variant of the $6j$-symbols of the positive representations of the modular double of $\mathrm{U}_q\mathfrak{sl}(2;\mathbb{R})$, which are the main building blocks of the $\mathrm{U}_{q\widetilde{q}}\mathfrak{sl}(2;\mathbb{R})$ Turaev-Viro invariants [23, 25] for hyperbolic 3-manifolds with cusp ends or with totally geodesic boundary.

For $b \in (0,1)$, let $Q = b + b^{-1}$. Let S_b be the *double sine function* defined for $z \in \mathbb{C}$ with $0 < \mathrm{Re}(z) < Q$ by

$$S_b(z) = \exp\left(\int_\Omega \frac{\sinh\left(\left(\frac{Q}{2} - z\right)t\right)}{4t\sinh(\frac{bt}{2})\sinh(\frac{t}{2b})}dt\right), \tag{2.11}$$

where the contour Ω goes along the real line and passes above the pole at the origin. For a six-tuple $(a_1,\dots,a_6) \in \mathbb{R}^6$, in the rest of this article we let

$$t_1 = a_1+a_2+a_3, \quad t_2 = a_1+a_5+a_6, \quad t_3 = a_2+a_4+a_6, \quad t_4 = a_3+a_4+a_5,$$
$$q_1 = a_1+a_2+a_4+a_5, \quad q_2 = a_1+a_3+a_4+a_6, \quad q_3 = a_2+a_3+a_5+a_6 \quad \text{and} \quad q_4 = 2Q.$$

A six-tuple $(a_1,\dots,a_6) \in \mathbb{R}^6$ is called *b-admissible* if

$$0 < q_j - t_i < Q$$

for all $i,j \in \{1,2,3,4\}$. In particular, if $(a_1,\dots,a_6)$ is b-admissible, then

$$\max\{t_1,t_2,t_3,t_4\} < \min\{q_1,q_2,q_3,q_4\}.$$

Definition 2.11. *[24, Definition 1.1]. The* b-6j symbol *for a b-admissible six-tuple $(a_1,\dots,a_6) \in \mathbb{R}^6$ is given by*

$$\begin{Bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Bmatrix}_b = \int_\Gamma \frac{\prod_{i=1}^4 S_b(u-t_i)\prod_{j=1}^4 S_b(q_j-u)}{\prod_{i=1}^4\prod_{j=1}^4\sqrt{S_b(q_j-t_i)}}d\mathrm{Im}u, \tag{2.12}$$

where, as a convention, the square root of the double sine function is defined as

$$\sqrt{S_b(z)} = \exp\left(\frac{1}{2}\int_\Omega \frac{\sinh\left(\left(\frac{Q}{2} - z\right)t\right)}{4t\sinh(\frac{bt}{2})\sinh(\frac{t}{2b})}dt\right),$$

and the contour Γ is any vertical line passing the interval $(\max\{t_i\},\min\{q_j\})$. See Figure 4.

Proposition 2.12. *[24, Proposition 3.1] The integral in (2.12) converges absolutely, and is independent of the choice of the vertical line Γ.*

The b-6j symbols satisfy the following Tetrahedral Symmetry and Reflection Symmetry.

Proposition 2.13. *[24, Proposition 3.3]*

$$\begin{Bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Bmatrix}_b = \begin{Bmatrix} a_2 & a_1 & a_3 \\ a_5 & a_4 & a_6 \end{Bmatrix}_b = \begin{Bmatrix} a_1 & a_3 & a_2 \\ a_4 & a_6 & a_5 \end{Bmatrix}_b = \begin{Bmatrix} a_1 & a_5 & a_6 \\ a_4 & a_2 & a_3 \end{Bmatrix}_b.$$

Proposition 2.14. *[24, Propositions 3.4] If $(a_1,\dots,a_6)$ is b-admissible, then $(Q-a_1,a_2,\dots,a_6)$ is also b-admissible, and*

$$\begin{Bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Bmatrix}_b = \begin{Bmatrix} Q-a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Bmatrix}_b.$$

Moreover, the same result holds if we change $a_k \mapsto Q - a_k$ for any $k \in \{1,\dots,6\}$.

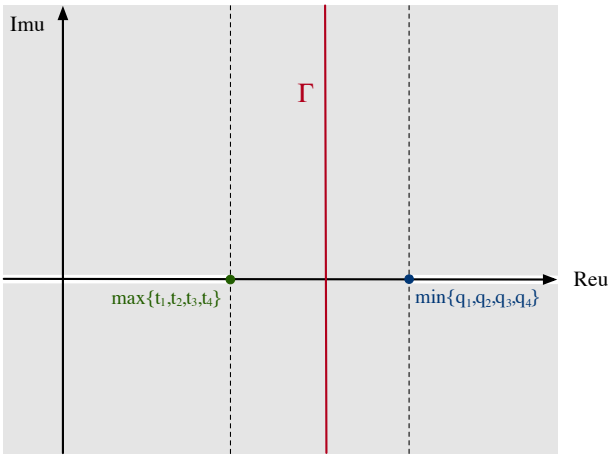


Figure 4: The contour Γ.

For a b-admissible six-tuple $(a_1, \dots, a_6)$, we will consider the following new set of variables, which will be intensively used later.

$$\alpha_k = 2\pi b a_k - \pi b^2 \text{ for } k \in \{1, \dots, 6\}, \quad \xi = \pi b u, \tag{2.13}$$

$$\tau_i = \pi b t_i - \frac{3\pi b^2}{2} \text{ for } i \in \{1, 2, 3, 4\}, \quad \text{and} \quad \eta_j = \pi b q_j - 2\pi b^2 \text{ for } j \in \{1, 2, 3, 4\}. \tag{2.14}$$

Then

$$\tau_1 = \frac{\alpha_1 + \alpha_2 + \alpha_3}{2}, \quad \tau_2 = \frac{\alpha_1 + \alpha_5 + \alpha_6}{2}, \quad \tau_3 = \frac{\alpha_2 + \alpha_4 + \alpha_6}{2}, \quad \tau_4 = \frac{\alpha_3 + \alpha_4 + \alpha_5}{2},$$
$$\eta_1 = \frac{\alpha_1 + \alpha_2 + \alpha_4 + \alpha_5}{2}, \quad \eta_2 = \frac{\alpha_1 + \alpha_3 + \alpha_4 + \alpha_6}{2}, \quad \eta_3 = \frac{\alpha_2 + \alpha_3 + \alpha_5 + \alpha_6}{2} \quad \text{and} \quad \eta_4 = 2\pi.$$

Let $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_6)$ and let

$$\begin{aligned} U_b(\boldsymbol{\alpha}, \xi) = &- \pi \mathbf{i} b^2 \sum_{i=1}^{4} \sum_{j=1}^{4} \log S_b\left(\frac{\eta_j - \tau_i}{\pi b} + \frac{b}{2}\right) \\ &+ 2\pi \mathbf{i} b^2 \sum_{i=1}^{4} \log S_b\left(\frac{\xi - \tau_i - 2\pi b^2}{\pi b} + \frac{b}{2}\right) + 2\pi \mathbf{i} b^2 \sum_{j=1}^{4} \log S_b\left(\frac{\eta_j - \xi + \frac{3\pi b^2}{2}}{\pi b} + \frac{b}{2}\right). \end{aligned} \tag{2.15}$$

Then we have

$$\begin{Bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Bmatrix}_b = \frac{1}{\pi \mathbf{i} b} \int_\Gamma \exp\left(\frac{U_b(\boldsymbol{\alpha}, \xi)}{2\pi \mathbf{i} b^2}\right) d\xi, \tag{2.16}$$

where Γ is a vertical straight line that intersects the real axis in the interval $\left(\max\{\tau_i\} + \frac{3\pi b^2}{2}, \min\{\eta_j\} + 2\pi b^2\right)$.

2.7 Double sine function and dilogarithm function

By the functional equation (see e.g. [37, A.15])

$$S_b\left(z + b^{\pm 1}\right) = 2 \sin\left(\pi b^{\pm 1} z\right) S_b(z), \tag{2.17}$$

the double sine function S_b extended to a meromorphic function on $\mathbb{C}$ with the set of poles $\{-nb - mb^{-1} \mid m, n \in \mathbb{Z}_{\geqslant 0}\}$ and the set of zeros $\{Q+nb+mb^{-1} \mid m, n \in \mathbb{Z}_{\geqslant 0}\}$. In the rest of this paper, we will intensively use the function $\log S_b\Big(\frac{x}{\pi b} + \frac{b}{2}\Big)$. For $-\frac{\pi b^2}{2} < \mathrm{Re} x < \pi + \frac{\pi b^2}{2}$ so that $0 < \mathrm{Re}\Big(\frac{x}{\pi b} + \frac{b}{2}\Big) < Q$, we let

$$\log S_b\Big(\frac{x}{\pi b} + \frac{b}{2}\Big) = \int_\Omega \frac{\sinh\Big(\big(\frac{1}{2b} - \frac{x}{\pi b}\big)t\Big)}{4t\sinh(\frac{bt}{2})\sinh(\frac{t}{2b})}dt.$$

The function $\log S_b\Big(\frac{x}{\pi b} + \frac{b}{2}\Big)$ can be holomorphically extended to the region $\mathbb{C}\smallsetminus(-\infty, 0] \cup [\pi, \infty)$.

Let $\log : \mathbb{C}\smallsetminus(-\infty, 0] \to \mathbb{C}$ be the standard logarithm function defined by

$$\log z = \log|z| + \mathbf{i}\arg z$$

with $-\pi < \arg z < \pi$. The dilogarithm function $\mathrm{Li}_2 : \mathbb{C}\smallsetminus(1, \infty) \to \mathbb{C}$ is defined by

$$\mathrm{Li}_2(z) = -\int_0^z \frac{\log(1-u)}{u}du$$

where the integral is along any path in $\mathbb{C}\smallsetminus(1, \infty)$ connecting 0 and z, which is holomorphic in $\mathbb{C}\smallsetminus[1, \infty)$ and continuous in $\mathbb{C}\smallsetminus(1, \infty)$. The following closely related complexified Lobachevsky function

$$L(x) = x^2 - \pi x + \frac{\pi^2}{6} - \mathrm{Li}_2\big(e^{2\mathbf{i}x}\big) \tag{2.18}$$

will be used intensively in the rest of the paper, which is continuous on $\mathbb{C}\smallsetminus(-\infty, 0) \cup (\pi, \infty)$ and holomorphic on $\mathbb{C}\smallsetminus(-\infty, 0] \cup [\pi, \infty)$. The name of the function comes from the fact that for $x \in [0, \pi]$,

$$L(x) = -2\mathbf{i}\Lambda(x), \tag{2.19}$$

where Λ is the Lobachevsky function defined by

$$\Lambda(x) = -\int_0^x \log|2\sin t|dt.$$

Proposition 2.15. *(1) [23, Proposition 2.1] Let $\nu_b(x)$ be such that*

$$2\pi\mathbf{i}b^2 \log S_b\Big(\frac{x}{\pi b} + \frac{b}{2}\Big) - L(x) = \nu_b(x)b^4.$$

Then for any $\epsilon > 0$, there exists a $B_\epsilon > 0$ independent of b such that for all $b \in (0, 1)$ and for all $x \in \mathbb{C}$ with $\epsilon < \mathrm{Re} x < \pi - \epsilon$,

$$|\nu_b(x)| < B_\epsilon.$$

(2) There exists a $B > 0$ such that for all $b \in (0, 1)$ and for all $x \in (0, \pi)$,

$$\left|2\pi\mathbf{i}b^2 \log S_b\Big(\frac{x}{\pi b} + \frac{b}{2}\Big) - L(x)\right| < Bb^2.$$

Proof. We only need to prove (2). By [13, (B.33)], for $z \in \mathbb{C}$ with $0 < \mathrm{Re} z < Q$, we have

$$\log S_b(z) = \int_0^\infty \left(\frac{\sinh\big((\frac{Q}{2} - z)t\big)}{2\sinh\big(\frac{bt}{2}\big)\sinh\big(\frac{t}{2b}\big)} - \frac{Q-2z}{t}\right)\frac{dt}{t}.$$

For $x \in \mathbb{C}$ with $0 < \mathrm{Re}x < \pi$, we have $0 < \mathrm{Re}\big(\frac{x}{\pi b} + \frac{b}{2}\big) < Q$; and by the change of variable $s = \frac{t}{2\pi b}$, we have

$$2\pi\mathbf{i}b^2 \log S_b\Big(\frac{x}{\pi b} + \frac{b}{2}\Big) = 2\pi\mathbf{i}\int_0^\infty \left(\frac{b^2 \sinh\big((\pi - 2x)s\big)}{2\sinh\big(\pi b^2 s\big)\sinh\big(\pi s\big)} - \frac{(\pi - 2x)}{2\pi^2 s}\right)\frac{ds}{s}. \tag{2.20}$$

Then for any $\epsilon > 0$ and $x \in \mathbb{C}$ with $\epsilon < \mathrm{Re}x < \pi - \epsilon$, by (1) and letting $b \to 0$, we have

$$L(x) = 2\pi\mathbf{i}\int_0^\infty \left(\frac{\sinh\big((\pi - 2x)s\big)}{2\pi s\sinh\big(\pi s\big)} - \frac{(\pi - 2x)}{2\pi^2 s}\right)\frac{ds}{s}.$$

Letting $\epsilon \to 0$, we have

$$L(x) = 2\pi\mathbf{i}\int_0^\infty \left(\frac{\sinh\big((\pi - 2x)s\big)}{2\pi s\sinh\big(\pi s\big)} - \frac{(\pi - 2x)}{2\pi^2 s}\right)\frac{ds}{s} \tag{2.21}$$

for all $x \in \mathbb{C}$ with $0 < \mathrm{Re}x < \pi$. By (2.20) and (2.21), we have

$$2\pi\mathbf{i}b^2 \log S_b\Big(\frac{x}{\pi b} + \frac{b}{2}\Big) - L(x) = \mathbf{i}\int_0^\infty \left(\frac{\pi b^2 s}{\sinh(\pi b^2 s)} - 1\right)\frac{\sinh\big((\pi - 2x)s\big)}{s^2\sinh\big(\pi s\big)}ds.$$

Now, since

$$0 < \frac{\pi b^2 s}{\sinh(\pi b^2 s)} < 1$$

for all $s \in (0, +\infty)$, and

$$\left|\frac{\sinh\big((\pi - 2x))s\big)}{\sinh\big(\pi s\big)}\right| < 1$$

for all $x \in (0, \pi)$ and $s \in (0, +\infty)$, we have

$$\begin{aligned}\left|2\pi\mathbf{i}b^2 \log S_b\Big(\frac{x}{\pi b} + \frac{b}{2}\Big) - L(x)\right| <& \int_0^\infty \left(\frac{1}{s^2} - \frac{\pi b^2}{s\sinh(\pi b^2 s)}\right)ds \\ =& \pi b^2 \int_0^\infty \left(\frac{1}{u^2} - \frac{1}{u\sinh u}\right)du\end{aligned}$$

for all $x \in (0, \pi)$, where the equality comes from the change of variable $u = \pi b^2 s$.

Finally, let

$$B = \pi\int_0^\infty \left(\frac{1}{u^2} - \frac{1}{u\sinh u}\right)du.$$

As $u \to 0^+$, we have

$$\frac{1}{u^2} - \frac{1}{u\sinh u} = \frac{1}{6} + O(u^2);$$

and as $u \to +\infty$, we have

$$\frac{1}{u^2} - \frac{1}{u\sinh u} = O(u^{-2}).$$

Hence the improper integral converges at both 0 and $+\infty$, and $B < +\infty$. This completes the proof. $\square$

3 Extended quantum $6j$-symbols

In this section, we let $r \geqslant 3$ be an odd integer, and let $q = e^{\frac{2\pi\mathbf{i}}{r}}$. For $n \in \mathbb{Z}_{\geqslant 0}$, we let

$$\{n\} = q^n - q^{-n} = 2\mathbf{i}\sin\frac{2n\pi}{r},$$

and let

$$\{n\}! = \prod_{k=1}^{n}\{k\} \quad \text{and} \quad \{0\}! = 0.$$

Lemma 3.1. *Let $b = \sqrt{\frac{2}{r}} \in (0,1)$, and let S_b be the double sine function defined in (2.11).*

(1) For $0 \leqslant n \leqslant r-2$,

$$\{n\}! = \mathbf{i}^n b^{-1} S_b\big((n+1)b\big). \tag{3.1}$$

(2) For $\frac{r-1}{2} \leqslant n \leqslant r-2$,

$$\{n\}! = \frac{(-1)^{n+\frac{r-1}{2}} 2\mathbf{i}^n b^{-1}}{S_b\big((r-n)b\big)}. \tag{3.2}$$

Proof. Letting $z = bk$ in (2.17), we have

$$2\sin\Big(\frac{2k\pi}{r}\Big) = 2\sin\big(\pi k b^2\big) = \frac{S_b\big((k+1)b\big)}{S_b(kb)},$$

from which and Lemma 3.2 below, (1) follows.

To see (2), by (1), we have

$$\{n\}! = \mathbf{i}^n b^{-1} \frac{S_b\Big(\big(n+1\big)b\Big)}{S_b\Big(\big(n+1-\frac{r}{2}\big)b\Big)} S_b\bigg(\Big(n+1-\frac{r}{2}\Big)b\bigg); \tag{3.3}$$

by (2.17) with $z = \big(n+1-\frac{r}{2}\big)b = (n+1)b - b^{-1}$, we have

$$\frac{S_b\Big(\big(n+1\big)b\Big)}{S_b\Big(\big(n+1-\frac{r}{2}\big)b\Big)} = 2\sin\bigg(\Big(n+1-\frac{r}{2}\Big)\pi\bigg) = (-1)^{n+\frac{r-1}{2}}2; \tag{3.4}$$

and by the functional equation (see e.g. [37, A.16])

$$S_b(z)S_b(Q-z) = 1, \tag{3.5}$$

we have

$$S_b\bigg(\Big(n+1-\frac{r}{2}\Big)b\bigg) = \frac{1}{S_b\big((r-n)b\big)}. \tag{3.6}$$

Putting (3.3), (3.4) and (3.6) together, we have (2). □

We consider (3.2) because the function S_b has poles on the right of Q, and to avoid these poles we need to move the variables there into $(0, Q)$.

Lemma 3.2. *For $b \in (0,1)$, $S_b(b) = b$.*

Proof. By (2.17), we have

$$\frac{S_b(z+b)}{S_b(z+b^{-1})} = \frac{\sin(\pi z b)}{\sin(\pi z b^{-1})};$$

and letting $z \to 0$, we have

$$\frac{S_b(b)}{S_b(b^{-1})} = b^2. \tag{3.7}$$

By (3.5), we have

$$S_b(b)S_b(b^{-1}) = 1. \tag{3.8}$$

Putting (3.7) and (3.8) together, we have that $S_b(b) = \pm b$.

It now suffices to show that $S_b(b) > 0$. To see this, by [37, A.7, A.13], we have $\overline{S_b(z)} = S_b(\overline{z})$. As a consequence, $S_b(z)$ is real for all $z \in (0, Q)$. Since S_b has no zeros in $(0, Q)$ and $S_b\big(\frac{Q}{2}\big) = 1 > 0$, we have $S_b(z) > 0$ for all $z \in (0, Q)$. In particular, $S_b(b) > 0$. □

As a direct consequence of Lemma 3.1, we have

Proposition 3.3. *Let $Q_4 = r-2$. Then the quantum $6j$-symbol of an r-hyperideal six-tuple $(a_1, \dots, a_6)$ can be computed as*

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{vmatrix} = \frac{(-1)^{\frac{r+1}{2}}2}{r} \sin\frac{2\pi}{r} \sum_{k=\max\{T_1,\dots,T_4\}}^{\min\{Q_1,\dots,Q_4\}} \frac{\prod_{i=1}^4 \prod_{j=1}^4 \sqrt{S_b\big((Q_j - T_i + 1)b\big)}}{\prod_{i=1}^4 S_b\big((k - T_i + 1)b\big) \prod_{j=1}^4 S_b\big((Q_j - k + 1)b\big)}.$$

Definition 3.4. *A six-tuple $(a_1, \dots, a_6)$ of real numbers in $[0, r-2]$ is called* r-hyperideal *if for $\{i, j, k\} = \{1, 2, 3\}, \{1, 5, 6\}, \{2, 4, 6\}$ and $\{3, 4, 5\}$,*

(1) $0 \leqslant a_i + a_j - a_k \leqslant r - 2$, *and*

(2) $r - 2 \leqslant a_i + a_j + a_k \leqslant 2(r-2)$.

Definition 3.5. *The* extended quantum $6j$-symbol *of an r-hyperideal six-tuple $(a_1, \dots, a_6) \in [0, r-2]^6$ is defined as*

$$\left\|\begin{matrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{matrix}\right\| \doteq \frac{(-1)^{\frac{r+1}{2}}2}{r} \sin\frac{2\pi}{r} \sum_k \frac{\prod_{i=1}^4 \prod_{j=1}^4 \sqrt{S_b\big((Q_j - T_i + 1)b\big)}}{\prod_{i=1}^4 S_b\big((k - T_i + 1)b\big) \prod_{j=1}^4 S_b\big((Q_j - k + 1)b\big)},$$

where $Q_4 = r - 2$, and the summation is over all the integers k in between $\max\{T_1, T_2, T_3, T_4\}$ and $\min\{Q_1, Q_2, Q_3, Q_4\}$. As a convention, if there is no integer in between, we define $\left\|\begin{matrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{matrix}\right\| \doteq 0$.

The extended quantum $6j$-symbols satisfy the following basic properties.

Proposition 3.6. *(1) For any r-admissible six-tuple $(a_1, \dots, a_6)$ of the hyperideal type,*

$$\left\|\begin{matrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{matrix}\right\| = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{vmatrix}.$$

(2) For any r-hyperideal six-tuple $(a_1, \dots, a_6)$ of real numbers,

$$\left\|\begin{matrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{matrix}\right\| = \left\|\begin{matrix} a_2 & a_1 & a_3 \\ a_5 & a_4 & a_6 \end{matrix}\right\| = \left\|\begin{matrix} a_1 & a_3 & a_2 \\ a_4 & a_6 & a_5 \end{matrix}\right\| = \left\|\begin{matrix} a_1 & a_5 & a_6 \\ a_4 & a_2 & a_3 \end{matrix}\right\|.$$

Proof. (1) follows directly from Proposition 3.3, and (2) follows directly from the definition. □

The following Proposition 3.7, whose proof is postponed to Section 6, will play the central role in the proof of Theorem 1.6 and Theorem 1.7.

Proposition 3.7. *There exists a $\delta_2 > 0$ such that for any r-hyperideal six-tuple $(a_1, \dots, a_6)$ of real numbers,*

$$\left| \left\| \begin{matrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{matrix} \right\| - \left\| \begin{matrix} r-2-a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{matrix} \right\| \right| < O\Big(e^{\frac{r}{2\pi}(v_8-\delta_2)}\Big),$$

where v_8 is the volume of the regular ideal octahedron. Moreover, the same result holds if we change $a_k \mapsto r-2-a_k$ for any $k \in \{1, \dots, 6\}$.

In the rest of this paper, we will intensively use the following change of variables

$$\begin{aligned}
\alpha_k &= \frac{2\pi(a_k+1)}{r} = \pi b^2 a_k + \pi b^2 \text{ for } k \in \{1, \dots, 6\}, \quad \xi = \frac{2\pi k}{r} = \pi b^2 k, \\
\tau_i &= \frac{2\pi T_i}{r} + \frac{3\pi}{r} = \pi b^2 T_i + \frac{3\pi b^2}{2} \text{ for } i \in \{1,2,3,4\}, \text{and} \\
\eta_j &= \frac{2\pi Q_j}{r} + \frac{4\pi}{r} = \pi b^2 Q_j + 2\pi b^2 \text{ for } j \in \{1,2,3,4\}.
\end{aligned}$$

Then

$$\begin{aligned}
&\tau_1 = \frac{\alpha_1+\alpha_2+\alpha_3}{2}, \quad \tau_2 = \frac{\alpha_1+\alpha_5+\alpha_6}{2}, \quad \tau_3 = \frac{\alpha_2+\alpha_4+\alpha_6}{2}, \quad \tau_4 = \frac{\alpha_3+\alpha_4+\alpha_5}{2}, \\
&\eta_1 = \frac{\alpha_1+\alpha_2+\alpha_4+\alpha_5}{2}, \ \eta_2 = \frac{\alpha_1+\alpha_3+\alpha_4+\alpha_6}{2}, \ \eta_3 = \frac{\alpha_2+\alpha_3+\alpha_5+\alpha_6}{2}, \quad \text{and} \quad \eta_4 = 2\pi.
\end{aligned}$$

In these new variables, we have

$$\left\| \begin{matrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{matrix} \right\| = e^{\left(\frac{r+1}{2}\right)\pi\mathbf{i}} \frac{2}{r} \sin\frac{2\pi}{r} \sum_k e^{\frac{r}{4\pi\mathbf{i}}\left(U_r(\alpha_1,\dots,\alpha_6,\xi)\right)}, \tag{3.9}$$

where the summation is over the integers k in the interval $[\max\{T_1, \dots, T_4\}, \min\{Q_1, \dots, Q_4\}]$, and

$$\begin{aligned}
U_r(\alpha_1, \dots, \alpha_6, \xi) =& \frac{1}{2}\sum_{i=1}^4\sum_{j=1}^4 2\pi\mathbf{i}b^2 \log S_b\left(\frac{\eta_j-\tau_i}{\pi b} + \frac{b}{2}\right) \\
&- \sum_{i=1}^4 2\pi\mathbf{i}b^2 \log S_b\left(\frac{\xi-\tau_i+2\pi b^2}{\pi b} + \frac{b}{2}\right) - \sum_{j=1}^4 2\pi\mathbf{i}b^2 \log S_b\left(\frac{\eta_j-\xi-\frac{3\pi b^2}{2}}{\pi b} + \frac{b}{2}\right).
\end{aligned} \tag{3.10}$$

In light of Proposition 2.15, we define

$$U(\alpha_1, \dots, \alpha_6, \xi) = \frac{1}{2}\sum_{i=1}^4\sum_{j=1}^4 L(\eta_j-\tau_i) - \sum_{i=1}^4 L(\xi-\tau_i) - \sum_{j=1}^4 L(\eta_j-\xi), \tag{3.11}$$

$$\kappa(\alpha_1, \dots, \alpha_6, \xi) = 16\pi^2 + 14\pi\sum_{i=1}^4 \tau_i - 56\pi\xi - 8\pi\mathbf{i}\sum_{i=1}^4 \log\left(1-e^{2\mathbf{i}(\xi-\tau_i)}\right) + 6\pi\mathbf{i}\sum_{j=1}^4 \log\left(1-e^{2\mathbf{i}(\eta_j-\xi)}\right), \tag{3.12}$$

$$\nu_r(\alpha_1, \dots, \alpha_6, \xi) = \left(U_r(\alpha_1, \dots, \alpha_6, \xi) - U(\alpha_1, \dots, \alpha_6, \xi) - \frac{\kappa(\alpha_1, \dots, \alpha_6, \xi)}{r}\right) r^2. \tag{3.13}$$

Then we have

$$\left\|\begin{matrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{matrix}\right\| = e^{\left(\frac{r+1}{2}\right)\pi\mathbf{i}}\frac{2}{r}\sin\frac{2\pi}{r}\sum_k e^{\frac{r}{4\pi\mathbf{i}}\left(U(\alpha_1,\ldots,\alpha_6,\xi)+\frac{\kappa(\alpha_1,\ldots,\alpha_6,\xi)}{r}+\frac{\nu_r(\alpha_1,\ldots,\alpha_6,\xi)}{r^2}\right)}, \tag{3.14}$$

where the summation is over the integers k in the interval $[\max\{T_1,\ldots,T_4\},\min\{Q_1,\ldots,Q_4\}]$ as in (3.9).

The new variables also satisfy $\alpha_k \in [\frac{2\pi}{r}, 2\pi - \frac{2\pi}{r}]$ for each $k \in \{1,\ldots,6\}$, and for $\{i,j,k\} = \{1,2,3\}, \{1,5,6\}, \{2,4,6\}$ and $\{3,4,5\}$, $\frac{2\pi}{r} \leqslant \alpha_i+\alpha_j-\alpha_k \leqslant 2\pi-\frac{2\pi}{r}$, and $2\pi+\frac{2\pi}{r} \leqslant \alpha_i+\alpha_j+\alpha_k \leqslant 4\pi-\frac{2\pi}{r}$. These conditions suggest us to make the following

Definition 3.8. *A six-tuple* $(\alpha_1,\ldots,\alpha_6) \in (0,2\pi)^6$ *is of the* hyperideal type *if for* $\{i,j,k\} = \{1,2,3\}$, $\{1,5,6\}$, $\{2,4,6\}$ *and* $\{3,4,5\}$,

(1) $0 < \alpha_i + \alpha_j - \alpha_k < 2\pi$, *and*

(2) $2\pi < \alpha_i + \alpha_j + \alpha_k < 4\pi$.

In the rest of this paper, we let

$$\mathrm{B} = \Big\{(\alpha_1,\ldots,\alpha_6,\xi) \in (0,2\pi)^6\times\mathbb{R} \;\Big|\; (\alpha_1,\ldots,\alpha_6) \text{ is of the hyperideal type, and } \max\{\tau_i\} < \xi < \min\{\eta_j\}\Big\};$$

and for $\epsilon > 0$, let

$$\mathrm{B}_\epsilon = \Big\{(\alpha_1,\ldots,\alpha_6,\xi) \in \mathrm{B} \;\Big|\; d\big((\alpha_1,\ldots,\alpha_6),\partial\mathrm{B}\big) > \epsilon\Big\}.$$

By (2.19), for $(\alpha_1,\ldots,\alpha_6,\xi)$ in the region B, we have

$$U(\alpha_1,\ldots,\alpha_6,\xi) = 2\mathbf{i}V(\alpha_1,\ldots,\alpha_6,\xi), \tag{3.15}$$

where $V : \mathrm{B} \to \mathbb{R}$ is given by

$$V(\alpha_1,\ldots,\alpha_6,\xi) \doteq -\frac{1}{2}\sum_{i=1}^4\sum_{j=1}^4\Lambda(\eta_j-\tau_i) + \sum_{i=1}^4\Lambda(\xi-\tau_i) + \sum_{j=1}^4\Lambda(\eta_j-\xi). \tag{3.16}$$

A result of Costantino [9] shows that for each $\boldsymbol{\alpha} = (\alpha_1,\ldots,\alpha_6)$ of the hyperideal type, there exists a unique $\xi_{\boldsymbol{\alpha}}$ so that $(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}) \in \mathrm{B}$ and

$$\frac{\partial V(\boldsymbol{\alpha},\xi)}{\partial\xi}\Big|_{\xi=\xi_{\boldsymbol{\alpha}}} = 0. \tag{3.17}$$

Indeed, he proves that for each fixed $\boldsymbol{\alpha}$, V is strictly concave down in ξ, i.e.,

$$\frac{\partial^2 V(\boldsymbol{\alpha},\xi)}{\partial\xi^2} < 0, \tag{3.18}$$

and with derivatives $\pm\infty$ at the boundary points of the interval of ξ, hence there is a unique critical point $\xi_{\boldsymbol{\alpha}}$ at which V achieves the absolute maximum. Moreover, by using the Murakami-Yano formula [30, 39] he shows that

$$V(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}) = \mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}), \tag{3.19}$$

where $\Delta_{\boldsymbol{\alpha}}$ is the truncated hyperideal tetrahedron with dihedral angles $|\alpha_1-\pi|,\ldots,|\alpha_6-\pi|$. As a consequence, for any $(\boldsymbol{\alpha},\xi) \in \mathrm{B}$, we have

$$V(\boldsymbol{\alpha},\xi) \leqslant V(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}) = \mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) \leqslant v_8 = V\Big(\boldsymbol{\pi},\frac{7\pi}{4}\Big), \tag{3.20}$$

where v_8 is the volume of the regular ideal octahedron; and the equality holds if and only if $(\boldsymbol{\alpha}, \xi) = \left(\boldsymbol{\pi}, \frac{7\pi}{4}\right)$. This, in particular, implies that

$$\max_{(\boldsymbol{\alpha},\xi)\in \mathrm{B}\setminus \mathrm{B}_\epsilon} V(\boldsymbol{\alpha}, \xi) < v_8 \tag{3.21}$$

for any sufficiently small $\epsilon > 0$; and for each $\boldsymbol{\alpha}$ of the hyperideal type,

$$\max_{\xi\in(\max\{\tau_i\},\min\{\eta_j\})\setminus[\xi_{\boldsymbol{\alpha}}-\epsilon,\xi_{\boldsymbol{\alpha}}+\epsilon]} V(\boldsymbol{\alpha}, \xi) < \mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) \tag{3.22}$$

for any sufficiently small $\epsilon > 0$.

The function U extends holomorphically to

$$\mathrm{B}_{\mathbb{C}} = \Big\{(\alpha_1, \dots, \alpha_6, \xi) \in \mathbb{C}^7 \;\Big|\; (\mathrm{Re}\alpha_1, \dots, \mathrm{Re}\alpha_6, \xi) \in \mathrm{B}\Big\}.$$

For $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_6) \in \mathbb{C}^6$ such that $(\mathrm{Re}\alpha_1, \dots, \mathrm{Re}\alpha_6)$ is of the hyperideal type, we let $\xi_{\boldsymbol{\alpha}}$ be such that

$$\frac{\partial U(\boldsymbol{\alpha}, \xi)}{\partial \xi}\Big|_{\xi=\xi_{\boldsymbol{\alpha}}} = 0. \tag{3.23}$$

Following the idea of [30], see also [2], it is proved that there is at most one $\xi_{\boldsymbol{\alpha}}$ such that $(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}}) \in \mathrm{B}_{\mathbb{C}}$; and it is not clear that $(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}})$ is always in $\mathrm{B}_{\mathbb{C}}$.

4 Computation of the Reshetikhin-Turaev invariants

4.1 The relative case

Let $L_{\mathrm{FSL}} = L_1 \cup \cdots \cup L_n$ be a fundamental shadow link in $M_c = \#^{c+1}(S^2 \times S^1)$, and let $U_{c+1} \subset S^3$ be the disjoint union of $c+1$ unknots with the 0-framings by doing surgery along which we get M_c. Let (M, L) be the pair obtained from (M_c, L_{FSL}) by doing a change-of-pair operation $T_{(L_{\mathrm{FSL}};L)}$ as introduced in Section 2.4, i.e., $M = (M_c)_{L_{\mathrm{FSL}}}$ and $L = \cup_{i=1}^n L_i^*$ and L_i^* is the framed unknot in $M_c \setminus L_{\mathrm{FSL}}$ with the core curve isotopic to the meridian of the tubular neighborhood of L_i. Suppose L_i^* has the framing q_i and L_i has the framing p_i for each $i \in \{1, \dots, n\}$. Let $\boldsymbol{b}^{(r)}$ be a sequence of colorings of L, where $b_i^{(r)} \in \{0, 2, \dots, r-3\}$ is an even integer for each i. Assume that for each i, either $b_i^{(r)} > \frac{r-1}{2}$ for all r or $b_i^{(r)} < \frac{r-1}{2}$ for all r. In the former case we let $\mu_i = 1$ and in the latter case we let $\mu_i = -1$, and we let

$$\theta_i^{(r)} = \mu_i\left(\frac{4\pi(b_i^{(r)}+1)}{r} - 2\pi\right) = \left|\frac{4\pi(b_i^{(r)}+1)}{r} - 2\pi\right|.$$

Let $\boldsymbol{\theta}^{(r)} = (\theta_1^{(r)}, \dots, \theta_n^{(r)})$, and let $M^{(r)} = M_{L_{\theta^{(r)}}}$ be the hyperbolic cone manifold consisting of M and a hyperbolic cone metric on M with singular locus L and cone angles $\boldsymbol{\theta}^{(r)}$, if it exists, and let $\mathrm{Vol}(M^{(r)})$ and $\mathrm{CS}(M^{(r)})$ respectively be the volume and the Chern-Simons invariant of $M^{(r)}$. For $i \in \{1, \dots, n\}$, let u_i and v_i respectively be the meridian and the preferred longitude of a tubular neighborhood of L_i, let $m_i = p_i u_i + v_i$ be the meridian of a tubular neighborhood of L_i^* and let $\mathbf{m} = (m_i)_{i\in\{1,\dots,n\}}$. Let $\rho_{M^{(r)}} : \pi_1(M \setminus L) \to \mathrm{PSL}(2;\mathbb{C})$ be the holonomy representation of the restriction of hyperbolic cone metric $M^{(r)}$ to $M \setminus L$, and let $\mathbb{T}_{(M\setminus L,\mathbf{m})}([\rho_{M^{(r)}}])$ be the Reidemeister torsion of $M \setminus L$ twisted by the adjoint action of $\rho_{M^{(r)}}$.

Also let

$$\theta_i = \lim_{r\to\infty} \theta_i^{(r)}$$

for $i \in \{1, \dots, n\}$, let $\boldsymbol{\theta} = (\theta_1, \dots, \theta_n)$, let M_{L_θ} be the hyperbolic cone manifold consisting of M and a hyperbolic cone metric on M with singular locus L and cone angles $\boldsymbol{\theta}$, if it exists, and let $\mathrm{Vol}(M_{L_\theta})$ and $\mathrm{CS}(M_{L_\theta})$ respectively be the volume and the Chern-Simons invariant of M_{L_θ}. Then Theorem 1.6 can be rephrased as follows.

Theorem 4.1. *There exists a $\delta > 0$ such that if*

$$\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) > \mathrm{Vol}(M_c \smallsetminus L_{FSL}) - \delta,$$

then the following hold:

(1) As r varies over all positive odd integers,

$$\mathrm{RT}_r\big(M, L, \boldsymbol{b}^{(r)}\big) = \frac{C_r}{\sqrt{\pm \mathbb{T}_{(M \smallsetminus L, \mathbf{m})}([\rho_{M^{(r)}}])}} e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M^{(r)}) + \mathbf{i}\mathrm{CS}(M^{(r)})\right)} \Big(1 + O\Big(\frac{1}{r}\Big)\Big),$$

where C_r is a quantity of norm 1, which is independent of the geometric structure on M.

(2) As r varies over all positive odd integers,

$$\lim_{r\to\infty} \frac{4\pi}{r} \log \mathrm{RT}_r\big(M, L, \boldsymbol{b}^{(r)}\big) = \mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) + \mathbf{i}\mathrm{CS}(M_{L_{\boldsymbol{\theta}}}) \mod \mathbf{i}\pi^2\mathbb{Z}.$$

Clearly, (2) is a direct consequence of (1); and the equivalence between (1) and the original statement of Conjecture 1.4 can be seen from Remark 1.5.

In the rest of this section, we will drop the superscript and write $\boldsymbol{b} = \boldsymbol{b}^{(r)}$ for the simplicity of the notations, and keep in mind that it is a quantity depending on r. Then the r-th relative Reshetikhin-Turaev invariant of (M, L) with coloring $\boldsymbol{b}$ can be computed as

$$\mathrm{RT}_r\big(M, L, \boldsymbol{b}\big) = A_{\boldsymbol{b},r} \sum_{\boldsymbol{a}} h_{\boldsymbol{b},r}(\boldsymbol{a}),$$

where

$$A_{\boldsymbol{b},r} = \left(\frac{2\sin\frac{2\pi}{r}}{\sqrt{r}}\right)^{n-c} e^{-\sigma(U_{c+1}\cup L)\left(-\frac{3}{r} - \frac{r+1}{4}\right)\pi\mathbf{i}} \prod_{i=1}^{n} q^{\frac{q_i b_i (b_i+2)}{2}},$$

the sum is over all multi-even integers $\boldsymbol{a} = (a_i)_{i\in\{1,\dots,n\}}$ in $\{0, 2, \dots, r-3\}$,

$$h_{\boldsymbol{b},r}(\boldsymbol{a}) = \prod_{i=1}^{n} (-1)^{\frac{\iota_i a_i}{2}} q^{\left(p_i + \frac{\iota_i}{2}\right)\frac{a_i(a_i+2)}{2}} [(a_i+1)(b_i+1)] \prod_{s=1}^{c} \begin{vmatrix} a_{s_1} & a_{s_2} & a_{s_3} \\ a_{s_4} & a_{s_5} & a_{s_6} \end{vmatrix},$$

and $a_{s_1}, \dots, a_{s_6}$ are the colors of the edges of the building block Δ_s inherited from the colors on L_{FSL}, with the convention that $\begin{vmatrix} a_{s_1} & a_{s_2} & a_{s_3} \\ a_{s_4} & a_{s_5} & a_{s_6} \end{vmatrix} = 0$ if the six-tuple $(a_{s_1}, \dots, a_{s_6})$ is not r-admissible. As a_i is even for each i, $(-1)^{(p-\iota_i+1)a_i} = 1$, multiplying which to $h_{\boldsymbol{b},r}(\boldsymbol{a})$ we have

$$h_{\boldsymbol{b},r}(\boldsymbol{a}) = \prod_{i=1}^{n} (-1)^{(p_i - \frac{\iota_i}{2})a_i} q^{\left(p_i + \frac{\iota_i}{2}\right)\frac{a_i(a_i+2)}{2}} (-1)^{a_i} [(a_i+1)(b_i+1)] \prod_{s=1}^{c} \begin{vmatrix} a_{s_1} & a_{s_2} & a_{s_3} \\ a_{s_4} & a_{s_5} & a_{s_6} \end{vmatrix}.$$

Now let

$$\mathrm{H}_r = \Big\{ \boldsymbol{a} \in [0, r-2]^n \;\Big|\; \text{for each building block } \Delta_s, (a_{s_1}, \dots, a_{s_6}) \text{ is } r\text{-hyperideal} \Big\}.$$

Then by Proposition 2.10 and the constant δ_1 therein, we have

$$\mathrm{RT}_r\big(M,L,\boldsymbol{b}\big)=A_{\boldsymbol{b},r}\sum_{\boldsymbol{a}\in \mathrm{H}_r\cap(2\mathbb{Z})^n} h_{\boldsymbol{b},r}(\boldsymbol{a})+O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{\mathrm{FSL}})-\delta_1\right)}\Big).$$

On H_r, we define the function

$$h_{\boldsymbol{b},r}(\boldsymbol{a})\doteq\prod_{i=1}^{n}(-1)^{(p_i-\frac{\iota_i}{2})a_i}q^{\left(p_i+\frac{\iota_i}{2}\right)\frac{a_i(a_i+2)}{2}}(-1)^{a_i}[(a_i+1)(b_i+1)]\prod_{s=1}^{c}\left\|\begin{matrix}a_{s_1}&a_{s_2}&a_{s_3}\\a_{s_4}&a_{s_5}&a_{s_6}\end{matrix}\right\|,$$

where $\left\|\begin{matrix}a_{s_1}&a_{s_2}&a_{s_3}\\a_{s_4}&a_{s_5}&a_{s_6}\end{matrix}\right\|$ is the extended quantum $6j$-symbol of $(a_{s_1},\dots,a_{s_6})$ defined in Definition 3.5. Then by Proposition 3.7 and the constant δ_2 therein, we have

$$\mathrm{RT}_r\big(M,L,\boldsymbol{b}\big)=\frac{A_{\boldsymbol{b},r}}{2^n}\sum_{\boldsymbol{a}\in \mathrm{H}_r\cap\mathbb{Z}^n} h_{\boldsymbol{b},r}(\boldsymbol{a})+O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{\mathrm{FSL}})-\delta_0\right)}\Big) \tag{4.1}$$

for any $\delta_0<\min\{\delta_1,\delta_2\}$.

Proposition 4.2.

$$\mathrm{RT}_r\big(M,L,\boldsymbol{b}\big)=B_r\sum_{\boldsymbol{a},\boldsymbol{k}}\Big(\sum_{\boldsymbol{\epsilon}} g^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}(\boldsymbol{a},\boldsymbol{k})\Big)+O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{FSL})-\delta_0\right)}\Big),$$

where

$$B_r=2^{-n}r^{-\frac{n+c}{2}}e^{\left(-\sigma(U_{c+1}\cup L)(-\frac{3}{r}-\frac{r+1}{4})-(\frac{r}{4}+1+\frac{1}{r})\sum_{i=1}^n(p_i+q_i)+(\frac{1}{2}-\frac{1}{2r})\sum_{i=1}^n\iota_i-\frac{r}{2}(n-c)-\frac{n}{2}\right)\pi\mathbf{i}},$$

$\boldsymbol{\epsilon}=(\epsilon_i)_{i\in\{1,\dots,n\}}\in\{-1,1\}^n$ *runs over all multi-signs,* $\boldsymbol{a}=(a_i)_{i\in\{1,\dots,n\}}$ *runs over multi-integers in* H_r, *and* $\mathbf{k}=(k_s)_{s\in\{1,\dots,c\}}$ *runs over all multi-integers with each* k_s *lying in between* $\max\{T_{s_1},T_{s_2},T_{s_3},T_{s_4}\}$ *and* $\min\{Q_{s_1},Q_{s_2},Q_{s_3},Q_{s_4}\}$, *with*

$$g^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}(\boldsymbol{a},\boldsymbol{k})=e^{\frac{r}{4\pi\mathbf{i}}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta},r}\left(\frac{2\pi(\boldsymbol{a}+\mathbf{1})}{r},\frac{2\pi\boldsymbol{k}}{r}\right)},$$

where $\frac{2\pi(\boldsymbol{a}+\mathbf{1})}{r}=\big(\frac{2\pi(a_i+1)}{r}\big)_{i\in\{1,\dots,n\}}$ *and* $\frac{2\pi\boldsymbol{k}}{r}=\big(\frac{2\pi k_s}{r}\big)_{s\in\{1,\dots,c\}}$, *and with* $\boldsymbol{\beta}=(\beta_i)_{i\in\{1,\dots,n\}}=\big(\frac{2\pi(b_i+1)}{r}\big)_{i\in\{1,\dots,n\}}$, $\boldsymbol{\alpha}=(\alpha_i)_{i\in\{1,\dots,n\}}$, $\boldsymbol{\xi}=(\xi_s)_{s\in\{1,\dots,c\}}$, *and*

$$\begin{aligned}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta},r}(\boldsymbol{\alpha},\boldsymbol{\xi})=&-\sum_{i=1}^n q_i(\beta_i-\pi)^2-\sum_{i=1}^n p_i(\alpha_i-\pi)^2-\sum_{i=1}^n 2\epsilon_i(\alpha_i-\pi)(\beta_i-\pi)\\&-\sum_{i=1}^n\frac{\iota_i}{2}(\alpha_i-\pi)^2+\sum_{s=1}^c U_r(\alpha_{s_1},\dots,\alpha_{s_6},\xi_s)+\Big(\sum_{i=1}^n\frac{\iota_i}{2}\Big)\pi^2.\end{aligned}$$

Proof. For an integer a and an even integer b, let $\alpha=\frac{2\pi(a+1)}{r}$ and $\beta=\frac{2\pi(b+1)}{r}$. Then we have

$$\begin{aligned}(-1)^a q^{(a+1)(b+1)}&=q^{(a+1-\frac{r}{2})(b+1-\frac{r}{2})+(a+1)r+\frac{br}{2}-\frac{r^2}{4}}=e^{\left(-\frac{r}{2}\right)\pi\mathbf{i}}e^{\frac{r}{4\pi\mathbf{i}}\left(-2(\alpha-\pi)(\beta-\pi)\right)},\\(-1)^a q^{-(a+1)(b+1)}&=q^{-(a+1-\frac{r}{2})(b+1-\frac{r}{2})-(a+1)r-\frac{br}{2}+\frac{r^2}{4}}=-e^{\left(-\frac{r}{2}\right)\pi\mathbf{i}}e^{\frac{r}{4\pi\mathbf{i}}\left(2(\alpha-\pi)(\beta-\pi)\right)}.\end{aligned}$$

As a consequence, we have

$$\prod_{i=1}^{n}(-1)^{a_i}[(a_i+1)(b_i+1)] = \frac{(-1)^{-\frac{r}{2}}}{\{1\}^n}\sum_{\boldsymbol{\epsilon}\in\{-1,1\}^n} e^{\frac{r}{4\pi\mathbf{i}}\sum_{i=1}^{n}\left(-2\epsilon_i(\alpha_i-\pi)(\beta_i-\pi)\right)}.$$

Then the result follows from (4.1), (3.9) and the following computations that for any integer a,

$$\begin{aligned}(-1)^a q^{\frac{a(a+2)}{2}} &= q^{\frac{1}{2}\left((a+1-\frac{r}{2})^2+2(a+1)r-\frac{r^2}{4}-r-1\right)} = e^{\left(-\frac{r}{4}-1-\frac{1}{r}\right)\pi\mathbf{i}} e^{\frac{r}{4\pi\mathbf{i}}\left(-(\alpha-\pi)^2\right)},\\ (-1)^{-\frac{a}{2}} q^{\frac{a(a+2)}{4}} &= q^{\frac{1}{4}\left((a+1-\frac{r}{2})^2-\frac{r^2}{4}+r-1\right)} = e^{\left(\frac{1}{2}-\frac{1}{2r}\right)\pi\mathbf{i}} e^{\frac{r}{4\pi\mathbf{i}}\left(-\frac{(\alpha-\pi)^2}{2}+\frac{\pi^2}{2}\right)}.\end{aligned}$$

□

Next we apply the Poisson Summation Formula. Notice that the summation in Proposition 4.2 is finite, and we need a summation over all the integer points to use the Poisson Summation Formula. For this reason, we consider the following regions and a bump function.

Let $\beta_i = \frac{2\pi(b_i+1)}{r}$ and $\alpha_i = \frac{2\pi(a_i+1)}{r}$ for $i \in \{1,\dots,n\}$, $\xi_s = \frac{2\pi k_s}{r}$ for $s \in \{1,\dots,c\}$, $\tau_{s_i} = \frac{2\pi T_{s_i}}{r} + \frac{3\pi}{r}$ for $i \in \{1,2,3,4\}$, and $\eta_{s_j} = \frac{2\pi Q_{s_j}}{r} + \frac{4\pi}{r}$ for $j \in \{1,2,3,4\}$. We let

$$\mathrm{D} = \Big\{(\boldsymbol{\alpha},\boldsymbol{\xi}) \;\Big|\; (\alpha_{s_1},\dots,\alpha_{s_6},\xi_s) \in \mathrm{B} \text{ for each } s \in \{1,\dots,c\}\Big\},$$

and for a sufficiently small $\epsilon > 0$, let

$$\mathrm{D}_\epsilon = \Big\{(\boldsymbol{\alpha},\boldsymbol{\xi}) \in \mathrm{D} \;\Big|\; d\big((\boldsymbol{\alpha},\boldsymbol{\xi}),\partial\mathrm{D}\big) > \epsilon\Big\},$$

where d is the Euclidean distance on $\mathbb{R}^{n+c}$. Let $\psi : \mathbb{R}^{n+c} \to \mathbb{R}$ be a C^∞-smooth bump function supported on $(\mathrm{D},\mathrm{D}_\epsilon)$, i.e.,

$$\begin{cases} \psi(\boldsymbol{\alpha},\boldsymbol{\xi}) = 1, & (\boldsymbol{\alpha},\boldsymbol{\xi}) \in \overline{\mathrm{D}_\epsilon} \\ 0 < \psi(\boldsymbol{\alpha},\boldsymbol{\xi}) < 1, & (\boldsymbol{\alpha},\boldsymbol{\xi}) \in \mathrm{D}\smallsetminus\overline{\mathrm{D}_\epsilon} \\ \psi(\boldsymbol{\alpha},\boldsymbol{\xi}) = 0, & (\boldsymbol{\alpha},\boldsymbol{\xi}) \notin \mathrm{D}, \end{cases}$$

let

$$f^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}(\boldsymbol{a},\boldsymbol{k}) = \psi\Big(\frac{2\pi(\boldsymbol{a}+\mathbf{1})}{r},\frac{2\pi\boldsymbol{k}}{r}\Big) g^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}(\boldsymbol{a},\boldsymbol{k}),$$

and let

$$f_{\boldsymbol{b},r} = \sum_{\boldsymbol{\epsilon}\in\{-1,1\}^n} f^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}.$$

Then

$$\mathrm{RT}_r\big(M,L,\boldsymbol{b}\big) = B_r \sum_{(\boldsymbol{a},\boldsymbol{k})\in\mathbb{Z}^{n+c}} f_{\boldsymbol{b},r}\big(\boldsymbol{a},\boldsymbol{k}\big) + \text{error term} + O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{\mathrm{FSL}})-\delta_0\right)}\Big),$$

where the error term comes from the use of the bump function. Since $f_{\boldsymbol{b},r}$ is C^∞-smooth and equals zero out of D, it is in the Schwartz space on $\mathbb{R}^{n+c}$. Then by the Poisson Summation Formula (see e.g. [36, Theorem 3.1]),

$$\sum_{(\boldsymbol{a},\boldsymbol{k})\in\mathbb{Z}^{n+c}} f_{\boldsymbol{b},r}\big(\boldsymbol{a},\boldsymbol{k}\big) = \sum_{(\boldsymbol{m},\boldsymbol{n})\in\mathbb{Z}^{n+c}} \widehat{f_{\boldsymbol{b},r}}(\boldsymbol{m},\boldsymbol{n}),$$

where $\boldsymbol{m} = (m_i)_{i\in\{1,\dots,n\}} \in \mathbb{Z}^n$, $\boldsymbol{n} = (n_s)_{s\in\{1,\dots,c\}} \in \mathbb{Z}^c$, and $\widehat{f_{\boldsymbol{b},r}}(\boldsymbol{m},\boldsymbol{n})$ is the $(\boldsymbol{m},\boldsymbol{n})$-th Fourier coefficient of $f_{\boldsymbol{b},r}$ defined by

$$\widehat{f_{\boldsymbol{b},r}}(\boldsymbol{m},\boldsymbol{n}) = \int_{\mathbb{R}^{n+c}} f_{\boldsymbol{b},r}\big(\boldsymbol{a},\boldsymbol{k}\big) e^{\sum_{i=1}^{n} 2\pi\mathbf{i} m_i a_i + \sum_{s=1}^{c} 2\pi\mathbf{i} n_s k_s} d\boldsymbol{a}d\boldsymbol{k},$$

where $d\boldsymbol{a}d\boldsymbol{k} = \prod_{i=1}^{n} da_i \prod_{s=1}^{c} dk_s$.

Proposition 4.3.

$$\mathrm{RT}_r\big(M,L,\boldsymbol{b}\big) = B_r \sum_{(\boldsymbol{m},\boldsymbol{n})\in\mathbb{Z}^{n+c}} \widehat{f_{\boldsymbol{b},r}}(\boldsymbol{m},\boldsymbol{n}) + \textit{error term} + O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{FSL})-\delta_0\right)}\Big), \tag{4.2}$$

where in terms of the variables $(\boldsymbol{\alpha},\boldsymbol{\xi})$*, the Fourier coefficients can be computed as*

$$\widehat{f_{\boldsymbol{b},r}}(\boldsymbol{m},\boldsymbol{n}) = \sum_{\boldsymbol{\epsilon}\in\{-1,1\}^n} \widehat{f^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}}(\boldsymbol{m},\boldsymbol{n})$$

with

$$\widehat{f^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}}(\boldsymbol{m},\boldsymbol{n}) = \Big(\frac{r}{2\pi}\Big)^{n+c}\int_{\mathrm{D}} \psi(\boldsymbol{\alpha},\boldsymbol{\xi}) e^{\frac{r}{4\pi\mathbf{i}}\left(\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta},r}(\boldsymbol{\alpha},\boldsymbol{\xi})-\sum_{i=1}^n 4\pi m_i\alpha_i-\sum_{s=1}^c 4\pi n_s\xi_s\right)} d\boldsymbol{\alpha}d\boldsymbol{\xi},$$

where $d\boldsymbol{\alpha}d\boldsymbol{\xi} = \prod_{i=1}^n d\alpha_i\prod_{s=1}^c d\xi_s$*, and* $\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta},r}(\boldsymbol{\alpha},\boldsymbol{\xi})$ *is as defined in Proposition 4.2.*

We will estimate the leading Fourier coefficients, the non-leading Fourier coefficients and the error term respectively in Sections 5.2, 5.3 and 5.4, and prove Theorem 4.1 (hence Theorem 1.6) in Section 5.5.

4.2 The closed case

In the setting of the previous section, the ambient manifold $M = (M_c)_{L_{\mathrm{FSL}}}$ is the closed oriented 3-manifold obtained from M_c by doing surgery along the framed fundamental shadow link L_{FSL}. As a consequence, $\mathrm{RT}_r(M) = \mathrm{RT}_r(M,L,\mathbf{0})$.

Let

$$\mathrm{H}_r = \Big\{\boldsymbol{a}\in[0,r-2]^n \;\Big|\; \text{for each building block } \Delta_s, (a_{s_1},\dots,a_{s_6}) \text{ is } r\text{-hyperideal}\Big\}$$

be as in the previous section. Then we have

Proposition 4.4.

$$\mathrm{RT}_r(M) = B_r\sum_{\boldsymbol{a},\boldsymbol{k}}\Big(\sum_{\boldsymbol{\epsilon}} g^{\boldsymbol{\epsilon}}_r(\boldsymbol{a},\boldsymbol{k})\Big) + O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{FSL})-\delta_0\right)}\Big),$$

where

$$B_r = 2^{-n}r^{-\frac{n+c}{2}}e^{\left(-\sigma(U_{c+1}\cup L)(-\frac{3}{r}-\frac{r+1}{4})-(\frac{r}{4}+1+\frac{1}{r})\sum_{i=1}^n p_i+(\frac{1}{2}-\frac{1}{2r})\sum_{i=1}^n \iota_i-\frac{r}{2}(n-c)-\frac{n}{2}\right)\pi\mathbf{i}},$$

$\boldsymbol{\epsilon} = (\epsilon_i)_{i\in\{1,\dots,n\}}\in\{-1,1\}^n$ *runs over all multi-signs,* $\boldsymbol{a} = (a_i)_{i\in\{1,\dots,n\}}$ *runs over multi-integers in* H_r*, and* $\boldsymbol{k} = (k_s)_{s\in\{1,\dots,c\}}$ *runs over all multi-integers with each* k_s *lying in between* $\max\{T_{s_1},T_{s_2},T_{s_3},T_{s_4}\}$ *and* $\min\{Q_{s_1},Q_{s_2},Q_{s_3},Q_{s_4}\}$*, with*

$$g^{\boldsymbol{\epsilon}}_r(\boldsymbol{a},\boldsymbol{k}) = e^{\sum_{i=1}^n\epsilon_i(\alpha_i-\pi)\mathbf{i}}\cdot e^{\frac{r}{4\pi\mathbf{i}}\mathcal{W}^{\boldsymbol{\epsilon}}_r\left(\frac{2\pi(\boldsymbol{a}+\mathbf{1})}{r},\frac{2\pi\boldsymbol{k}}{r}\right)},$$

where $\frac{2\pi(\boldsymbol{a}+\mathbf{1})}{r} = \big(\frac{2\pi(a_i+1)}{r}\big)_{i\in\{1,\dots,n\}}$*,* $\frac{2\pi\boldsymbol{k}}{r} = \big(\frac{2\pi k_s}{r}\big)_{s\in\{1,\dots,c\}}$*, and with* $\boldsymbol{\alpha} = (\alpha_i)_{i\in\{1,\dots,n\}}$*,* $\boldsymbol{\xi} = (\xi_s)_{s\in\{1,\dots,c\}}$*, and*

$$\begin{aligned}\mathcal{W}^{\boldsymbol{\epsilon}}_r(\boldsymbol{\alpha},\boldsymbol{\xi}) =& -\sum_{i=1}^n p_i(\alpha_i-\pi)^2 + \sum_{i=1}^n 2\epsilon_i\pi(\alpha_i-\pi)\\ &-\sum_{i=1}^n\frac{\iota_i}{2}(\alpha_i-\pi)^2 + \sum_{s=1}^c U_r(\alpha_{s_1},\dots,\alpha_{s_6},\xi_s) + \Big(\sum_{i=1}^n\frac{\iota_i}{2}\Big)\pi^2.\end{aligned}$$

Proof. Since the coloring $\boldsymbol{b} = \mathbf{0}$, we have $\beta_i = \frac{2\pi}{r}$ for each $i \in \{1, \dots, n\}$. Then the result follows directly from Proposition 4.2. □

To use the Poisson Summation Formula, we let

$$\mathrm{D} = \Big\{(\boldsymbol{\alpha}, \boldsymbol{\xi}) \;\Big|\; (\alpha_{s_1}, \dots, \alpha_{s_6}, \xi_s) \in \mathrm{B} \text{ for each } s \in \{1, \dots, c\}\Big\},$$

for a sufficiently small $\epsilon > 0$ let

$$\mathrm{D}_\epsilon = \Big\{(\boldsymbol{\alpha}, \boldsymbol{\xi}) \in \mathrm{D} \;\Big|\; d\big((\boldsymbol{\alpha}, \boldsymbol{\xi}), \partial\mathrm{D}\big) > \epsilon\Big\},$$

and let $\psi : \mathbb{R}^{n+c} \to \mathbb{R}$ be a C^∞-smooth bump function supported on $(\mathrm{D}, \mathrm{D}_\epsilon)$ as in the previous section. Let

$$f_r^{\boldsymbol{\epsilon}}(\boldsymbol{a}, \boldsymbol{k}) = \psi\Big(\frac{2\pi(\boldsymbol{a} + \mathbf{1})}{r}, \frac{2\pi\boldsymbol{k}}{r}\Big) g_r^{\boldsymbol{\epsilon}}(\boldsymbol{a}, \boldsymbol{k}),$$

and let $f_r = \sum_{\boldsymbol{\epsilon} \in \{-1,1\}^n} f_r^{\boldsymbol{\epsilon}}$. Then

$$\mathrm{RT}_r(M) = B_r \sum_{(\boldsymbol{a}, \boldsymbol{k}) \in \mathbb{Z}^{n+c}} f_r\big(\boldsymbol{a}, \boldsymbol{k}\big) + \text{error term} + O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{\mathrm{FSL}}) - \delta_0\right)}\Big),$$

where the error term comes from the use of the bump function; and by using the Poisson Summation Formula as in Proposition 4.3, we have

Proposition 4.5.

$$\mathrm{RT}_r(M) = B_r \sum_{(\boldsymbol{m}, \boldsymbol{n}) \in \mathbb{Z}^{n+c}} \widehat{f_r}(\boldsymbol{m}, \boldsymbol{n}) + \textit{error term} + O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{FSL}) - \delta_0\right)}\Big), \tag{4.3}$$

where in terms of the variables $(\boldsymbol{\alpha}, \boldsymbol{\xi})$, the Fourier coefficients can be computed as

$$\widehat{f_r}(\boldsymbol{m}, \boldsymbol{n}) = \sum_{\boldsymbol{\epsilon} \in \{-1,1\}^n} \widehat{f_r^{\boldsymbol{\epsilon}}}(\boldsymbol{m}, \boldsymbol{n})$$

with

$$\widehat{f_r^{\boldsymbol{\epsilon}}}(\boldsymbol{m}, \boldsymbol{n}) = \Big(\frac{r}{2\pi}\Big)^{n+c} \int_{\mathrm{D}} \psi(\boldsymbol{\alpha}, \boldsymbol{\xi}) e^{\sum_{i=1}^n \epsilon_i(\alpha_i - \pi)\mathbf{i}} e^{\frac{r}{4\pi\mathbf{i}}\left(\mathcal{W}_r^{\boldsymbol{\epsilon}}(\boldsymbol{\alpha}, \boldsymbol{\xi}) - \sum_{i=1}^n 4\pi m_i \alpha_i - \sum_{s=1}^c 4\pi n_s \xi_s\right)} d\boldsymbol{\alpha}_I d\boldsymbol{\xi},$$

where $d\boldsymbol{\alpha} d\boldsymbol{\xi} = \prod_{i=1}^n d\alpha_i \prod_{s=1}^c d\xi_s$, and $\mathcal{W}_r^{\boldsymbol{\epsilon}}(\boldsymbol{\alpha}, \boldsymbol{\xi})$ is as defined in Proposition 4.4.

We will estimate the leading Fourier coefficients, the non-leading Fourier coefficients and the error term respectively in Sections 5.2, 5.3 and 5.4, and prove Theorem 1.7 in Section 5.5.

5 Asymptotics

The goal of this section is to prove Theorem 1.6 (by proving Theorem 4.1) and Theorem 1.7. The argument follows almost verbatim that in [41, 23]. We will point out the main difference and give more details there. The main tool we use is still the following Saddle Point Approximations.

Proposition 5.1. *[41, Proposition 5.1] Let $D_{\boldsymbol{z}}$ be a region in $\mathbb{C}^n$ and let $D_{\boldsymbol{a}}$ be a region in $\mathbb{R}^k$. Let $f(\boldsymbol{z},\boldsymbol{a})$ and $g(\boldsymbol{z},\boldsymbol{a})$ be complex valued functions on $D_{\boldsymbol{z}}\times D_{\boldsymbol{a}}$ which are holomorphic in $\boldsymbol{z}$ and smooth in $\boldsymbol{a}$. For each positive integer r, let $f_r(\boldsymbol{z},\boldsymbol{a})$ be a complex valued function on $D_{\boldsymbol{z}}\times D_{\boldsymbol{a}}$ holomorphic in $\boldsymbol{z}$ and smooth in $\boldsymbol{a}$. For a fixed $\boldsymbol{a}\in D_{\boldsymbol{a}}$, let $f^{\boldsymbol{a}}$, $g^{\boldsymbol{a}}$ and $f_r^{\boldsymbol{a}}$ be the holomorphic functions on $D_{\boldsymbol{z}}$ defined by $f^{\boldsymbol{a}}(\boldsymbol{z})=f(\boldsymbol{z},\boldsymbol{a})$, $g^{\boldsymbol{a}}(\boldsymbol{z})=g(\boldsymbol{z},\boldsymbol{a})$ and $f_r^{\boldsymbol{a}}(\boldsymbol{z})=f_r(\boldsymbol{z},\boldsymbol{a})$. Suppose $\{\boldsymbol{a}_r\}$ is a convergent sequence in $D_{\boldsymbol{a}}$ with $\lim_r \boldsymbol{a}_r=\boldsymbol{a}_0$, $f_r^{\boldsymbol{a}_r}$ is of the form*

$$f_r^{\boldsymbol{a}_r}(\boldsymbol{z})=f^{\boldsymbol{a}_r}(\boldsymbol{z})+\frac{\upsilon_r(\boldsymbol{z},\boldsymbol{a}_r)}{r^2},$$

$\{S_r\}$ is a sequence of embedded real n-dimensional closed disks in $D_{\boldsymbol{z}}$ sharing the same boundary and converging to an embedded n-dimensional disk S_0, and $\boldsymbol{c}_r$ is a point on S_r such that $\{\boldsymbol{c}_r\}$ is convergent in $D_{\boldsymbol{z}}$ with $\lim_r \boldsymbol{c}_r=\boldsymbol{c}_0$. If for each r

(i) $\boldsymbol{c}_r$ *is a critical point of $f^{\boldsymbol{a}_r}$ in $D_{\boldsymbol{z}}$,*

(ii) $\mathrm{Re}f^{\boldsymbol{a}_r}(\boldsymbol{c}_r)>\mathrm{Re}f^{\boldsymbol{a}_r}(\boldsymbol{z})$ *for all $\boldsymbol{z}\in S_r\setminus\{\boldsymbol{c}_r\}$,*

(iii) the domain $\{\boldsymbol{z}\in D_{\boldsymbol{z}}\mid \mathrm{Re}f^{\boldsymbol{a}_r}(\boldsymbol{z})<\mathrm{Re}f^{\boldsymbol{a}_r}(\boldsymbol{c}_r)\}$ deformation retracts to $S_r\setminus\{\boldsymbol{c}_r\}$,

(iv) the Hessian matrix $\mathrm{Hess}(f^{\boldsymbol{a}_0})$ of $f^{\boldsymbol{a}_0}$ at $\boldsymbol{c}_0$ is non-singular,

(v) $|g^{\boldsymbol{a}_r}(\boldsymbol{c}_r)|$ is bounded from below by a positive constant independent of r, and

(vi) $|\upsilon_r(\boldsymbol{z},\boldsymbol{a}_r)|$ is bounded from above by a constant independent of r on $D_{\boldsymbol{z}}$,

then

$$\int_{S_r} g^{\boldsymbol{a}_r}(\boldsymbol{z})e^{rf_r^{\boldsymbol{a}_r}(\boldsymbol{z})}d\boldsymbol{z}=\Big(\frac{2\pi}{r}\Big)^{\frac{n}{2}}\frac{g^{\boldsymbol{a}_r}(\boldsymbol{c}_r)}{\sqrt{\det\big(-\mathrm{Hess}(f^{\boldsymbol{a}_r})(\boldsymbol{c}_r)\big)}}e^{rf^{\boldsymbol{a}_r}(\boldsymbol{c}_r)}\Big(1+O\Big(\frac{1}{r}\Big)\Big).$$

Proposition 5.2. *[40, Proposition 6.1] Let D be a region in $\mathbb{C}^n$ and let $f(\boldsymbol{z})$ and $g(\boldsymbol{z})$ be holomorphic functions on D independent of r. Let $f_r(\boldsymbol{z})$ be a holomorphic function of the form*

$$f_r(\boldsymbol{z})=f(\boldsymbol{z})+\frac{\upsilon_r(\boldsymbol{z})}{r^2}.$$

Let S be an embedded real n-dimensional closed disk in D and let $\mathbf{c}$ be a point on S. If

(i) $\mathbf{c}$ *is a critical point of f in D,*

(ii) $\mathrm{Re}(f)(\boldsymbol{c})>\mathrm{Re}(f)(\boldsymbol{z})$ *for all $\boldsymbol{z}\in S\setminus\{\boldsymbol{c}\}$,*

(iii) the domain $\{\boldsymbol{z}\in D\mid \mathrm{Re}f(\boldsymbol{z})<\mathrm{Re}f(\boldsymbol{c})\}$ deformation retracts to $S\setminus\{(\boldsymbol{c})\}$,

(iv) the Hessian matrix $\mathrm{Hess}(f)(\boldsymbol{c})$ of f at $\mathbf{c}$ is non-singular,

(v) $g(\boldsymbol{c})\neq 0$, and

(vi) $|\upsilon_r(\boldsymbol{z})|$ is bounded from above by a constant independent of r in D,

then

$$\int_S g(\boldsymbol{z})e^{rf_r(\boldsymbol{z})}d\boldsymbol{z}=\Big(\frac{2\pi}{r}\Big)^{\frac{n}{2}}\frac{g(\boldsymbol{c})}{\sqrt{\det\big(-\mathrm{Hess}(f)(\boldsymbol{c})\big)}}e^{rf(\boldsymbol{c})}\Big(1+O\Big(\frac{1}{r}\Big)\Big).$$

5.1 Properties of relevant functions

In light of Proposition 2.15, we consider the functions $\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}}$ and $\mathcal{W}^{\boldsymbol{\epsilon}}$ on D, respectively defined by

$$\begin{aligned}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}}(\boldsymbol{\alpha},\boldsymbol{\xi}) = &-\sum_{i=1}^{n} q_i(\beta_i-\pi)^2 - \sum_{i=1}^{n} p_i(\alpha_i-\pi)^2 - \sum_{i=1}^{n} 2\epsilon_i(\alpha_i-\pi)(\beta_i-\pi)\\ &-\sum_{i=1}^{n}\frac{\iota_i}{2}(\alpha_i-\pi)^2 + \sum_{s=1}^{c} U(\alpha_{s_1},\dots,\alpha_{s_6},\xi_s) + \Big(\sum_{i=1}^{n}\frac{\iota_i}{2}\Big)\pi^2\end{aligned}$$

and

$$\begin{aligned}\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{\alpha},\boldsymbol{\xi}) = &-\sum_{i=1}^{n} p_i(\alpha_i-\pi)^2 + \sum_{i=1}^{n} 2\epsilon_i\pi(\alpha_i-\pi)\\ &-\sum_{i=1}^{n}\frac{\iota_i}{2}(\alpha_i-\pi)^2 + \sum_{s=1}^{c} U(\alpha_{s_1},\dots,\alpha_{s_6},\xi_s) + \Big(\sum_{i=1}^{n}\frac{\iota_i}{2}\Big)\pi^2,\end{aligned}$$

where the function U is as defined in (3.11); and by abuse of notations consider the functions κ and ν_b on D, respectively defined by

$$\kappa(\boldsymbol{\alpha},\boldsymbol{\xi}) = \sum_{s=1}^{c}\kappa(\alpha_{s_1},\dots,\alpha_{s_6},\xi_s)$$

and

$$\nu_r(\boldsymbol{\alpha},\boldsymbol{\xi}) = \sum_{s=1}^{c}\nu_r(\alpha_{s_1},\dots,\alpha_{s_6},\xi_s),$$

where κ and ν_r on the right hand side are as defined in (3.12) and (3.13). Notice that all the functions above extend analytically to

$$\mathrm{D}_{\mathbb{C}} = \Big\{(\boldsymbol{\alpha},\boldsymbol{\xi})\in\mathbb{C}^{n+c}\;\Big|\;(\mathrm{Re}\boldsymbol{\alpha},\mathrm{Re}\boldsymbol{\xi})\in\mathrm{D}\Big\},$$

where $\mathrm{Re}\boldsymbol{\alpha} = (\mathrm{Re}\alpha_i)_{i\in\{1,\dots,n\}}$ and $\mathrm{Re}\boldsymbol{\xi} = (\mathrm{Re}\xi_s)_{s\in\{1,\dots,c\}}$.

For $\epsilon>0$, we denote by $B_\epsilon(\boldsymbol{z}_0)$ the ϵ-neighborhood of $\boldsymbol{z}_0 = \big(\pi,\dots,\pi,\frac{7\pi}{4},\dots,\frac{7\pi}{4}\big)$ in $\mathbb{C}^{n+c}$ in the L^∞ distance, that is

$$B_\epsilon(\boldsymbol{z}_0) = \Big\{(\boldsymbol{\alpha},\boldsymbol{\xi})\in\mathbb{C}^{n+c}\;\Big|\;d_{L^\infty}\big((\boldsymbol{\alpha},\boldsymbol{\xi}),\boldsymbol{z}_0\big)<\epsilon\Big\},$$

where d_{L^∞} is the L^∞-distance on $\mathbb{C}^{n+c}$.

Proposition 5.3. *There exists an $\epsilon'>0$ independent of M_c, L_{FSL} and L such that for any $\boldsymbol{\beta}\in[0,2\pi]^n$ and $\boldsymbol{\epsilon}\in\{-1,1\}^n$, $\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}}$ is strictly concave down in $\{\mathrm{Re}(\alpha_i)\}_{i\in\{1,\dots,n\}}$ and in $\{\mathrm{Re}(\xi_s)\}_{s\in\{1,\dots,c\}}$, and is strictly concave up in $\{\mathrm{Im}(\alpha_i)\}_{i\in\{1,\dots,n\}}$ and in $\{\mathrm{Im}(\xi_s)\}_{s\in\{1,\dots,c\}}$ on $B_{\epsilon'}(\boldsymbol{z}_0)$.*

Proof. In the special case $(\alpha_1,\dots,\alpha_6,\xi) = (\pi,\dots,\pi,\frac{7\pi}{4})$, by a direct computation, the Hessian matrix of $\mathrm{Im}U$ is strictly negative definite in $\mathrm{Re}\alpha_i$'s and in $\mathrm{Re}\xi$. Then by continuity, there exists an $\epsilon'>0$ such that if $d_{L^\infty}\big((\alpha_1,\dots,\alpha_6,\xi),(\pi,\dots,\pi,\frac{7\pi}{4})\big)<\epsilon'$, the Hessian matrix of $\mathrm{Im}U$ is strictly negative definite, i.e., $\mathrm{Im}U$ is strictly concave down in $\mathrm{Re}\alpha_i$'s and in $\mathrm{Re}\xi$. This implies that $\sum_{s=1}^{c}\mathrm{Im}U(\boldsymbol{\alpha}_s,\xi_s)$ is strictly concave down in $\{\mathrm{Re}(\alpha_i)\}_{i\in\{1,\dots,n\}}$ and in $\{\mathrm{Re}(\xi_s)\}_{s\in\{1,\dots,c\}}$ on $B_{\epsilon'}(\boldsymbol{z}_0)$. As all the other terms in $\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}}$ are linear functions of $\{\mathrm{Re}(\alpha_i)\}_{i\in\{1,\dots,n\}}$ and $\{\mathrm{Re}(\xi_s)\}_{s\in\{1,\dots,c\}}$, $\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}}$ is strictly concave down in $\{\mathrm{Re}(\alpha_i)\}_{i\in\{1,\dots,n\}}$ and in $\{\mathrm{Re}(\xi_s)\}_{s\in\{1,\dots,c\}}$ on $B_{\epsilon'}(\boldsymbol{z}_0)$; and by the harmonicity, it is strictly concave up in $\{\mathrm{Im}(\alpha_i)\}_{i\in\{1,\dots,n\}}$ and in $\{\mathrm{Im}(\xi_s)\}_{s\in\{1,\dots,c\}}$ on $B_{\epsilon'}(\boldsymbol{z}_0)$. □

For $\boldsymbol{\beta} \in (0, 2\pi)^n$, $\boldsymbol{\epsilon} = (\epsilon_i)_{i\in\{1,\dots,n\}} \in \{-1, 1\}^n$ and $i \in \{1, \dots, n\}$, let

$$\alpha_i^{\boldsymbol{\epsilon}} = \pi + \frac{\epsilon_i \mu_i}{2} \mathbf{i} \mathrm{H}(u_i) \tag{5.1}$$

with $\mathrm{H}(u_i)$ the logarithmic holonomy of u_i of the hyperbolic cone manifold $M_{L_{\boldsymbol{\theta}}}$, and let

$$\boldsymbol{\alpha}^{\boldsymbol{\epsilon}} = \big(\alpha_i^{\boldsymbol{\epsilon}}\big)_{i\in\{1,\dots,n\}}. \tag{5.2}$$

For $s \in \{1, \dots, c\}$, let $\xi_s^{\boldsymbol{\epsilon}} = \xi_{(\alpha_{s_1}^{\boldsymbol{\epsilon}},\dots,\alpha_{s_6}^{\boldsymbol{\epsilon}})}$ be as defined in (3.23), and let

$$\boldsymbol{\xi}^{\boldsymbol{\epsilon}} = \big(\xi_s^{\boldsymbol{\epsilon}}\big)_{s\in\{1,\dots,c\}}.$$

Keep in mind that both $\boldsymbol{\alpha}^{\boldsymbol{\epsilon}}$ and $\boldsymbol{\xi}^{\boldsymbol{\epsilon}}$ depend on $\boldsymbol{\beta}$.

Also, for $\boldsymbol{\sigma} \in \{0, 1\}^n$, $\boldsymbol{\epsilon} = (\epsilon_i)_{i\in\{1,\dots,n\}} \in \{-1, 1\}^n$ and $i \in \{1, \dots, n\}$, let

$$\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} = \pi - \frac{(-1)^{\sigma_i}\epsilon_i}{2} \mathbf{i} \mathrm{H}(u_i) \tag{5.3}$$

with $\mathrm{H}(u_i)$ the logarithmic holonomy of u_i of the hyperbolic metric on M, and let

$$\boldsymbol{\alpha}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} = \big(\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}\big)_{i\in\{1,\dots,n\}}. \tag{5.4}$$

For $s \in \{1, \dots, c\}$, let $\xi_s^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} = \xi_{(\alpha_{s_1}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}},\dots,\alpha_{s_6}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}})}$ be as defined in (3.23), and let

$$\boldsymbol{\xi}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} = \big(\xi_s^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}\big)_{s\in\{1,\dots,c\}}.$$

Proposition 5.4. *There exists an $\epsilon'' > 0$ independent of M_c, L_{FSL} and L and depending only on ϵ' in Proposition 5.3 such that:*

(1) If the L^∞-distance

$$d_{L^\infty}(\boldsymbol{\alpha}^{\boldsymbol{\epsilon}}, \boldsymbol{\pi}) < \epsilon'',$$

then

$$\boldsymbol{z}^{\boldsymbol{\epsilon}} = \big(\boldsymbol{\alpha}^{\boldsymbol{\epsilon}}, \boldsymbol{\xi}^{\boldsymbol{\epsilon}}\big) \in B_{\epsilon'}(\boldsymbol{z}_0),$$

which is a critical point of $\mathcal{W}_{\boldsymbol{\beta}}^{\boldsymbol{\epsilon}}$ with critical value

$$\mathcal{W}_{\boldsymbol{\beta}}^{\boldsymbol{\epsilon}}(\boldsymbol{z}^{\boldsymbol{\epsilon}}) = \mathbf{i}\Big(\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) + \mathbf{i}\mathrm{CS}(M_{L_{\boldsymbol{\theta}}})\Big)$$

for a suitable choice of the value of the Chern-Simons invariant $\mathrm{CS}(M_{L_{\boldsymbol{\theta}}})$.

(2) If the L^∞-distance

$$d_{L^\infty}(\boldsymbol{\alpha}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}, \boldsymbol{\pi}) < \epsilon'',$$

then

$$\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} = \big(\boldsymbol{\alpha}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}, \boldsymbol{\xi}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}\big) \in B_{\epsilon'}(\boldsymbol{z}_0),$$

which is a critical point of $\mathcal{W}^{\boldsymbol{\epsilon}} - 4\pi \sum_{i=1}^n \sigma_i\epsilon_i\alpha_i$ with critical value

$$\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) - 4\pi \sum_{i=1}^n \sigma_i\epsilon_i\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} = \mathbf{i}\Big(\mathrm{Vol}(M) + \mathbf{i}\mathrm{CS}(M)\Big) - 4\pi^2 \sum_{i=1}^n \sigma_i\epsilon_i$$

for a suitable choice of the value of the Chern-Simons invariant $\mathrm{CS}(M)$.

Proof. In the special case that $(\alpha_1,\dots,\alpha_6)=(\pi,\dots,\pi)$, by solving equation (3.23) directly, we have $\xi_{(\pi,\dots,\pi)}=\frac{7\pi}{4}$, and $(\pi,\dots,\pi,\frac{7\pi}{4})\in B_{\epsilon'}(\boldsymbol{z}_0)$. Then by continuity, there exists an $\epsilon''>0$ such that if $d_{L^\infty}\big((\alpha_1,\dots,\alpha_6),(\pi,\dots,\pi)\big)<\epsilon''$, then $d_{L^\infty}\big((\alpha_1,\dots,\alpha_6,\xi_{(\alpha_1,\dots,\alpha_6)}),(\pi,\dots,\pi,\frac{7\pi}{4})\big)<\epsilon'$. As a consequence, if $d_{L^\infty}(\boldsymbol{\alpha}^{\boldsymbol{\epsilon}},\boldsymbol{\pi})<\epsilon''$, then $(\boldsymbol{\alpha}^{\boldsymbol{\epsilon}},\boldsymbol{\xi}^{\boldsymbol{\epsilon}})\in B_{\epsilon'}(\boldsymbol{z}_0)$ as desired in (1); and if $d_{L^\infty}(\boldsymbol{\alpha}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}},\boldsymbol{\pi})<\epsilon''$, then $(\boldsymbol{\alpha}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}},\boldsymbol{\xi}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}})\in B_{\epsilon'}(\boldsymbol{z}_0)$ as desired in (2).

The rest of the proof of (1) follows verbatim that of [41, Propositions 4.1 and 5.2]; and we include a sketch of the proof of (2), which is by essentially the same argument, for the readers' convenience. As proved in [41, Proposition 4.1], the function

$$\mathcal{U}(\boldsymbol{\alpha})\doteq-\sum_{i=1}^{n}\frac{\iota_i}{2}(\alpha_i-\pi)^2+\sum_{s=1}^{c}U_r(\alpha_{s_1},\dots,\alpha_{s_6},\xi_{(\alpha_{s_1},\dots,\alpha_{s_6})})+\Big(\sum_{i=1}^{n}\frac{\iota_i}{2}\Big)\pi^2$$

coincides with the Neumann-Zagier potential function Φ recalled in Section 2.2; and by (2.3), we have

$$\frac{\partial\mathcal{U}(\boldsymbol{\alpha})}{\partial\mathrm{H}(u_i)}=\frac{\mathrm{H}(v_i)}{2}. \tag{5.5}$$

Then by the chain rule, one verifies that for each $s\in\{1,\dots,c\}$,

$$\frac{\partial}{\partial\xi_s}\Big|_{\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}}\Big(\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{\alpha},\boldsymbol{\xi})-4\pi\sum_{i=1}^{n}\sigma_i\epsilon_i\alpha_i\Big)=\frac{\partial}{\partial\xi_s}\Big|_{\xi_s^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}}U(\alpha_{s_1}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}},\dots,\alpha_{s_6}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}},\xi_s)=0;$$

and by (5.5) and the chain rule, for each $i\in\{1,\dots,n\}$,

$$\begin{aligned}\frac{\partial}{\partial\alpha_i}\Big|_{\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}}\Big(\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{\alpha},\boldsymbol{\xi})-4\pi\sum_{i=1}^{n}\sigma_i\epsilon_i\alpha_i\Big)=&(-1)^{\sigma_i}\epsilon_i\mathbf{i}\Big(p_i\mathrm{H}(u_i)+\mathrm{H}(v_i)-(-1)^{\sigma_i}\big(2-4\sigma_i\big)\pi\mathbf{i}\Big)\\=&(-1)^{\sigma_i}\epsilon_i\mathbf{i}\Big(p_i\mathrm{H}(u_i)+\mathrm{H}(v_i)-2\pi\mathbf{i}\Big)=0,\end{aligned}$$

where the last equality comes from Thurston's Hyperbolic Gluing Equations. Hence $\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}$ is a critical point of $\mathcal{W}^{\boldsymbol{\epsilon}}-4\pi\sum_{i=1}^{n}\sigma_i\epsilon_i\alpha_i$.

For each $i\in I$, let $\gamma_i=-u_i$ be the curve on the boundary of a tubular neighborhood of L_i^* that is isotopic to the core of L_i^* and with the orientation so that $(p_iu_i+v_i)\cdot\gamma_i=1$. Then

$$\mathrm{H}(v_i)=2\pi\mathbf{i}-p_i\mathrm{H}(u_i)\quad\text{and}\quad\mathrm{H}(\gamma_i)=-\mathrm{H}(u_i);$$

and plugging in (5.3), we have

$$-\frac{\mathrm{H}(u_i)\mathrm{H}(v_i)}{4}+\frac{2\pi\mathbf{i}\mathrm{H}(\gamma_i)}{4}=-p_i(\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}-\pi)^2+2(-1)^{\sigma_1}\epsilon_i\pi(\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}-\pi).$$

If $\sigma_1=0$, then we have

$$-p_i(\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}-\pi)^2+2\epsilon_i\pi(\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}-\pi)-4\pi\sigma_i\epsilon_i\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}=-\frac{\mathrm{H}(u_i)\mathrm{H}(v_i)}{4}+\frac{2\pi\mathbf{i}\mathrm{H}(\gamma_i)}{4};$$

and if $\sigma_i=1$, then we have

$$-p_i(\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}-\pi)^2+2\epsilon_i\pi(\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}-\pi)-4\pi\sigma_i\epsilon_i\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}=-\frac{\mathrm{H}(u_i)\mathrm{H}(v_i)}{4}+\frac{2\pi\mathbf{i}\mathrm{H}(\gamma_i)}{4}-4\epsilon_i\pi^2.$$

Then by (2.5), we have

$$\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}})-4\pi\sum_{i=1}^{n}\sigma_i\epsilon_i\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}=\mathbf{i}\Big(\mathrm{Vol}(M)+\mathbf{i}\mathrm{CS}(M)\Big)\mod\pi^2\mathbb{Z};$$

and with the choice of the value of the Chern-Simons invariant so that

$$\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}^{\mathbf{0},\boldsymbol{\epsilon}}) = \mathbf{i}\Big(\mathrm{Vol}(M) + \mathbf{i}\mathrm{CS}(M)\Big),$$

we have

$$\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) - 4\pi\sum_{i=1}^{n}\sigma_i\epsilon_i\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} = \mathbf{i}\Big(\mathrm{Vol}(M) + \mathbf{i}\mathrm{CS}(M)\Big) - 4\pi^2\sum_{i=1}^{n}\sigma_i\epsilon_i.$$

This completes the proof of (2). □

Proposition 5.5. *(1) For each $\boldsymbol{\epsilon} \in \{-1,1\}^n$, the determinant of the Hessian matrix of $\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}}$ at the critical point $\boldsymbol{z}^{\epsilon}$ equals*

$$\frac{\det\big(-\mathrm{Hess}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}}(\boldsymbol{z}^{\boldsymbol{\epsilon}})\big)}{\exp\big(\frac{\kappa(\boldsymbol{z}^{\boldsymbol{\epsilon}})}{2\pi\mathbf{i}}\big)} = \pm 2^{n+c}\cdot\mathbb{T}_{(M\smallsetminus L,\mathbf{m})}([\rho_{M^{(r)}}]).$$

As a consequence, $\boldsymbol{z}^{\boldsymbol{\epsilon}}$ is a nondegenerate critical point of $\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}}$.

(2) For each $\boldsymbol{\sigma} \in \{0,1\}^n$ and $\boldsymbol{\epsilon} \in \{-1,1\}^n$, the determinant of the Hessian matrix of $\mathcal{W}^{\boldsymbol{\epsilon}}$ at the critical point $\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}$ of $\mathcal{W}^{\boldsymbol{\epsilon}} - 4\pi\sum_{i=1}^{n}\sigma_i\epsilon_i\alpha_i$ equals

$$\frac{\det\big(-\mathrm{Hess}\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}})\big)}{\exp\big(\frac{\kappa(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}})}{2\pi\mathbf{i}}\big)} = \pm 2^{n+c}\cdot\mathbb{T}_{(M\smallsetminus L,\mathbf{m})}([\rho_{M\smallsetminus L}]),$$

where $\rho_{M\smallsetminus L}$ is the holonomy representation of the restriction of the hyperbolic metric on M to $M\smallsetminus L$. As a consequence, $\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}$ is a nondegenerate critical point of $\mathcal{W}^{\boldsymbol{\epsilon}} - 4\pi\sum_{i=1}^{n}\sigma_i\epsilon_i\alpha_i$

Proof. The proof follows from the same algebraic computation as in the proof of [23, Formula (5.11)]. □

5.2 Asymptotics of the leading Fourier coefficients

Proposition 5.6. *Let ϵ' and ϵ'' respectively be the constants in Proposition 5.3 and Proposition 5.4, and let*

$$B_{\epsilon',\epsilon''}(\boldsymbol{z}_0) = B_{\min\{\epsilon',\epsilon''\}}(\boldsymbol{\pi})\times B_{\epsilon'}\Big(\frac{7\boldsymbol{\pi}}{4}\Big).$$

(1) Let $\boldsymbol{\alpha}^{\boldsymbol{\epsilon}}$ be as defined in (5.1) and (5.2). If $d_{L^\infty}(\boldsymbol{\alpha}^{\boldsymbol{\epsilon}},\boldsymbol{\pi}) < \min\{\epsilon',\epsilon''\}$ and

$$\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) > \max_{\mathrm{D}\smallsetminus B_{\epsilon',\epsilon''}(\boldsymbol{z}_0)}\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}},$$

then

$$\widehat{f^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}}(\mathbf{0},\mathbf{0}) = \frac{e^{\frac{(n+c)\pi\mathbf{i}}{4}}r^{\frac{n+c}{2}}}{\sqrt{\pm\mathbb{T}_{(M\smallsetminus L,\mathbf{m})}([\rho_{M_{L_{\boldsymbol{\theta}}}}])}}e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}})+\mathbf{i}\mathrm{CS}(M_{L_{\boldsymbol{\theta}}})\right)}\Big(1+O\Big(\frac{1}{r}\Big)\Big).$$

(2) Let $\boldsymbol{\alpha}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}$ be as defined in (5.3) and (5.4), and let $\boldsymbol{\sigma\epsilon} \doteq (\sigma_i\epsilon_i)_{i\in\{1,\dots,n\}}$. If $d_{L^\infty}(\boldsymbol{\alpha}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}},\boldsymbol{\pi}) < \min\{\epsilon',\epsilon''\}$ and

$$\mathrm{Vol}(M) > \max_{\mathrm{D}\smallsetminus B_{\epsilon',\epsilon''}(\boldsymbol{z}_0)}\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}, \tag{5.6}$$

then

$$\widehat{f^{\boldsymbol{\epsilon}}_{r}}(\boldsymbol{\sigma\epsilon},\mathbf{0}) = e^{\frac{(n+c)\pi\mathbf{i}}{4}}r^{\frac{n+c}{2}}\frac{(-1)^{\sum_{i=1}^{n}\sigma_i}e^{(-1)^{\sigma_i}\sum_{i=1}^{n}\frac{\mathrm{H}(u_i)}{2}}}{\sqrt{\pm\mathbb{T}_{(M\smallsetminus L,\mathbf{m})}([\rho_{M\smallsetminus L}])}}e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M)+\mathbf{i}\mathrm{CS}(M)\right)}\Big(1+O\Big(\frac{1}{r}\Big)\Big),$$

where $\rho_{M\smallsetminus L}$ is the holonomy representation of the restriction of the hyperbolic metric on M to $M\smallsetminus L$.

Proof. The proof of (1) follows verbatim that of [41, Proposition 5.5], namely, deform the contour of the integral to satisfy all the conditions of Proposition 5.1. The proof of (2) is by essentially the same argument, and we include a sketch here for the readers' convenience.

As depicted in Figure 5, let

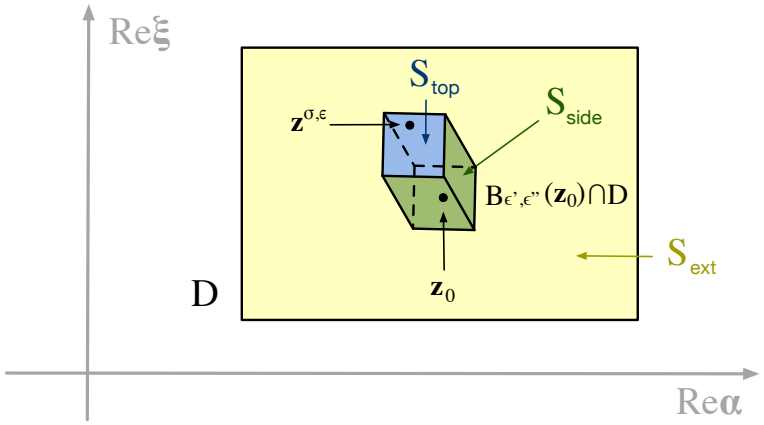


Figure 5: The deformed surface S, where S_{top} is the "top" of the rectangular cylinder colored in blue, S_{side} is the "side" of the rectangular cylinder colored in green, $B_{\epsilon',\epsilon''}(\boldsymbol{z}_0) \cap \mathrm{D}$ is the "bottom" of the rectangular cylinder, and S_{ext} is the exterior of $B_{\epsilon',\epsilon''}(\boldsymbol{z}_0)$ in D colored in light yellow.

$$\begin{aligned}
\mathrm{S}_{\text{top}} =& \Big\{ \boldsymbol{z} + \mathrm{Im}\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} \;\Big|\; \boldsymbol{z} \in B_{\epsilon',\epsilon''}(\boldsymbol{z}_0) \cap \mathrm{D} \Big\}, \\
\mathrm{S}_{\text{side}} =& \Big\{ \boldsymbol{z} + t \cdot \mathbf{i}\mathrm{Im}\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} \;\Big|\; \boldsymbol{z} \in \partial B_{\epsilon',\epsilon''}(\boldsymbol{z}_0) \cap \mathrm{D}, t \in [0,1] \Big\}, \\
\mathrm{S}_{\text{ext}} =& \mathrm{D} \setminus B_{\epsilon',\epsilon''}(\boldsymbol{z}_0);
\end{aligned}$$

and let

$$\mathrm{S} = \mathrm{S}_{\text{top}} \cup \mathrm{S}_{\text{side}} \cup \mathrm{S}_{\text{ext}}.$$

Then all the conditions of Proposition 5.2 are satisfied with $D = \mathrm{D}_{\mathbb{C}}$, $f(\boldsymbol{z}) = \frac{\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}) - 4\pi \sum_{i=1}^n \sigma_i \epsilon_i \alpha_i}{4\pi \mathbf{i}}$, $g(\boldsymbol{z}) = e^{\sum_{i=1}^n \epsilon_i(\alpha_i - \pi)\mathbf{i} + \frac{\kappa(\boldsymbol{z})}{4\pi \mathbf{i}}}$, $f_r(\boldsymbol{z}) = \frac{\mathcal{W}_r^{\boldsymbol{\epsilon}}(\boldsymbol{z}) - 4\pi \sum_{i=1}^n \sigma_i \epsilon_i \alpha_i}{4\pi \mathbf{i}}$, $\upsilon_r(\boldsymbol{z}) = \frac{\nu_r(\boldsymbol{z})}{4\pi \mathbf{i}}$, $S = \mathrm{S}$ and $\boldsymbol{c} = \boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}$. Indeed, (i) follows from Proposition 5.4 (2), (ii) and (iii) follow from Proposition 5.3 and assumption (5.6), (iv) follows from Proposition 5.5, (v) follows from that g is an exponential function, and (vi) follows from Proposition 2.15 (1). See the proof of [41, Proposition 5.5] for more details of the argument. As a consequence, we have

$$\begin{aligned}
&\widehat{f_r^{\boldsymbol{\epsilon}}}(\boldsymbol{\sigma}\boldsymbol{\epsilon}, \mathbf{0}) \\
=& \frac{\left(\frac{r}{2\pi}\right)^{\frac{n+c}{2}} e^{\sum_{i=1}^n \epsilon_i(\alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} - \pi)\mathbf{i} + \frac{\kappa(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}})}{4\pi \mathbf{i}}}}{\sqrt{\det\left(-\mathrm{Hess}\left(\frac{\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}) - 4\pi \sum_{i=1}^n \sigma_i \epsilon_i \alpha_i}{4\pi \mathbf{i}} \right)\Big|_{\boldsymbol{z}=\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}} \right)}} e^{\frac{r}{4\pi \mathbf{i}}\left(\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) - 4\pi \sum_{i=1}^n \sigma_i \epsilon_i \alpha_i^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} \right)} \left(1 + O\left(\frac{1}{r}\right)\right) \\
=& e^{-\frac{(n+c)\pi \mathbf{i}}{4}} r^{\frac{n+c}{2}} \frac{(-1)^{\sum_{i=1}^n \sigma_i} e^{(-1)^{\sigma_i} \sum_{i=1}^n \frac{\mathrm{H}(u_i)}{2}}}{\sqrt{\pm \mathbb{T}_{(M \setminus L, \mathbf{m})}([\rho_{M \setminus L}])}} e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M) + \mathbf{i}\mathrm{CS}(M) \right)} \left(1 + O\left(\frac{1}{r}\right)\right),
\end{aligned}$$

where the last equality comes from Proposition 5.4 (2) and Proposition 5.5 (2). □

5.3 Estimate of the other Fourier coefficients

Proposition 5.7. *There exists an $\epsilon''' > 0$ such that:*

(1) If $d_{L^\infty}(\boldsymbol{z}^{\boldsymbol{\epsilon}}, \boldsymbol{z}_0) < \epsilon'''$ and

$$\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) > \max_{\mathrm{D}\setminus B_{\epsilon'''}(\boldsymbol{z}_0)} \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}},$$

then there exists a $\delta' > 0$ such that for any $(\boldsymbol{m}, \boldsymbol{n}) \neq (\mathbf{0}, \mathbf{0})$ and $k > n + c$,

$$\left|\widehat{f^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}}(\boldsymbol{m}, \boldsymbol{n})\right| < \frac{O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}})-\delta'\right)}\Big)}{\Big(\sum_{i=1}^n m_i^2 + \sum_{s=1}^c n_s^2\Big)^k}.$$

(2) If $d_{L^\infty}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}, \boldsymbol{z}_0) < \epsilon'''$ and

$$\mathrm{Vol}(M) > \max_{\mathrm{D}\setminus B_{\epsilon'''}(\boldsymbol{z}_0)} \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}, \tag{5.7}$$

then there exists a $\delta' > 0$ such that for any $(\boldsymbol{m}, \boldsymbol{n}) \neq (\boldsymbol{\sigma}\boldsymbol{\epsilon}, \mathbf{0})$ and $k > n + c$

$$\left|\widehat{f^{\boldsymbol{\epsilon}}_{r}}(\boldsymbol{m}, \boldsymbol{n})\right| < \frac{O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M)-\delta'\right)}\Big)}{\Big(\sum_{i=1}^n m_i^2 + \sum_{s=1}^c n_s^2\Big)^k}.$$

Proof. We will prove (2) below; and that of (1) follows verbatim.

By the Integration by Parts as on [32, Page 703], one has

$$\widehat{f^{\boldsymbol{\epsilon}}_{r}}(\boldsymbol{m}, \boldsymbol{n}) = \frac{r^{n+c-k}}{\Big(\sum_{i=1}^n m_i^2 + \sum_{s=1}^c n_s^2\Big)^k} \int_{\mathrm{D}} \Psi(\boldsymbol{\alpha}, \boldsymbol{\xi}) e^{\frac{r}{4\pi\mathbf{i}}\left(\mathcal{W}^{\boldsymbol{\epsilon}}_r(\boldsymbol{\alpha},\boldsymbol{\xi}) - \sum_{i=1}^n 4\pi m_i\alpha_i - \sum_{s=1}^c 4\pi n_s\xi_s\right)} d\boldsymbol{\alpha} d\boldsymbol{\xi},$$

where $\Psi(\boldsymbol{\alpha}, \boldsymbol{\xi})$ is a polynomial of the partial derivatives of order at most $2k$ of $\psi(\boldsymbol{\alpha}, \boldsymbol{\xi}) e^{\sum_{i=1}^n \epsilon_i(\alpha_i - \pi)\mathbf{i}}$ and $\mathcal{W}^{\boldsymbol{\epsilon}}_r$; and it suffices to bound each $\mathcal{W}^{\boldsymbol{\epsilon}}_r(\boldsymbol{z}) - 4\pi\sum_{i=1}^n m_i\alpha_i - 4\pi\sum_{s=1}^c n_s\xi_s$.

For the simplicity of notations, for $\boldsymbol{\epsilon} \in \{-1, 1\}^n$ and $(\boldsymbol{m}, \boldsymbol{n}) \in \mathbb{Z}^{n+c}$, we let $\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}$ be the function given by

$$\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}(\boldsymbol{z}) \doteq \mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}) - 4\pi\sum_{i=1}^n m_i\alpha_i - 4\pi\sum_{s=1}^c n_s\xi_s.$$

We first observe that for all $\boldsymbol{\epsilon} \in \{-1, 1\}^n$ and $(\boldsymbol{m}, \boldsymbol{n}) \in \mathbb{Z}^{n+c}$, the functions $\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}$ share the same second derivatives, as they only differ by a linear function. As a consequence, there is an $N > 0$ such that for any $\boldsymbol{\epsilon} \in \{-1, 1\}^n$, $(\boldsymbol{m}, \boldsymbol{n}) \in \mathbb{Z}^{n+c}$, unit vector $\mathbf{u} \in \mathbb{C}^{n+c}$ and $\boldsymbol{z} \in B_{\epsilon',\epsilon''}(\boldsymbol{z}_0)$, the second directional derivative

$$\left|D_{\mathbf{u}} D_{\mathbf{u}} \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}(\boldsymbol{z})\right| < N.$$

As a consequence, for any $\boldsymbol{z}_1, \boldsymbol{z}_2 \in B_{\epsilon',\epsilon''}(\boldsymbol{z}_0)$ and for the unit vector $\mathbf{u} = \frac{\boldsymbol{z}_1 - \boldsymbol{z}_2}{|\boldsymbol{z}_1 - \boldsymbol{z}_2|}$, the directional derivative

$$\left|D_{\mathbf{u}} \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}(\boldsymbol{z}_1) - D_{\mathbf{u}} \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}(\boldsymbol{z}_2)\right| < N|\boldsymbol{z}_1 - \boldsymbol{z}_2| \leqslant N\sqrt{2n + 2c} \cdot d_{L^\infty}(\boldsymbol{z}_1, \boldsymbol{z}_2). \tag{5.8}$$

Next, let

$$\epsilon''' = \min\Big\{\epsilon', \epsilon'', \frac{\pi}{4N(n + c)}\Big\}.$$

Then as long as $d_{L^\infty}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}, \boldsymbol{z}_0) < \epsilon'''$, we have $\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}} \in B_{\epsilon',\epsilon''}(\boldsymbol{z}_0)$; and for any $\boldsymbol{z} \in B_{\epsilon'''}(\boldsymbol{z}_0)$,

$$d_{L^\infty}(\boldsymbol{z}, \boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) < 2\epsilon''' \leqslant \frac{\pi}{2N(n+c)}. \tag{5.9}$$

By Proposition 5.4 (2), the total derivative

$$D\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) = \mathbf{0};$$

and by (5.8) and (5.9), we have

$$\left|D_{\mathbf{u}}\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z})\right| < \frac{\pi}{\sqrt{2n+2c}} \tag{5.10}$$

for any $\boldsymbol{z} \in B_{\epsilon'''}(\boldsymbol{z}_0)$ and $\mathbf{u} = \frac{\boldsymbol{z}-\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}}{|\boldsymbol{z}-\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}|}$, the unit vector along the direction of the line segment connecting $\boldsymbol{z}$ and $\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}$.

Now, since $(\boldsymbol{m}, \boldsymbol{n}) \neq (\boldsymbol{\sigma\epsilon}, \mathbf{0})$, at least one of the following four cases happens:

(1) $m_i > \max\{0, \epsilon_i\}$ for some $i \in \{1, \ldots, n\}$.

(2) $m_i < \min\{0, \epsilon_i\}$ for some $i \in \{1, \ldots, n\}$.

(3) $n_s > 0$ for some $s \in \{1, \ldots, c\}$.

(4) $n_s < 0$ for some $s \in \{1, \ldots, c\}$.

In case (1), we let $\mathbf{u}_i \in \mathbb{C}^{n+c}$ be the vector with the i-th component 1 and all the other components 0; let

$$\begin{aligned}
\mathrm{S}_{\text{top}} =& \Big\{ \boldsymbol{z} + \epsilon'''\mathbf{i}\mathbf{u}_i \;\Big|\; \boldsymbol{z} \in B_{\epsilon'''}(\boldsymbol{z}_0) \cap \mathrm{D} \Big\}, \\
\mathrm{S}_{\text{side}} =& \Big\{ \boldsymbol{z} + t \cdot \mathbf{i}\mathbf{u}_i \;\Big|\; \boldsymbol{z} \in \partial B_{\epsilon'''}(\boldsymbol{z}_0) \cap \mathrm{D}, t \in [0, \epsilon'''] \Big\}, \\
\mathrm{S}_{\text{ext}} =& \mathrm{D} \setminus B_{\epsilon'''}(\boldsymbol{z}_0);
\end{aligned}$$

and let

$$\mathrm{S} = \mathrm{S}_{\text{top}} \cup \mathrm{S}_{\text{side}} \cup \mathrm{S}_{\text{ext}}.$$

For $\boldsymbol{z} \in \mathrm{S}_{\text{top}}$, as $\mathrm{S}_{\text{top}} \subset B_{\epsilon'''}(\boldsymbol{z}_0)$, by the Mean Value Theorem we have

$$\begin{aligned}
\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}) \leqslant& \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) + \left|D_{\mathbf{u}}\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}')\right| \cdot |\boldsymbol{z} - \boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}| \\
<& \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) + 2\pi\epsilon''',
\end{aligned}$$

where $\boldsymbol{z}'$ is some point on the line segment from $\boldsymbol{z}$ to $\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}$ and $\mathbf{u}$ is the unit vector along the direction of this line segment, and the last inequality comes from (5.10) and that $|\boldsymbol{z} - \boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}|$ is less than the diameter of $B_{\epsilon'''}(\boldsymbol{z}_0)$, which equals $2\epsilon'''\sqrt{2n+2c}$. As a consequence,

$$\begin{aligned}
\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}(\boldsymbol{z}) =& \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}) - 4\pi m_i\epsilon''' \\
=& \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}) - 4\pi m_i\epsilon''' + 4\pi\sigma_i\epsilon_i\epsilon''' \\
<& \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) + 2\pi\epsilon''' - 4\pi m_i\epsilon''' + 4\pi\sigma_i\epsilon_i\epsilon''' \\
=& \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) - 2\pi\epsilon''' - 4\pi(m_i - \sigma_i\epsilon_i - 1)\epsilon''' \\
\leqslant& \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) - 2\pi\epsilon''',
\end{aligned} \tag{5.11}$$

where the last equality comes from that $m_i \geqslant \epsilon_i \sigma_i + 1$.

For $\boldsymbol{z} \in \mathrm{S}_{\text{side}}$, let

$$\delta'' = \mathrm{Vol}(M) - \max_{\mathrm{D} \smallsetminus B_{\epsilon'''}(\boldsymbol{z}_0)} \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}} > 0.$$

Then by assumption (5.7) and Proposition 5.4 (2), for $\boldsymbol{z}' \in \partial B_{\epsilon'''}(\boldsymbol{z}_0) \cap \mathrm{D}$, we have

$$\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}(\boldsymbol{z}') = \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}') \leqslant \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) - \delta''. \tag{5.12}$$

By Proposition 5.3, we have that $\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}(\boldsymbol{z}' + t \cdot \mathbf{i}\mathbf{u}_i)$ is strictly concave up in t for any $\boldsymbol{z}' \in \partial B_{\epsilon'''}(\boldsymbol{z}_0)$; and by (5.11) and (5.12), we have

$$\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}(\boldsymbol{z}) < \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) - \min\{2\pi\epsilon''', \delta''\}. \tag{5.13}$$

For $\boldsymbol{z} \in \mathrm{S}_{\text{ext}}$, by the assumption (5.7) and Proposition 5.4 (2), we have

$$\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}(\boldsymbol{z}) = \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}(\boldsymbol{z}) \leqslant \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) - \delta''. \tag{5.14}$$

Finally, let

$$\delta' = \min\{2\pi\epsilon''', \delta''\};$$

and notice that, as both ϵ''' and δ'' are independent of $(\boldsymbol{m}, \boldsymbol{n})$, δ' is so as well. Putting (5.11), (5.13) and (5.14) together, we have

$$\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{m},\boldsymbol{n}}(\boldsymbol{z}) < \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\sigma\epsilon},\mathbf{0}}(\boldsymbol{z}^{\boldsymbol{\sigma},\boldsymbol{\epsilon}}) - \delta' = \mathrm{Vol}(M) - \delta'$$

on S, where the equality comes from Proposition 5.4 (2). Then the result follows from Proposition 2.15 (1).

In case (2), we deform $B_{\epsilon'''}(\boldsymbol{z}_0) \cap \mathrm{D}$ along the $-\mathbf{i}\mathbf{u}_i$ direction by ϵ'''; and in case (3) and case (4), we deform $B_{\epsilon'''}(\boldsymbol{z}_0) \cap \mathrm{D}$ respectively along the $\pm\mathbf{i}\mathbf{u}_{n+s}$ directions by ϵ'''. Then the rest of the proof follows verbatim that of case (1). □

5.4 Estimate of the error term

Proposition 5.8. *(1) If*

$$\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) > \max_{\partial \mathrm{D}} \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}},$$

then for any

$$\delta' < \frac{1}{2}\Big(\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) - \max_{\partial \mathrm{D}} \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}}\Big),$$

there exists an $\epsilon'''' > 0$ such that the error term in (4.2) is less than $O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) - \delta'\right)}\Big)$ on $\mathrm{D} \smallsetminus \mathrm{D}_{\epsilon''''}$.

(2) If

$$\mathrm{Vol}(M) > \max_{\partial \mathrm{D}} \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}},$$

then for any

$$\delta' < \frac{1}{2}\Big(\mathrm{Vol}(M) - \max_{\partial \mathrm{D}} \mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}\Big),$$

there exists an $\epsilon'''' > 0$ such that the error term in (4.3) is less than $O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M) - \delta'\right)}\Big)$ on $\mathrm{D} \smallsetminus \mathrm{D}_{\epsilon''''}$.

Proof. By Proposition 2.15 (2), the relation (2.19), and the continuity of the Lobachevsky function Λ, we can choose a sufficiently small $\epsilon'''' > 0$ so that if $\big(\frac{2\pi(\boldsymbol{a}+\mathbf{1})}{r}, \frac{2\pi\boldsymbol{k}}{r}\big) \in \mathrm{D}\smallsetminus\mathrm{D}_{\epsilon''''}$, then for (1),

$$\Big|g^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}(\boldsymbol{a},\boldsymbol{k})\Big| < O\Big(e^{\frac{r}{4\pi}\left(\max_{\partial\mathrm{D}}\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}_{\boldsymbol{\beta}}+\delta'\right)}\Big) < O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}})-\delta'\right)}\Big),$$

and for (2),

$$\Big|g^{\boldsymbol{\epsilon}}_{r}(\boldsymbol{a},\boldsymbol{k})\Big| < O\Big(e^{\frac{r}{4\pi}\left(\max_{\partial\mathrm{D}}\mathrm{Im}\mathcal{W}^{\boldsymbol{\epsilon}}+\delta'\right)}\Big) < O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M)-\delta'\right)}\Big).$$

Let ψ be the bump function supported on $(\mathrm{D},\mathrm{D}_{\epsilon''''})$. Then the error term in (4.2) is less than $O\big(e^{\frac{r}{4\pi}(\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}})-\delta')}\big)$, and the error term in (4.3) is less than $O\big(e^{\frac{r}{4\pi}(\mathrm{Vol}(M)-\delta')}\big)$. □

5.5 Proof of Theorem 1.6 and Theorem 1.7

Proof of Theorem 1.6. We will prove Theorem 1.6 by proving Theorem 4.1.

Let δ_1 and δ_2 respectively be as in Propositions 2.10 and 3.7, and let

$$\delta \in (0, \min\{\delta_1, \delta_2\})$$

be sufficiently small so that if

$$\mathrm{Vol}(M_{L_{\boldsymbol{\theta}}}) > \mathrm{Vol}(M_{\mathrm{FSL}}) - \delta,$$

then all the conditions of Proposition 5.3 and (1) of Propositions 5.4, 5.6, 5.7 and 5.8 are satisfied. The existence of such δ is guaranteed by Thurston's Hyperbolic Dehn-filling Theorem (see e.g. [26, Section 15.4.5]). Then by Propositions 5.3, 5.4, 5.5, 5.6, 5.7 and 5.8,

$$\begin{aligned}\mathrm{RT}_r\big(M,L,\boldsymbol{b}\big) =& B_r \sum_{\boldsymbol{\epsilon}\in\{-1,1\}^n} \widehat{f^{\boldsymbol{\epsilon}}_{\boldsymbol{b},r}}(\mathbf{0},\mathbf{0}) + O\Big(e^{\frac{r}{4\pi}(\mathrm{Vol}(M_{\mathrm{FSL}})-\delta)}\Big)\\ =& \frac{C_r}{\sqrt{\pm\mathbb{T}_{(M\smallsetminus L,\mathbf{m})}([\rho_{M^{(r)}}])}} e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M^{(r)})+\mathbf{i}\mathrm{CS}(M^{(r)})\right)}\Big(1+O\Big(\frac{1}{r}\Big)\Big),\end{aligned}$$

where

$$C_r = e^{\left(\frac{r+1}{4}\sigma(U_{c+1}\cup L)-(\frac{r}{4}+1)\sum_{i=1}^n(p_i+q_i)+\frac{1}{2}\sum_{i=1}^n\iota_i-\frac{r}{2}(n-c)-\frac{3n}{4}-\frac{c}{4}\right)\pi\mathbf{i}}$$

is a complex number of norm 1 independent of the geometry on (M,L). □

Proof of Theorem 1.7. Let δ_1 and δ_2 respectively be as in Propositions 2.10 and 3.7, and let

$$\delta \in (0, \min\{\delta_1, \delta_2\})$$

be sufficiently small so that if

$$\mathrm{Vol}(M) > \mathrm{Vol}(M_{\mathrm{FSL}}) - \delta,$$

then all the conditions of Proposition 5.3 and (2) of Propositions 5.4, 5.6, 5.7 and 5.8 are satisfied. Again, the existence of δ is guaranteed by Thurston's Hyperbolic Dehn-filling Theorem (see e.g. [26, Section

15.4.5]). Then by Propositions 5.3, 5.4, 5.5, 5.6, 5.7 and 5.8,

$$\begin{aligned}
\mathrm{RT}_r\big(M\big) =& B_r \sum_{\boldsymbol{\sigma}\in\{0,1\}^n,\boldsymbol{\epsilon}\in\{-1,1\}^n} \widehat{f_r^{\boldsymbol{\epsilon}}}(\boldsymbol{\sigma}\epsilon,\mathbf{0}) + O\Big(e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M_{\mathrm{FSL}})-\delta\right)}\Big)\\
=& C_r \frac{\sum_{\boldsymbol{\sigma}\in\{0,1\}^n}(-1)^{\sum_{i=1}^n \sigma_i} e^{(-1)^{\sigma_i}\sum_{i=1}^n \frac{\mathrm{H}(u_i)}{2}}}{\sqrt{\pm\mathbb{T}_{(M\smallsetminus L,\mathbf{m})}([\rho_{M^{(r)}}])}} e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M^{(r)})+\mathbf{i}\mathrm{CS}(M^{(r)})\right)}\Big(1+O\Big(\frac{1}{r}\Big)\Big)\\
=& C_r \frac{\prod_{i=1}^n 2\sinh\frac{\mathrm{H}(u_i)}{2}}{\sqrt{\pm\mathbb{T}_{(M\smallsetminus L,\mathbf{m})}([\rho_{M^{(r)}}])}} e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M^{(r)})+\mathbf{i}\mathrm{CS}(M^{(r)})\right)}\Big(1+O\Big(\frac{1}{r}\Big)\Big)\\
=& \frac{C_r}{\sqrt{\pm\mathrm{Tor}(M)}} e^{\frac{r}{4\pi}\left(\mathrm{Vol}(M)+\mathbf{i}\mathrm{CS}(M)\right)}\Big(1+O\Big(\frac{1}{r}\Big)\Big),
\end{aligned}$$

where

$$C_r = e^{\left(\frac{r+1}{4}\sigma(U_{c+1}\cup L)-(\frac{r}{4}+1)\sum_{i=1}^n p_i+\frac{1}{2}\sum_{i=1}^n \iota_i-\frac{r}{2}(n-c)-\frac{3n}{4}-\frac{c}{4}\right)\pi\mathbf{i}}$$

is a complex number of norm 1 independent of the geometry on M, and the last equality comes from the Surgery Formula of the adjoint twisted Reidemeister torsion (see [35, Theorem 4.1]). □

6 Weak reflection symmetry of extended quantum $6j$-symbols

Proposition 3.7 is a consequence of Proposition 6.1 (2) and Theorem 6.2 below.

To state them, we recall that a six-tuple $(a_1,\dots,a_6)\in[0,r-2]^6$ is r-hyperideal if for $\{i,j,k\}=\{1,2,3\}$, $\{1,5,6\}$, $\{2,4,6\}$ and $\{3,4,5\}$,

(1) $0\leqslant a_i+a_j-a_k\leqslant r-2$, and

(2) $r-2\leqslant a_i+a_j+a_k\leqslant 2(r-2)$;

and a six-tuple $(\alpha_1,\dots,\alpha_6)\in(0,2\pi)^6$ is of the hyperideal type if for $\{i,j,k\}=\{1,2,3\}$, $\{1,5,6\}$, $\{2,4,6\}$ and $\{3,4,5\}$,

(1) $0<\alpha_i+\alpha_j-\alpha_k<2\pi$, and

(2) $2\pi<\alpha_i+\alpha_j+\alpha_k<4\pi$.

For $(a_1,\dots,a_6)\in[0,r-2]^6$ and each $k\in\{1,\dots,6\}$, we call

$$\alpha_k=\frac{2\pi(a_k+1)}{r}$$

the corresponding α_k. Then $(a_1,\dots,a_6)$ being r-hyperideal implies that the corresponding $(\alpha_1,\dots,\alpha_6)$ is of the hyperideal type. In the rest of this paper, we let

$$\mathrm{H}=\Big\{(\alpha_1,\dots,\alpha_6)\in(0,2\pi)^6\ \Big|\ (\alpha_1,\dots,\alpha_6)\text{ is of the hyperideal type}\Big\};$$

and for $\epsilon_0>0$ sufficiently small, let

$$\mathrm{H}_{\epsilon_0}=\Big\{(\alpha_1,\dots,\alpha_6)\in\mathrm{H}\ \Big|\ d\big((\alpha_1,\dots,\alpha_6),\partial\mathrm{H}\big)>\epsilon_0\Big\}.$$

For $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_6) \in \mathrm{H}$ and each $k \in \{1, \dots, 6\}$, let

$$\theta_k = |\alpha_k - \pi|.$$

Then by [4], $(\theta_1, \dots, \theta_6)$ form the set of dihedral angles of a truncated hyperideal tetrahedron $\Delta_{\boldsymbol{\alpha}}$; and we denote the hyperbolic volume of $\Delta_{\boldsymbol{\alpha}}$ by $\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})$.

Proposition 6.1. *(1) For $\epsilon_0 > 0$ sufficiently small and any r-hyperideal six-tuple $(a_1, \dots, a_6)$ with the corresponding $(\alpha_1, \dots, \alpha_6) \in \mathrm{H} \smallsetminus \mathrm{H}_{\epsilon_0}$,*

$$\left| \begin{Vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Vmatrix} \right| < O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})+\delta\right)}\Big)$$

for any $\delta > 0$.

(2) For $\epsilon_0 > 0$ sufficiently small, there exists a $\delta_3 > 0$ depending only on ϵ such that for any r-hyperideal six-tuple $(a_1, \dots, a_6)$ with the corresponding $(\alpha_1, \dots, \alpha_6) \in \mathrm{H} \smallsetminus \mathrm{H}_{\epsilon_0}$,

$$\left| \begin{Vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Vmatrix} \right| < O\Big(e^{\frac{r}{2\pi}\left(v_8-\delta_3\right)}\Big),$$

where v_8 is the volume of the regular ideal hyperbolic octahedron.

Proposition 6.1 will be proved in Section 6.1.

Theorem 6.2 (Weak Reflection Symmetry)**.** *For $\epsilon_0 > 0$ sufficiently small, there exists a $\delta_4 > 0$ depending only on ϵ_0 such that for any r-hyperideal six-tuple $(a_1, \dots, a_6)$ with the corresponding $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_6) \in \mathrm{H}_{\epsilon_0}$, we have*

$$\left| \begin{Vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Vmatrix} - \begin{Vmatrix} r-2-a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Vmatrix} \right| < O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_4\right)}\Big).$$

Moreover, the same result holds if we change $a_k \mapsto r - 2 - a_k$ for any $k \in \{1, \dots, 6\}$.

Theorem 6.2 is a consequence of Proposition 6.3, Proposition 6.4 and Proposition 6.5 below.

To state them, we let $\epsilon > 0$ be sufficiently small so that for each $\boldsymbol{\alpha} \in \mathrm{H}_{\epsilon_0}$, the interval $[\xi_{\boldsymbol{\alpha}} - 3\epsilon, \xi_{\boldsymbol{\alpha}} + 3\epsilon]$ entirely lies in $(\max\{\tau_i\}, \min\{\eta_j\})$. Recall that

$$\tau_1 = \frac{\alpha_1+\alpha_2+\alpha_3}{2}, \quad \tau_2 = \frac{\alpha_1+\alpha_5+\alpha_6}{2}, \quad \tau_3 = \frac{\alpha_2+\alpha_4+\alpha_6}{2}, \quad \tau_4 = \frac{\alpha_3+\alpha_4+\alpha_5}{2},$$
$$\eta_1 = \frac{\alpha_1+\alpha_2+\alpha_4+\alpha_5}{2}, \quad \eta_2 = \frac{\alpha_1+\alpha_3+\alpha_4+\alpha_6}{2}, \quad \eta_3 = \frac{\alpha_2+\alpha_3+\alpha_5+\alpha_6}{2} \quad \text{and} \quad \eta_4 = 2\pi;$$

and keep in mind that τ_i's and η_j's are quantities depending on $\boldsymbol{\alpha}$. For $k \in \mathbb{Z}$, we call

$$\xi = \frac{2k\pi}{r}$$

the corresponding ξ.

Proposition 6.3. *There exists a $\delta_5 > 0$ depending only on ϵ such that for any r-hyperideal six-tuple $(a_1, \dots, a_6)$ with the corresponding $(\alpha_1, \dots, \alpha_6) \in \mathrm{H}_{\epsilon_0}$ and integer k with the corresponding $\xi \in (\max\{\tau_i\}, \min\{\eta_j\}) \smallsetminus [\xi_{\boldsymbol{\alpha}} - \epsilon, \xi_{\boldsymbol{\alpha}} + \epsilon]$, we have*

$$\left| e^{\frac{r}{4\pi\mathbf{i}} U_r(\boldsymbol{a},k)} \right| < O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_5\right)}\Big).$$

Proposition 6.3 will be proved in Section 6.1.

For the r-hyperideal six-tuple $(a_1,\dots,a_6)$ with the corresponding $(\alpha_1,\dots,\alpha_6)\in \mathrm{H}_{\epsilon_0}$ and integer k with the corresponding $\xi\in[\xi_{\boldsymbol{\alpha}}-\epsilon,\xi_{\boldsymbol{\alpha}}+\epsilon]$, we will use the Poisson Summation Formula. Let $\psi:\mathbb{R}\to\mathbb{R}$ be a C^∞-smooth bump function supported on $([\xi_{\boldsymbol{\alpha}}-2\epsilon,\xi_{\boldsymbol{\alpha}}+2\epsilon],[\xi_{\boldsymbol{\alpha}}-\epsilon,\xi_{\boldsymbol{\alpha}}+\epsilon])$, i.e.,

$$\begin{cases} \psi(\xi)=1, & \xi\in[\xi_{\boldsymbol{\alpha}}-\epsilon,\xi_{\boldsymbol{\alpha}}+\epsilon]\\ 0<\psi(\xi)<1, & \xi\in[\xi_{\boldsymbol{\alpha}}-2\epsilon,\xi_{\boldsymbol{\alpha}}+2\epsilon]\smallsetminus[\xi_{\boldsymbol{\alpha}}-\epsilon,\xi_{\boldsymbol{\alpha}}+\epsilon]\\ \psi(\xi)=0, & \xi\notin[\xi_{\boldsymbol{\alpha}}-2\epsilon,\xi_{\boldsymbol{\alpha}}+2\epsilon]. \end{cases}$$

Then by (3.9) and Proposition 6.3, we have

$$\left\|\begin{matrix} a_1 & a_2 & a_3\\ a_4 & a_5 & a_6\end{matrix}\right\| = (-1)^{\frac{r+1}{2}}\frac{2}{r}\sin\frac{2\pi}{r}\sum_{k\in\mathbb{Z}}\psi(\xi)e^{\frac{r}{4\pi\mathbf{i}}\left(U_r(\alpha_1,\dots,\alpha_6,\xi)\right)}+O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_5\right)}\Big); \tag{6.1}$$

and by the Poisson Summation Formula, we have

$$\left\|\begin{matrix} a_1 & a_2 & a_3\\ a_4 & a_5 & a_6\end{matrix}\right\| = \frac{(-1)^{\frac{r+1}{2}}}{\pi}\sin\frac{2\pi}{r}\sum_{n\in\mathbb{Z}}\mathrm{I}_{\boldsymbol{a}}(n)+O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_5\right)}\Big), \tag{6.2}$$

where the n-th Fourier coefficient

$$\mathrm{I}_{\boldsymbol{a}}(n)=\int_{\mathbb{R}}\psi(\xi)\exp\left(\frac{r}{4\pi\mathbf{i}}\Big(U_r(\alpha_1,\dots,\alpha_6,\xi)-4\pi n\xi\Big)\right)d\xi. \tag{6.3}$$

Proposition 6.4. *There exists a $\delta_6>0$ such that for any $n\neq 0$,*

$$\big|\mathrm{I}_{\boldsymbol{a}}(n)\big|<\frac{1}{k^2}\cdot O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_6\right)}\Big).$$

Proposition 6.4 will be proved in Section 6.1.

Proposition 6.5. *For $\boldsymbol{a}=(a_1,\dots,a_6)$, let $\boldsymbol{a}'$ be obtained from $\boldsymbol{a}$ by changing $a_k\mapsto r-2-a_k$ for some $k\in\{1,\dots,6\}$. Then there exists a $\delta_7>0$ such that*

$$\big|\mathrm{I}_{\boldsymbol{a}}(0)-\mathrm{I}_{\boldsymbol{a}'}(0)\big|<O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_7\right)}\Big).$$

Proposition 6.5 will be proved in Section 6.2.

6.1 The routine part

The goal of this section is to prove Propositions 6.1, 6.3 and 6.4

Proof of Proposition 6.1. For (1), by Proposition 2.15 (2), the relation (2.19), (3.19) and (3.20), for any $\delta>0$, we have

$$\Big|e^{\frac{r}{4\pi\mathbf{i}}U_r(\boldsymbol{\alpha},\xi)}\Big|<O\Big(e^{\frac{r}{2\pi}\left(V(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}})+\delta\right)}\Big)=O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})+\delta\right)}\Big).$$

Therefore, by (3.9), we have

$$\left|\left\|\begin{matrix} a_1 & a_2 & a_3\\ a_4 & a_5 & a_6\end{matrix}\right\|\right|\leqslant\frac{2}{r}\sin\frac{2\pi}{r}\sum_k\Big|e^{\frac{r}{4\pi\mathbf{i}}U_r(\boldsymbol{\alpha},\xi)}\Big|<O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})+\delta\right)}\Big)$$

for any $\delta_3 < \delta$, where the sum is over all the integers in between $\max\{T_i\}$ and $\min\{Q_j\}$.

For (2), by (3.21), for $\epsilon > 0$ sufficiently small, we have

$$\max_{(\boldsymbol{\alpha},\xi)\in \mathrm{B}\smallsetminus \mathrm{B}_\epsilon} V(\boldsymbol{\alpha},\xi) < v_8.$$

Let

$$\delta = \frac{1}{2}\Big(v_8 - \max_{(\boldsymbol{\alpha},\xi)\in \mathrm{B}\smallsetminus \mathrm{B}_\epsilon} V(\boldsymbol{\alpha},\xi)\Big) > 0.$$

If $\frac{2\pi(\boldsymbol{a}+\mathbf{1})}{r} \in \mathrm{H}\smallsetminus \mathrm{H}_\epsilon$, then $\big(\frac{2\pi(\boldsymbol{a}+\mathbf{1})}{r}, \frac{2\pi k}{r}\big) \in \mathrm{B}\smallsetminus \mathrm{B}_\epsilon$ for any $k \in [\max\{T_i\}, \min\{Q_j\}] \cap \mathbb{Z}$; and by (1), we have

$$\Big|e^{\frac{r}{4\pi\mathbf{i}}U_r(\boldsymbol{\alpha},\xi)}\Big| < O\Big(e^{\frac{r}{2\pi}\left(\max_{(\boldsymbol{\alpha},\xi)\in \mathrm{B}\smallsetminus \mathrm{B}_\epsilon} V(\boldsymbol{\alpha},\xi)+\delta\right)}\Big) < O\Big(e^{\frac{r}{2\pi}\left(v_8-\delta\right)}\Big).$$

Therefore, by (3.9), we have

$$\left|\begin{Vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Vmatrix}\right| \leqslant \frac{2}{r}\sin\frac{2\pi}{r}\sum_k \Big|e^{\frac{r}{4\pi\mathbf{i}}U_r(\boldsymbol{\alpha},\xi)}\Big| < O\Big(e^{\frac{r}{2\pi}\left(v_8-\delta_3\right)}\Big)$$

for any $\delta_3 < \delta$, where the sum is over all the integers in between $\max\{T_i\}$ and $\min\{Q_j\}$. □

Proof of Proposition 6.3. By (3.19) and (3.22), let

$$\delta' = \frac{1}{2}\min_{\boldsymbol{\alpha}\in\overline{\mathrm{H}_{\epsilon_0}}}\Big(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) - \max_{\xi\in(\max\{\tau_i\},\min\{\eta_j\})\smallsetminus[\xi_{\boldsymbol{\alpha}}-\epsilon,\xi_{\boldsymbol{\alpha}}+\epsilon]} V(\boldsymbol{\alpha},\xi)\Big) > 0.$$

Then by Proposition 2.15 (2) and the relation (2.19), for any $\delta_5 < \delta'$, we have

$$\Big|e^{\frac{r}{4\pi\mathbf{i}}U_r(\boldsymbol{a},k)}\Big| < O\Big(e^{\frac{r}{2\pi}\left(\max_{\xi\in(\max\{\tau_i\},\min\{\eta_j\})\smallsetminus[\xi_{\boldsymbol{\alpha}}-\epsilon,\xi_{\boldsymbol{\alpha}}+\epsilon]} V(\boldsymbol{\alpha},\xi)+\delta_5\right)}\Big) = O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_5\right)}\Big)$$

for each integer k so that the corresponding $\xi \in (\max\{\tau_i\}, \min\{\eta_j\})\smallsetminus[\xi_{\boldsymbol{\alpha}}-\epsilon, \xi_{\boldsymbol{\alpha}}+\epsilon]$. □

Proof of Proposition 6.4. By the Integration by Parts as on [32, Page 703], one has

$$\mathrm{I}_{\boldsymbol{a}}(n) = \frac{1}{n^2}\int_{\mathbb{R}} \Psi(\xi) e^{\frac{r}{4\pi\mathbf{i}}\left(U_r(\boldsymbol{\alpha},\xi)-4\pi n\xi\right)} d\xi,$$

where $\Psi(\xi)$ is a polynomial of the partial derivatives of at most the second order of $\psi(\xi)$ and U_r, and it suffices to bound each $U_r(\boldsymbol{\alpha},\xi) - 4\pi n\xi$.

By (3.17) and (3.18), there exists an $\epsilon' > 0$ sufficiently small such that for any $\boldsymbol{\alpha} \in \overline{\mathrm{H}_{\epsilon_0}}$ and for any $\xi \in \mathbb{C}$ with $\mathrm{Re}\xi \in [\xi_{\boldsymbol{\alpha}} - \epsilon, \xi_{\boldsymbol{\alpha}} + \epsilon]$ and $|\mathrm{Im}\xi| < \epsilon'$, we have

$$\left|\frac{\partial \mathrm{Im}U(\boldsymbol{\alpha},\xi)}{\partial \mathrm{Im}\xi}\right| < 2\pi, \tag{6.4}$$

and

$$\frac{\partial^2 \mathrm{Im}U(\boldsymbol{\alpha},\xi)}{\partial \mathrm{Re}\xi^2} < 0. \tag{6.5}$$

Now, in the case that $n > 0$, we let

$$\begin{aligned}
\mathrm{C}_{\text{top}} =& \Big\{\xi + \epsilon'\mathbf{i} \;\Big|\; \xi \in [\xi_{\boldsymbol{\alpha}} - \epsilon, \xi_{\boldsymbol{\alpha}} + \epsilon]\Big\},\\
\mathrm{C}_{\text{side}} =& \Big\{\xi' + t\mathbf{i} \;\Big|\; \xi' = \xi_{\boldsymbol{\alpha}} \pm \epsilon, t \in [0, \epsilon']\Big\},\\
\mathrm{C}_{\text{ext}} =& \mathbb{R}\smallsetminus[\xi_{\boldsymbol{\alpha}} - \epsilon, \xi_{\boldsymbol{\alpha}} + \epsilon],
\end{aligned}$$

and let

$$\mathrm{C} = \mathrm{C}_{\text{top}} \cup \mathrm{C}_{\text{side}} \cup \mathrm{C}_{\text{ext}}.$$

For $\xi \in \mathrm{C}_{\text{top}}$, by the Mean Value Theorem, (6.4) and (3.20), we have

$$\begin{aligned}\mathrm{Im}U(\boldsymbol{\alpha}, \xi) \leqslant & \mathrm{Im}U(\boldsymbol{\alpha}, \mathrm{Re}\xi) + \left| \frac{\partial \mathrm{Im}U(\boldsymbol{\alpha}, \xi)}{\partial \mathrm{Im}\xi}\Big|_{\xi=\xi'} \right| \cdot |\mathrm{Im}\xi'| \\ < & \mathrm{Im}U(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}}) + 2\pi\epsilon',\end{aligned}$$

where ξ' is some point on the line segment connecting ξ and $\mathrm{Re}\xi$. Then as $n \geqslant 1$,

$$\begin{aligned}\mathrm{Im}\Big(U(\boldsymbol{\alpha}, \xi) - 4\pi n\xi\Big) < & \mathrm{Im}U(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}}) + 2\pi\epsilon' - 4\pi n\epsilon' \\ \leqslant & \mathrm{Im}U(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}}) - 2\pi\epsilon' \\ = & 2\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) - 2\pi\epsilon'.\end{aligned} \tag{6.6}$$

For $\xi \in \mathrm{C}_{\text{side}}$, by (3.15), (3.19) and (3.22), let

$$\delta' = \frac{1}{2} \min_{\boldsymbol{\alpha} \in \overline{\mathrm{H}_{\epsilon_0}}} \Big(2\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) - \max_{\xi = \xi_{\boldsymbol{\alpha}} \pm \epsilon} \mathrm{Im}U(\boldsymbol{\alpha}, \xi)\Big) > 0.$$

Then by (3.22), for $\xi' = \xi_{\boldsymbol{\alpha}} \pm \epsilon$, we have

$$\mathrm{Im}\Big(U(\boldsymbol{\alpha}, \xi') - 4\pi n\xi'\Big) = \mathrm{Im}U(\boldsymbol{\alpha}, \xi') < 2\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) - 2\delta'. \tag{6.7}$$

By (3.18) and the harmonicity of $\mathrm{Im}U$, we have that $\mathrm{Im}U(\boldsymbol{\alpha}, \xi' + t\mathbf{i})$ is strictly concave up in t; and by (6.6) and (6.7), we have

$$\mathrm{Im}\Big(U(\boldsymbol{\alpha}, \xi) - 4\pi n\xi\Big) < 2\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) - 2\min\{\pi\epsilon', \delta'\}. \tag{6.8}$$

For $\xi \in \mathrm{C}_{\text{ext}}$, by (3.22), we have

$$\mathrm{Im}\Big(U(\boldsymbol{\alpha}, \xi) - 4\pi n\xi\Big) = \mathrm{Im}U(\boldsymbol{\alpha}, \xi) < 2\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) - 2\delta'. \tag{6.9}$$

Finally, let

$$\delta_6 = \min\{\pi\epsilon', \delta'\};$$

and notice that, as both ϵ' and δ' are independent of n, δ_6 is so as well. Putting (6.6), (6.8) and (6.9) together, we have

$$\mathrm{Im}\Big(U(\boldsymbol{\alpha}, \xi) - 4\pi n\xi\Big) < 2\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) - 2\delta_6.$$

on C. Then the result follows from Proposition 2.15 (1).

In the case that $n < 0$, let

$$\mathrm{C}_{\text{top}} = \Big\{\xi' - \epsilon\mathbf{i} \;\Big|\; \xi \in [\xi_{\boldsymbol{\alpha}} - \epsilon, \xi_{\boldsymbol{\alpha}} + \epsilon]\Big\} \quad \text{and} \quad \mathrm{C}_{\text{side}} = \Big\{\xi' - t\mathbf{i} \;\Big|\; \xi' = \xi_{\boldsymbol{\alpha}} \pm \epsilon, t \in [0, \epsilon']\Big\},$$

and the rest of the proof follows verbatim that of the previous case. □

6.2 The challenging part

The goal of this section is to prove Proposition 6.5. The strategy is as follows. We will show that the 0-th Fourier coefficient of the extended quantum $6j$-symbols and the b-$6j$ symbols recalled in Section 2.6 satisfy the same real Gevrey-1 estimate (see Proposition 6.11), hence share the same reflection symmetry after the truncated Borel-Laplace transforms (see Proposition 6.12). This will induce the desired weak reflection symmetry for the 0-th Fourier coefficient of the extended quantum $6j$-symbols. The rest of this section is organized as follows. In Subsection 6.2.1, we collect the basic properties of real Gevrey-1 estimates and truncated Borel-Laplace transforms, with the proofs postponed to Appendix A. In Subsection 6.2.2, we establish a real Gevrey-1 estimate for the double sine function; and in Subsection 6.2.3 we use this estimate to derive the corresponding real Gevrey-1 estimates of the 0-th Fourier coefficient of the extended quantum $6j$-symbols and the b-$6j$ symbols. Finally, we complete the proof of Proposition 6.5 in Subsection 6.2.4.

6.2.1 Real Gevrey-1 estimate and truncated Borel-Laplace transform

For $x_0 > 0$, let f be a complex valued function on $(0, x_0)$. Let

$$\widehat{f}(x) = \sum_{n=0}^{\infty} a_n x^n$$

be a formal power series in x. Here we do not assume the continuity of f, nor the convergence of $\widehat{f}$. For $T > 0$, and let g be a complex valued integrable function on $[0, T]$.

Definition 6.6. *(1) If there exist constants $C > 0$ and $\sigma > 0$ such that*

$$\left| f(x) - \sum_{n=0}^{N-1} a_n x^n \right| \leqslant C\sigma^N N! |x|^N \tag{6.10}$$

for each $N \geqslant 0$ and $x \in (0, x_0)$, then we say that the function f satisfies a (C, σ) real Gevrey-1 estimate*, and call the formal power series $\widehat{f}$ a* Gevrey-1 expansion *of f*

(2) The Borel transform *$\mathcal{B}\widehat{f}$ of the formal power series $\widehat{f}$ is the following formal power series*

$$\mathcal{B}\widehat{f}(z) = \sum_{n=0}^{\infty} \frac{a_n}{n!} z^n.$$

(3) The truncated Laplace transform *of g truncated at T is the function $\mathcal{L}_T g$ defined on $(0, \infty)$ by*

$$\mathcal{L}_T g(x) = \frac{1}{x} \int_0^T e^{-\frac{t}{x}} g(t) dt.$$

Lemma 6.7. *Let $x_0 > 0$, and let f be a complex valued function on $(0, x_0)$. Suppose f satisfies a (C, σ) real Gevrey - 1 estimate for some $C > 0$ and $\sigma > 0$. Then:*

(1) For each $n \geqslant 0$,

$$|a_n| \leqslant C\sigma^n n!, \tag{6.11}$$

and the Borel transform $\mathcal{B}\widehat{f}(z)$ of $\widehat{f}$ defines a holomorphic function on $\left\{ z \in \mathbb{C} \;\middle|\; |z| < \sigma^{-1} \right\}$.

(2) For each $T \in (0, \sigma^{-1})$, there exists a $D > 0$ depending only on C, σ and T such that for each $x \in (0, \min\{x_0, T\})$,

$$\left| f(x) - \mathcal{L}_T \mathcal{B} \widehat{f}(x) \right| \leqslant D \sqrt{\frac{T}{x}} e^{-\frac{T}{x}}.$$

Lemma 6.8. *Let $x_0 > 0$, and let f be a complex valued function on $(0, x_0)$. Suppose f satisfies a (C, σ) real Gevrey - 1 estimate for some $C > 0$ and $\sigma > 0$. Then $e^{f(x)}$ satisfies a (C_1, σ_1) real Gevrey - 1 estimate*

$$\left| e^{f(x)} - \sum_{n=0}^{N-1} b_n x^n \right| \leqslant C_1 \sigma_1^N N! |x|^N$$

for some $C_1 > 0$ depending only on C, σ and x_0 and some σ_1 depending only on σ. Moreover, if

$$\widehat{f}(x) = \sum_{n=0}^{\infty} a_n x^n$$

is a Gevrey - 1 expansion of f, then for each $n \geqslant 0$,

$$b_n = e^{a_0} \sum_{k=0}^{n} \sum_{\substack{m_1, \ldots, m_n \geqslant 0 \\ m_1 + \cdots + m_n = k \\ m_1 + 2m_2 + \cdots + n m_n = n}} \frac{1}{m_1! \cdots m_n!} \prod_{j=1}^{n} a_j^{m_j}.$$

Lemma 6.9. *Let $\rho > 0$ and $\rho_0 \in (0, \rho)$, and let $g(u)$ be a holomorphic function defined for $u \in \mathbb{C}$ with $|u| < \rho$ satisfying*

$$\inf_{u \in [-\rho_0, \rho_0]} \mathrm{Re} g(u) \geqslant c \quad \text{and} \quad \sup_{|u| < \rho} |g(u)| \leqslant G$$

for some $c > 0$ and $G > 0$. Let $x_0 > 0$, and let $f(u, x)$ be a function defined for u with $|u| < \rho$ and $x \in (0, x_0)$. Suppose that $f(u, x)$ is continuous in $u \in [-\rho_0, \rho_0]$, and satisfies a (C, σ) real Gevrey-1 estimate

$$\left| f(u, x) - \sum_{n=0}^{N-1} a_n(u) x^n \right| \leqslant C \sigma^N N! x^N$$

for each u with $|u| < \rho$ and for some $C > 0$ and $\sigma > 0$; and suppose for each $n \geqslant 0$ that $a_n(u)$ is a holomorphic function on the disk $|u| < \rho$. Then the integral

$$I(x) = \frac{1}{\sqrt{2\pi x}} \int_{-\rho_0}^{\rho_0} e^{-\frac{g(u)u^2}{x}} f(u, x) du$$

satisfies a (C_2, σ_2) real Gevrey-1 estimate

$$\left| I(x) - \sum_{n=0}^{N-1} c_n x^n \right| \leqslant C_2 \sigma_2^N N! x^N$$

for some $C_2 > 0$ depending only on C, c and G, and some $\sigma_2 > 0$ depending only on σ, c, G, ρ_0 and ρ, and

$$c_n = \frac{1}{\sqrt{2}} \sum_{\substack{j,k \geqslant 0 \\ j+k=n}} \frac{F_j^{(2k)}(0)}{4^k k!},$$

where for each $j \geqslant 0$, F_j is a holomorphic function defined in a neighborhood of 0 such that if $v = u\sqrt{g(u)}$, then

$$F_j(v) = \frac{a_j(u)}{\left(u\sqrt{g(u)}\right)'}.$$

6.2.2 Gevrey-1 estimate of the double sine function

To have a uniform treatment of the extended quantum $6j$-symbols and the b-$6j$ symbols, we will consider the function $s_\hbar$ defined for $\hbar \in (-1,1)$ and $z \in \mathbb{C}$ with $0 < \mathrm{Re} z < \pi$ as follows. Let Ω be the contour going along the real line and passing above the origin. For $\hbar \neq 0$, we let

$$s_\hbar(z) \doteq \pi\hbar \int_\Omega \frac{\sinh\big((\pi + \pi\hbar - 2z)x\big)}{4x\sinh(\pi\hbar x)\sinh(\pi x)} dx;$$

and let

$$s_0(z) \doteq \int_\Omega \frac{\sinh\big((\pi - 2z)x\big)}{4x^2\sinh(\pi x)} dx.$$

Directly from the definition, we have

$$s_\hbar\Big(z + \frac{\pi\hbar}{2}\Big) = s_{-\hbar}\Big(z - \frac{\pi\hbar}{2}\Big).$$

Let

$$S_b(z) = \exp\left(\int_\Omega \frac{\sinh\Big(\big(\frac{Q}{2} - z\big)t\Big)}{4t\sinh(\frac{bt}{2})\sinh(\frac{t}{2b})} dt\right)$$

be the double sine function defined in (2.11), and let

$$L(z) = z^2 - \pi z + \frac{\pi^2}{6} - \mathrm{Li}_2\big(e^{2\mathbf{i}z}\big)$$

be the complexified Lobachevsky function defined in (2.18). Then for $\hbar > 0$ and $b = \sqrt{\hbar}$,

$$2\pi\mathbf{i}b^2 \log S_b\Big(\frac{z}{\pi b} + \frac{b}{2}\Big) = 2\mathbf{i}s_\hbar\Big(z + \frac{\pi\hbar}{2}\Big) = 2\mathbf{i}s_{-\hbar}\Big(z - \frac{\pi\hbar}{2}\Big); \tag{6.12}$$

and

$$L(z) = 2\mathbf{i}s_0(z). \tag{6.13}$$

Proposition 6.10. *Let $M > 0$, $\Lambda > 0$ and $\epsilon \in (0, \frac{\pi}{2})$. Suppose that*

$$\epsilon \leqslant \mathrm{Re} z \leqslant \pi - \epsilon, \quad \big|\mathrm{Im} z\big| \leqslant M \quad \textit{and} \quad |\lambda| \leqslant \Lambda.$$

Then there exists a $C > 0$ depending only on M and Λ, and a $\sigma > 0$ depending only on ϵ, such that for every integer $N \geqslant 1$ and real number $\hbar$ with $0 < |\hbar| < \frac{\epsilon}{2\pi(\Lambda+1)}$,

$$\left|s_\hbar\big(z + \lambda\pi\hbar\big) - \frac{1}{2\mathbf{i}}\sum_{n=0}^{N-1} \frac{\pi^n}{n!} B_n(\lambda) L^{(n)}(z)\hbar^n\right| \leqslant C\sigma^N N!|\hbar|^N, \tag{6.14}$$

where B_n is the n-th Bernoulli polynomial characterized by

$$\frac{te^{\lambda t}}{e^t - 1} = \sum_{n=0}^{\infty} \frac{B_n(\lambda)}{n!} t^n$$

for $t \in \mathbb{C}$ with $|t| < 2\pi$.

Proof. Let

$$A_\Lambda = \max_{|t|=4,\lambda\in[-\Lambda,\Lambda]} \left|\frac{te^{\lambda t}}{e^t-1}\right|,$$

and let

$$C_\Lambda = 2A_\Lambda + \frac{2}{1-e^{-2}}.$$

We first show that for all $t \in \mathbb{R}$ and $N \geqslant 1$,

$$\left|\frac{te^{\lambda t}}{e^t-1} - \sum_{n=0}^{N-1} \frac{B_n(\lambda)}{n!}t^n\right| \leqslant C_\Lambda e^{\Lambda|t|}\left(\frac{|t|}{2}\right)^N. \tag{6.15}$$

To see this, by Cauchy's Estimate, we have

$$\frac{|B_n(\lambda)|}{n!} \leqslant \frac{A_\Lambda}{4^n}. \tag{6.16}$$

As a consequence, for $t \in \mathbb{C}$ with $|t| < 2$, we have

$$\left|\frac{te^{\lambda t}}{e^t-1} - \sum_{n=0}^{N-1} \frac{B_n(\lambda)}{n!}t^n\right| \leqslant A_\Lambda \sum_{n=N}^{\infty}\left(\frac{|t|}{4}\right)^n \leqslant 2A_\Lambda e^{\Lambda|t|}\left(\frac{|t|}{2}\right)^N. \tag{6.17}$$

Now for $t \in \mathbb{R}$ with $|t| \geqslant 2$, we have

$$\left|\frac{te^{\lambda t}}{e^t-1}\right| \leqslant \frac{|t|e^{\Lambda|t|}}{1-e^{-2}} \leqslant \frac{2e^{\Lambda|t|}}{1-e^{-2}}\left(\frac{|t|}{2}\right)^N;$$

and by (6.16), we have

$$\left|\sum_{n=0}^{N-1} \frac{B_n(\lambda)}{n!}t^n\right| \leqslant A_\Lambda \sum_{n=0}^{N-1} 2^{-n}\left(\frac{|t|}{2}\right)^n \leqslant 2A_\Lambda e^{\Lambda|t|}\left(\frac{|t|}{2}\right)^N. \tag{6.18}$$

Then (6.15) follows from (6.17) and (6.18).

Next, to simplify the notation, let the remainder

$$R_N(t,\lambda) = \frac{te^{\lambda t}}{e^t-1} - \sum_{n=0}^{N-1}\frac{B_n(\lambda)}{n!}t^n.$$

Then we observe that for $N \geqslant 3$,

$$\begin{aligned} &s_\hbar\big(z+\lambda\pi\hbar\big) - \frac{1}{2\mathbf{i}}\sum_{n=0}^{N-1}\frac{\pi^n}{n!}B_n(\lambda)L^{(n)}(z)\hbar^n \\ =&\frac{1}{8}\int_\Omega \frac{e^{(\pi-2z)x}R_N(-2\pi\hbar x,\lambda) - e^{-(\pi-2z)x}R_N(2\pi\hbar x,\lambda)}{x^2\sinh(\pi x)}dx \\ =&\frac{1}{8}\int_{\mathbb{R}} \frac{e^{(\pi-2z)x}R_N(-2\pi\hbar x,\lambda) - e^{-(\pi-2z)x}R_N(2\pi\hbar x,\lambda)}{x^2\sinh(\pi x)}dx, \end{aligned} \tag{6.19}$$

where the first equality comes from $\sinh\big((\pi-2z)x\big) = \frac{1}{2}\big(e^{(\pi-2z)x} - e^{-(\pi-2z)x}\big)$, (6.13) and its consequence that

$$\frac{L^{(n)}(z)}{2\mathbf{i}} = \frac{1}{8}\int_\Omega \frac{(-2x)^n e^{(\pi-2z)x} - (2x)^n e^{-(\pi-2z)x}}{x^2\sinh(\pi x)}dx$$

for $n \geqslant 0$, and the second equality comes from (6.15) that the integrand is of order $O\big(x^{N-3}\big)$ hence 0 is a removable singularity of the integrand. For $x > 0$, we have

$$\frac{\big|e^{(\pi-2z)x}\big| + \big|e^{-(\pi-2z)x}\big|}{\big|\sinh(\pi x)\big|} \leqslant \frac{4e^{-2\epsilon x}}{1-e^{-2\pi x}} \leqslant 4e^{-2\epsilon x}\bigg(1+\frac{1}{2\pi x}\bigg).$$

Together with (6.15) and (6.19), we have

$$\begin{aligned}\bigg|s_\hbar\big(z+\lambda\pi\hbar\big) - \frac{1}{2\mathbf{i}}\sum_{n=0}^{N-1}\frac{\pi^n}{n!}B_n(\lambda)L^{(n)}(z)\hbar^n\bigg| \leqslant& C_\Lambda \pi^N|\hbar|^N\int_0^\infty x^{N-2}e^{2(\pi|\hbar|\Lambda-\epsilon)x}\bigg(1+\frac{1}{2\pi x}\bigg)dx\\ \leqslant& C_\Lambda\pi^N|\hbar|^N\int_0^\infty x^{N-2}e^{-\epsilon x}\bigg(1+\frac{1}{2\pi x}\bigg)dx\\ =& C_\Lambda\pi^N|\hbar|^N\Big((N-2)!\epsilon^{1-N}+\frac{1}{2\pi}(N-3)!\epsilon^{2-N}\Big)\\ \leqslant& C_\Lambda\pi^N\epsilon^{-N}N!|\hbar|^N,\end{aligned}$$

where the second inequality comes from that $|\hbar| < \frac{\epsilon}{2\pi(\Lambda+1)} < \frac{\epsilon}{2\pi\Lambda}$, and the last inequality comes from that $\epsilon < \frac{\pi}{2}$ so that $\epsilon < \frac{(N-1)N}{2}$ and $\frac{\epsilon^2}{2\pi} < \frac{(N-2)(N-1)N}{2}$. Then (6.14) holds for $N \geqslant 3$ with $C = C_\Lambda$ and $\sigma = \pi\epsilon^{-1}$.

For $N = 2$, we have

$$\begin{aligned}&\bigg|s_\hbar\big(z+\lambda\pi\hbar\big) - \frac{1}{2\mathbf{i}}\sum_{n=0}^{1}\frac{\pi^n}{n!}B_n(\lambda)L^{(n)}(z)\hbar^n\bigg|\\ \leqslant&\bigg|s_\hbar\big(z+\lambda\pi\hbar\big) - \frac{1}{2\mathbf{i}}\sum_{n=0}^{2}\frac{\pi^n}{n!}B_n(\lambda)L^{(n)}(z)\hbar^n\bigg| + \frac{\pi^2}{2}\Big|B_2(\lambda)\Big|\big|\cot z\big||\hbar|^2\\ \leqslant& C_\Lambda\pi^3\epsilon^{-3}3!|\hbar|^3 + \frac{\pi^2}{2}\Big|B_2(\lambda)\Big|\big|\cot z\big||\hbar|^2.\end{aligned}$$

Since $|\mathrm{Im} z| \leqslant M$, there is a $C_M > 0$ such that

$$\big|\cot z\big| \leqslant C_M\epsilon^{-2};$$

and since $B_2(\lambda)$ is bounded on $[-\Lambda,\Lambda]$ and $\frac{|\hbar|}{\epsilon} < \frac{1}{2\pi(\Lambda+1)} < 1$, we have (6.14) for $N = 2$, by increasing $C = C_\Lambda$ if necessary and with $\sigma = \pi\epsilon^{-1}$.

For $N = 1$, we have

$$\begin{aligned}&\bigg|s_\hbar\big(z+\lambda\pi\hbar\big) - \frac{1}{2\mathbf{i}}B_0(\lambda)L(z)\bigg|\\ \leqslant&\bigg|s_\hbar\big(z+\lambda\pi\hbar\big) - \frac{1}{2\mathbf{i}}\sum_{n=0}^{1}\frac{\pi^n}{n!}B_n(\lambda)L^{(n)}(z)\hbar^n\bigg| + \pi\Big|B_1(\lambda)\Big|\big|\log\big(2\sin z\big)\big||\hbar|\\ \leqslant& C\pi^2\epsilon^{-2}2!|\hbar|^2 + \pi\Big|B_1(\lambda)\Big|\big|\log\big(2\sin z\big)\big||\hbar|.\end{aligned}$$

Since $|\mathrm{Im} z| \leqslant M$, there is a $C_M > 0$ such that

$$\big|\log\big(2\sin z\big)\big| \leqslant C_M\epsilon^{-1};$$

and since $B_1(\lambda)$ is bounded on $[-\Lambda,\Lambda]$ and $\frac{|\hbar|}{\epsilon} < \frac{1}{2\pi(\Lambda+1)} < 1$, we have (6.14) for $N = 1$, by increasing C further if necessary and with $\sigma = \pi\epsilon^{-1}$. □

6.2.3 Extended quantum $6j$-symbols and b-$6j$symbols

The goal of this subsection is to understand the relationship between the 0-th Fourier coefficient of the extended quantum $6j$-symbols and the b-$6j$ symbols.

For $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_6) \in \mathrm{H}$, $\xi \in (\max\{\tau_i\}, \min\{\eta_j\})$, and $h \in (-1, 1)$ with $|\hbar|$ sufficiently small, define the function

$$V_\hbar(\boldsymbol{\alpha}, \xi) \doteq \frac{1}{2}\sum_{i=1}^{4}\sum_{j=1}^{4} s_\hbar\Big(\eta_j - \tau_i + \frac{\pi\hbar}{2}\Big) - \sum_{i=1}^{4} s_\hbar\Big(\xi - \tau_i + \frac{5\pi\hbar}{2}\Big) - \sum_{j=1}^{4} s_\hbar\big(\eta_j - \xi - \pi\hbar\big).$$

Let $\epsilon_0 > 0$ be as in Proposition 6.1 and Theorem 6.2, and let $\boldsymbol{a}$ by an r-hyperideal six-tuple with the corresponding $\boldsymbol{\alpha} \in \mathrm{H}_{\epsilon_0}$. Then by (6.3), (3.10) and (6.12), for $\hbar = b^2 = \frac{2}{r}$, we have that the 0-th Fourier coefficient of the extended quantum $6j$-symbol

$$\mathrm{I}_{\boldsymbol{a}}(0) = \int_{\mathbb{R}} \psi(\xi) \exp\left(\frac{V_\hbar(\alpha_1, \dots, \alpha_6, \xi)}{\pi\hbar}\right) d\xi. \tag{6.20}$$

For $b \in (0, 1)$, let $(a_1, \dots, a_6) \in \mathbb{R}^6$ be a b-admissible six-tuple, and recall that for each $k \in \{1, \dots, 6\}$, $\alpha_k = 2\pi b a_k - \pi b^2$. Then by (2.16) and (6.12), for $\hbar = b^2$, we have that the b-$6j$ symbol

$$\begin{Bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Bmatrix}_b = \frac{1}{\pi \mathbf{i} b} \int_\Gamma \exp\left(\frac{V_{-\hbar}(\alpha_1, \dots, \alpha_6, \xi)}{-\pi\hbar}\right) d\xi, \tag{6.21}$$

where Γ is a vertical straight line that intersects the real axis in the interval $\big(\max\{\tau_i\} + \frac{3\pi\hbar}{2}, \min\{\eta_j\} + 2\pi\hbar\big)$.

Recall that $\epsilon > 0$ is sufficiently small so that for each $\boldsymbol{\alpha} \in \mathrm{H}_{\epsilon_0}$, the interval $[\xi_{\boldsymbol{\alpha}} - 3\epsilon, \xi_{\boldsymbol{\alpha}} + 3\epsilon]$ entirely lies in $(\max\{\tau_i\}, \min\{\eta_j\})$; and also recall that $\Delta_{\boldsymbol{\alpha}}$ is the truncated hyperideal tetrahedron with dihedral angles $|\alpha_1 - \pi|, \dots, |\alpha_6 - \pi|$ with volume $\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})$.

We let

$$\mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar) = \frac{e^{-\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}}}{\sqrt{2\pi\hbar}} \int_{\xi_{\boldsymbol{\alpha}} - \epsilon}^{\xi_{\boldsymbol{\alpha}} + \epsilon} \exp\left(\frac{V_\hbar(\alpha_1, \dots, \alpha_6, \xi)}{\pi\hbar}\right) d\xi; \tag{6.22}$$

and let

$$\mathcal{I}^b_{\boldsymbol{\alpha}}(\hbar) = \frac{e^{\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}}}{\mathbf{i}\sqrt{2\pi\hbar}} \int_{\xi_{\boldsymbol{\alpha}} - \mathbf{i}\epsilon}^{\xi_{\boldsymbol{\alpha}} + \mathbf{i}\epsilon} \exp\left(\frac{V_{-\hbar}(\alpha_1, \dots, \alpha_6, \xi)}{-\pi\hbar}\right) d\xi, \tag{6.23}$$

where the integral is along the vertical line connecting $\xi_{\boldsymbol{\alpha}} \pm \mathbf{i}\epsilon$.

Proposition 6.11. *There exist functions $c_n, d_n : \mathrm{H}_{\epsilon_0} \to \mathbb{R}$ for all integers $n \geqslant 0$, and constants $C > 0$, $\sigma > 0$ and $\hbar_0 > 0$ depending only on ϵ, such that the following properties hold:*

(1) For all $\boldsymbol{\alpha} \in \mathrm{H}_{\epsilon_0}$, $N \geqslant 0$ and $\hbar \in (0, \hbar_0)$,

$$\left|\mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar) - \sum_{n=0}^{N-1} c_n(\boldsymbol{\alpha})\hbar^n\right| \leqslant C\sigma^N N!\hbar^N$$

and

$$\left|\mathcal{I}^b_{\boldsymbol{\alpha}}(\hbar) - \sum_{n=0}^{N-1} d_n(\boldsymbol{\alpha})\hbar^n\right| \leqslant C\sigma^N N!\hbar^N,$$

i.e., both $\mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar)$ and $\mathcal{I}^b_{\boldsymbol{\alpha}}(\hbar)$ satisfy a (C, σ) Gevrey-1 estimate. Moreover,

$$d_n(\boldsymbol{\alpha}) = (-1)^n c_n(\boldsymbol{\alpha})$$

for all $\boldsymbol{\alpha} \in \mathrm{H}_{\epsilon_0}$ and $n \geqslant 0$.

(2) Consider the formal power series

$$\widehat{\mathcal{I}^r_{\boldsymbol{\alpha}}}(\hbar)=\sum_{n=0}^{\infty}c_n(\boldsymbol{\alpha})\hbar^n \quad \textit{and} \quad \widehat{\mathcal{I}^b_{\boldsymbol{\alpha}}}(\hbar)=\sum_{n=0}^{\infty}d_n(\boldsymbol{\alpha})\hbar^n.$$

Then for all $\boldsymbol{\alpha}\in \mathrm{H}_{\epsilon_0}$, both of the Borel transforms $\mathcal{B}\widehat{\mathcal{I}^r_{\boldsymbol{\alpha}}}$ and $\mathcal{B}\widehat{\mathcal{I}^b_{\boldsymbol{\alpha}}}$ have radius of convergence at least σ^{-1}. Moreover,

$$\mathcal{B}\widehat{\mathcal{I}^r_{\boldsymbol{\alpha}}}(t)=\mathcal{B}\widehat{\mathcal{I}^b_{\boldsymbol{\alpha}}}(-t)$$

for all $t\in\mathbb{C}$ with $|t|<\sigma^{-1}$.

(3) For each $T\in(0,\sigma^{-1})$, there exists a $D>0$ depending only on ϵ and T such that for each $\hbar\in(0,\min\{\hbar_0,T\})$,

$$\left|\mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar)-\mathcal{L}_T\mathcal{B}\widehat{\mathcal{I}^r_{\boldsymbol{\alpha}}}(\hbar)\right|\leqslant D\sqrt{\frac{T}{\hbar}}e^{-\frac{T}{\hbar}}.$$

Proof. By Lemma 6.7, (2) and (3) follow directly from (1).

To see (1), for $\boldsymbol{\alpha}\in\mathrm{H}_{\epsilon_0}$, let $s=\xi-\xi_{\boldsymbol{\alpha}}$. Then by Proposition 6.10 with $M=2\epsilon$ and $\Lambda=3$, there exist $C_0>0$ and $\sigma_0>0$ depending only on ϵ such that for each $N\geqslant 1$, $\hbar\in\mathbb{R}$ with $|\hbar|\in(0,\frac{\epsilon}{8\pi})$, and $s\in\mathbb{C}$ with $|s|<2\epsilon$,

$$\left|V_\hbar\big(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}+s\big)-\sum_{n=0}^{N-1}v_n(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}+s)\hbar^n\right|\leqslant C_0\sigma_0^N N!|\hbar|^N,$$

where for each $n\geqslant 0$,

$$\begin{aligned}&v_n(\boldsymbol{\alpha},\xi)\\ =&\frac{\pi^n}{2\mathbf{i}n!}\bigg(\frac{B_n\big(\frac{1}{2}\big)}{2}\sum_{i=1}^{4}\sum_{j=1}^{4}L^{(n)}(\eta_j-\tau_i)-B_n\Big(\frac{5}{2}\Big)\sum_{i=1}^{4}L^{(n)}(\xi-\tau_i)-B_n(-1)\sum_{j=1}^{4}L^{(n)}(\eta_j-\xi)\bigg)\end{aligned}$$

is holomorphic in ξ. As a consequence, we have

$$\begin{aligned}\left|\frac{V_\hbar\big(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}+s\big)-V_0\big(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}+s\big)}{\pi\hbar}-\frac{1}{\pi}\sum_{n=0}^{N-1}v_{n+1}(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}+s)\hbar^n\right|\leqslant&\frac{C_0}{\pi}\sigma_0^{N+1}(N+1)!|\hbar|^N\\ \leqslant&\frac{C_0\sigma_0}{\pi}(2\sigma_0)^N N!|\hbar|^N,\end{aligned}$$

where the last inequality comes from that $N+1\leqslant 2^N$ for $N\geqslant 0$. Then by Lemma 6.8, there exist $C_1>0$ and $\sigma_1>0$ depending only on ϵ such that for each $N\geqslant 0$, $\hbar\in\mathbb{R}$ with $|\hbar|\in(0,\frac{\epsilon}{8\pi})$, and $s\in\mathbb{C}$ with $|s|<2\epsilon$,

$$\left|\exp\left(\frac{V_\hbar\big(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}+s\big)-V_0\big(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}+s\big)}{\pi\hbar}\right)-\sum_{n=0}^{N-1}b_n(\boldsymbol{\alpha},\xi_{\boldsymbol{\alpha}}+s)\hbar^n\right|\leqslant C_1\sigma_1^N N!|\hbar|^N, \tag{6.24}$$

where for each $n\geqslant 0$,

$$b_n(\boldsymbol{\alpha},\xi)=e^{\frac{v_1(\boldsymbol{\alpha},\xi)}{\pi}}\sum_{k=0}^{n}\sum_{\substack{m_1,\dots,m_n\geqslant 0\\ m_1+\cdots+m_n=k\\ m_1+2m_2+\cdots+nm_n=n}}\frac{1}{m_1!\cdots m_n!}\prod_{j=1}^{n}\left(\frac{v_{j+1}(\boldsymbol{\alpha},\xi)}{\pi}\right)^{m_j}$$

is holomorphic in ξ.

Next, for $\boldsymbol{\alpha} \in \mathrm{H}_{\epsilon_0}$ and $s \in \mathbb{C}$ with $|s| < 2\epsilon$, we let

$$g_{\boldsymbol{\alpha}}(s) \doteq \frac{V_0(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}}) - V_0(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + s)}{s^2}$$

for $s \neq 0$, and let

$$g_{\boldsymbol{\alpha}}(0) \doteq -\frac{1}{2} \frac{\partial^2 V_0(\boldsymbol{\alpha}, \xi)}{\partial \xi^2}\Big|_{\xi=\xi_{\boldsymbol{\alpha}}}.$$

Since $\frac{\partial V_0(\boldsymbol{\alpha}, \xi)}{\partial \xi}\Big|_{\xi=\xi_{\boldsymbol{\alpha}}} = 0$, $g_{\boldsymbol{\alpha}}$ is continuous at 0, and hence is holomorphic on the disk $|s| < 2\epsilon$. Then by the compactness of the region $\big\{(\boldsymbol{\alpha}, \xi) \in \mathbb{R}^6 \times \mathbb{C} \;\big|\; \boldsymbol{\alpha} \in \overline{\mathrm{H}_{\epsilon_0}}, |\xi - \xi_{\boldsymbol{\alpha}}| \leqslant 2\epsilon\big\}$, there exists a $G > 0$ depending only on ϵ such that

$$\sup_{|s| \leqslant 2\epsilon} |g_{\boldsymbol{\alpha}}(s)| \leqslant \pi G \tag{6.25}$$

for all $\boldsymbol{\alpha} \in \mathrm{H}_{\epsilon_0}$. Also, by Taylor's Formula and that $\frac{\partial V_0(\boldsymbol{\alpha}, \xi)}{\partial \xi}\big|_{\xi=\xi_{\boldsymbol{\alpha}}} = 0$, we have

$$g_{\boldsymbol{\alpha}}(s) = -\int_0^1 (1-t) \frac{\partial^2 V_0(\boldsymbol{\alpha}, \xi)}{\partial \xi^2}\Big|_{\xi=\xi_{\boldsymbol{\alpha}}+ts} dt. \tag{6.26}$$

By the fact that $\mathrm{Re}V_0(\boldsymbol{\alpha}, \xi)$ is strictly concave down in $\mathrm{Re}\xi$ (see (3.18) or [9]) and is strictly concave up in $\mathrm{Im}\xi$ when sufficiently close to $\xi_{\boldsymbol{\alpha}}$ (see [24, Proposition 3.11]), and by the compactness of the region $\big\{(\boldsymbol{\alpha}, \xi) \in \mathbb{R}^6 \times \mathbb{C} \;\big|\; \boldsymbol{\alpha} \in \overline{\mathrm{H}_{\epsilon_0}}, |\xi - \xi_{\boldsymbol{\alpha}}| \leqslant \epsilon\big\}$, there exists a $c > 0$ depending only on ϵ such that

$$\sup_{s \in \mathbb{R}, |s| \leqslant \epsilon} \mathrm{Re}\bigg(\frac{\partial^2 V_0(\boldsymbol{\alpha}, \xi)}{\partial \xi^2}\Big|_{\xi=\xi_{\boldsymbol{\alpha}}+ts}\bigg) = \sup_{s \in \mathbb{R}, |s| \leqslant \epsilon} \frac{\partial^2 \mathrm{Re}V_0(\boldsymbol{\alpha}, \xi)}{\partial \mathrm{Re}\xi^2}\Big|_{\xi=\xi_{\boldsymbol{\alpha}}+ts} \leqslant -2\pi c$$

and

$$\sup_{s \in \mathbf{i}\mathbb{R}, |s| \leqslant \epsilon} \mathrm{Re}\bigg(\frac{\partial^2 V_0(\boldsymbol{\alpha}, \xi)}{\partial \xi^2}\Big|_{\xi=\xi_{\boldsymbol{\alpha}}+ts}\bigg) = -\inf_{s \in \mathbf{i}\mathbb{R}, |s| \leqslant \epsilon} \frac{\partial^2 \mathrm{Re}V_0(\boldsymbol{\alpha}, \xi)}{\partial \mathrm{Im}\xi^2}\Big|_{\xi=\xi_{\boldsymbol{\alpha}}+ts} \leqslant -2\pi c.$$

Together with (6.26), we have

$$\inf_{s \in \mathbb{R} \cup \mathbf{i}\mathbb{R}, |s| \leqslant \epsilon} \mathrm{Re} g_{\boldsymbol{\alpha}}(s) \geqslant 2\pi c \int_0^1 (1-t) dt = \pi c. \tag{6.27}$$

On the one hand, letting $s = u$, we have

$$\mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\epsilon}^{\epsilon} e^{-\frac{u^2 g_{\boldsymbol{\alpha}}(u)}{\pi\hbar}} \exp\left(\frac{V_\hbar(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + u) - V_0(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + u)}{\pi\hbar}\right) du. \tag{6.28}$$

Then by (6.24), (6.25) and (6.27), all the conditions of Lemma 6.9 are satisfied with $x = \hbar$, $\rho = 2\epsilon$, $\rho_0 = \epsilon$, $c = c$, $G = G$, $g(u) = \frac{g_{\boldsymbol{\alpha}}(u)}{\pi}$, $f(u, x) = \exp\big(\frac{V_\hbar(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}}+u) - V_0(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}}+u)}{\pi\hbar}\big)$ and $I(x) = \mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar)$. Then by Lemma 6.9, $\mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar)$ satisfies a (C, σ) Gevrey-1 estimate, where $C > 0$ and $\sigma > 0$ depend only on C_1, σ_1, c, G and ϵ, which in turn depend only on ϵ, and

$$c_n(\boldsymbol{\alpha}) = \frac{1}{\sqrt{2}} \sum_{\substack{j,k \geqslant 0 \\ j+k=n}} \frac{F^{(2k)}_{r,j}(0)}{4^k k!}, \tag{6.29}$$

where for each $j \geqslant 0$, $F_{r,j}$ is a holomorphic function defined in a neighborhood of 0 such that if $v_r = v_r(u) \doteq u\sqrt{\frac{g_{\boldsymbol{\alpha}}(u)}{\pi}}$, then

$$F_{r,j}(v_r) = \frac{b_j\big(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + u\big)}{\Big(u\sqrt{\frac{g_{\boldsymbol{\alpha}}(u)}{\pi}}\Big)'} = \frac{b_j\big(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + u\big)}{v_r'(u)}. \tag{6.30}$$

On the other hand, letting $s = \mathbf{i}u$, we have

$$\mathcal{I}^b_{\boldsymbol{\alpha}}(\hbar) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\epsilon}^{\epsilon} e^{-\frac{u^2 g_{\boldsymbol{\alpha}}(\mathbf{i}u)}{\pi\hbar}} \exp\left(\frac{V_{-\hbar}(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + \mathbf{i}u) - V_0(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + \mathbf{i}u)}{-\pi\hbar}\right) du. \tag{6.31}$$

Then by (6.24), (6.25) and (6.27), all the conditions of Lemma 6.9 are satisfied with $x = \hbar$, $\rho = 2\epsilon$, $\rho_0 = \epsilon$, $c = c$, $G = G$, $g(u) = \frac{g_{\boldsymbol{\alpha}}(\mathbf{i}u)}{\pi}$, $f(u, x) = \exp\big(\frac{V_{-\hbar}(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + \mathbf{i}u) - V_0(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + \mathbf{i}u)}{-\pi\hbar}\big)$ and $I(x) = \mathcal{I}^b_{\boldsymbol{\alpha}}(\hbar)$. Then by Lemma 6.9, $\mathcal{I}^b_{\boldsymbol{\alpha}}(\hbar)$ satisfies a (C, σ) Gevrey-1 estimate for the same C and σ as above, and

$$d_n(\boldsymbol{\alpha}) = \frac{1}{\sqrt{2}} \sum_{\substack{j,k \geqslant 0 \\ j+k=n}} \frac{F^{(2k)}_{b,j}(0)}{4^k k!}, \tag{6.32}$$

where for each $j \geqslant 0$, $F_{b,j}$ is a holomorphic function defined in a neighborhood of 0 such that if $v_b = v_b(u) \doteq u\sqrt{\frac{g_{\boldsymbol{\alpha}}(\mathbf{i}u)}{\pi}}$, then

$$F_{b,j}(v_b) = \frac{(-1)^j b_j\big(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + \mathbf{i}u\big)}{\Big(u\sqrt{\frac{g_{\boldsymbol{\alpha}}(\mathbf{i}u)}{\pi}}\Big)'} = \frac{(-1)^j b_j\big(\boldsymbol{\alpha}, \xi_{\boldsymbol{\alpha}} + \mathbf{i}u\big)}{v_b'(u)}. \tag{6.33}$$

Since $\mathbf{i}v_b(u) = v_r(\mathbf{i}u) = \mathbf{i}u\sqrt{\frac{g_{\boldsymbol{\alpha}}(\mathbf{i}u)}{\pi}}$, we have $v_b'(u) = \big(v_r(\mathbf{i}u)\big)'$. Together with (6.30) and (6.33), we have

$$F_{b,j}\big(v_b(u)\big) = (-1)^j F_{r,j}\big(v_r(\mathbf{i}u)\big) = (-1)^j F_{r,j}\big(\mathbf{i}v_b(u)\big).$$

Therefore, for each $k \geqslant 0$, we have

$$F^{(2k)}_{b,j}(0) = (-1)^{j+k} F^{(2k)}_{r,j}(0);$$

and together with (6.29) and (6.32), we have

$$d_n(\boldsymbol{\alpha}) = (-1)^n c_n(\boldsymbol{\alpha}).$$

□

6.2.4 Proof of Proposition 6.5

The proof of Proposition 6.5 relies on Proposition 6.12 and Lemma 6.15 below.

Proposition 6.12. *For $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_6) \in \mathrm{H}_{\epsilon_0}$, let $\boldsymbol{\alpha}' \in \mathrm{H}_{\epsilon_0}$ be obtained from $\boldsymbol{\alpha}$ by changing $\alpha_k \mapsto 2\pi - \alpha_k$ for some $k \in \{1, \ldots, 6\}$. Let T be as in Proposition 6.11 (3). Then*

$$\mathcal{L}_T \mathcal{B}\widehat{\mathcal{I}^r_{\boldsymbol{\alpha}}}(\hbar) = \mathcal{L}_T \mathcal{B}\widehat{\mathcal{I}^r_{\boldsymbol{\alpha}'}}(\hbar).$$

To prove Proposition 6.12, we need the following

Lemma 6.13. *Let $\hbar = b^2$. Then there is a $\delta' > 0$ such that for each b-admissible $\boldsymbol{a} = (a_1, \dots, a_6)$ with the corresponding $\boldsymbol{\alpha} \in \mathrm{H}_{\epsilon_0}$, the b-6j symbol satisfies*

$$\left| \sqrt{\frac{\pi}{2}} e^{\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}} \begin{Bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Bmatrix}_b - \mathcal{I}^b_{\boldsymbol{\alpha}}(\hbar) \right| < O\Big(e^{\frac{r}{2\pi}\left(-\delta'\right)}\Big).$$

Proof. Let $U_b(\boldsymbol{\alpha}, \xi)$ be as defined (2.15). Then by (6.12), we have

$$U_b(\boldsymbol{\alpha}, \xi) = -2\mathbf{i} V_{-\hbar}(\boldsymbol{\alpha}, \xi);$$

and by Proposition 2.15 (2), we have

$$\left| \pi\mathbf{i}b \begin{Bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Bmatrix}_b - \frac{\mathbf{i}\sqrt{2\pi\hbar}}{e^{\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}}} \mathcal{I}^b_{\boldsymbol{\alpha}}(\hbar) \right| \leqslant \int_{\Gamma \smallsetminus [\xi_{\boldsymbol{\alpha}} - \mathbf{i}\epsilon, \xi_{\boldsymbol{\alpha}} + \mathbf{i}\epsilon]} \left| \psi(\xi) e^{\frac{1}{2\pi\mathbf{i}b^2} U_b(\boldsymbol{\alpha}, \xi)} \right| d\xi,$$

where Γ here is chosen to pass $\xi_{\boldsymbol{\alpha}}$. Let ϵ_1 be as in the proof of [24, Theorem 1.5], and let $\delta' = \epsilon_1$. Then by (II) and (III) and letting $d = \epsilon$ in the proof of [24, Theorem 1.5], we have

$$\int_{\Gamma \smallsetminus [\xi_{\boldsymbol{\alpha}} - \mathbf{i}\epsilon, \xi_{\boldsymbol{\alpha}} + \mathbf{i}\epsilon]} \left| \psi(\xi) e^{\frac{1}{2\pi\mathbf{i}b^2} U_b(\boldsymbol{\alpha}, \xi)} \right| d\xi < O\Big(e^{\frac{r}{2\pi}\left(-\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) - \delta'\right)}\Big),$$

from which the result follows. □

Proof of Proposition 6.13. By Lemma 6.13, $\frac{e^{\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}}}{\sqrt{2\pi\hbar}} \begin{Bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{Bmatrix}_b$ and $\mathcal{I}^b_{\boldsymbol{\alpha}}(\hbar)$ satisfy the same Gevrey-1 estimate, hence share the same Borel transform. If we change $a_k \mapsto Q - a_k$, then the corresponding $\alpha_k \mapsto 2\pi - \alpha_k$. Then by Proposition 2.14 and the fact that $\Delta_{\boldsymbol{\alpha}} = \Delta_{\boldsymbol{\alpha}'}$, we have

$$d_n(\boldsymbol{\alpha}) = d_n(\boldsymbol{\alpha}') \tag{6.34}$$

for all $n \geqslant 0$. As a consequence,

$$\mathcal{B}\widehat{\mathcal{I}^b_{\boldsymbol{\alpha}}}(-\hbar) = \mathcal{B}\widehat{\mathcal{I}^b_{\boldsymbol{\alpha}'}}(-\hbar),$$

and by Proposition 6.11 (2),

$$\mathcal{B}\widehat{\mathcal{I}^r_{\boldsymbol{\alpha}}}(\hbar) = \mathcal{B}\widehat{\mathcal{I}^r_{\boldsymbol{\alpha}'}}(\hbar),$$

from which the result follows. □

Remark 6.14. By (6.34) and Proposition 6.11 (1), we also get

$$c_n(\boldsymbol{\alpha}) = c_n(\boldsymbol{\alpha}')$$

for all $n \geqslant 0$, which gives an affirmative answer to [5, Conjecture 3].

Lemma 6.15. *Let $\hbar = \frac{2}{r}$ and let $\delta_5 > 0$ be as in Proposition 6.3. Then for each r-hyperideal $\boldsymbol{a}$ with the corresponding $\boldsymbol{\alpha} \in \mathrm{H}_{\epsilon_0}$, the 0-th Fourier coefficient of the extended quantum 6j-symbol satisfies*

$$\left| \mathrm{I}_{\boldsymbol{a}}(0) - \sqrt{2\pi\hbar} e^{\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}} \mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar) \right| < O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}}) - \delta_5\right)}\Big).$$

Proof. By Proposition 2.15 (2), the relation (6.12) and the choice of δ_5 in Proposition 6.3, we have

$$\begin{aligned}\Big|\mathrm{I}_{\boldsymbol{a}}(0)-\sqrt{2\pi\hbar}e^{\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}}\mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar)\Big| =&\left|\int_{\mathbb{R}\smallsetminus[\xi_{\boldsymbol{\alpha}}-\epsilon,\xi_{\boldsymbol{\alpha}}+\epsilon]}\psi(\xi)e^{\frac{r}{4\pi\mathbf{i}}U_r(\boldsymbol{\alpha},\xi)}d\xi\right|\\ \leqslant&\int_{[\xi_{\boldsymbol{\alpha}}-2\epsilon,\xi_{\boldsymbol{\alpha}}+2\epsilon]\smallsetminus[\xi_{\boldsymbol{\alpha}}-\epsilon,\xi_{\boldsymbol{\alpha}}+\epsilon]}\left|e^{\frac{r}{4\pi\mathbf{i}}U_r(\boldsymbol{\alpha},\xi)}\right|d\xi\\ <&O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_5\right)}\Big).\end{aligned}$$

□

Proof of Proposition 6.5. By Proposition 6.12 we have

$$\begin{aligned}\mathrm{I}_{\boldsymbol{a}}(0)-\mathrm{I}_{\boldsymbol{a}'}(0)=&\Big(\mathrm{I}_{\boldsymbol{a}}(0)-\sqrt{2\pi\hbar}e^{\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}}\mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar)\Big)+\sqrt{2\pi\hbar}e^{\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}}\Big(\mathcal{I}^r_{\boldsymbol{\alpha}}(\hbar)-\mathcal{L}_T\mathcal{B}\widehat{\mathcal{I}^r_{\boldsymbol{\alpha}}}(\hbar)\Big)\\ &-\sqrt{2\pi\hbar}e^{\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}}\Big(\mathcal{I}^r_{\boldsymbol{\alpha}'}(\hbar)-\mathcal{L}_T\mathcal{B}\widehat{\mathcal{I}^r_{\boldsymbol{\alpha}'}}(\hbar)\Big)-\Big(\mathrm{I}_{\boldsymbol{a}'}(0)-\sqrt{2\pi\hbar}e^{\frac{\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})}{\pi\hbar}}\mathcal{I}^r_{\boldsymbol{\alpha}'}(\hbar)\Big).\end{aligned}$$

Let δ_5 be as in Lemma 6.15, let T be as in Proposition 6.11, and let

$$\delta_7<\min\{\delta_5,\pi T\}.$$

Then by Lemma 6.15 and Proposition 6.11 (3), each of the four summands above has order less than $O\big(e^{\frac{r}{2\pi}(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_7)}\big)$, from which the result follows. □

6.3 Proof of Theorem 6.2 and Proposition 3.7

Proof of Theorem 6.2. Let δ_5, δ_6 and δ_7 respectively be as in Proposition 6.3, Proposition 6.4 and Proposition 6.5, and let

$$\delta_4<\min\{\delta_5,\delta_6,\delta_7\}.$$

Then by (6.2), Proposition 6.3, Proposition 6.4 and Proposition 6.5, we have

$$\begin{aligned}&\left|\left\|\begin{matrix}a_1&a_2&a_3\\a_4&a_5&a_6\end{matrix}\right\|-\left\|\begin{matrix}r-2-a_1&a_2&a_3\\a_4&a_5&a_6\end{matrix}\right\|\right|\\ &\qquad<\sin\frac{2\pi}{r}\Big|\mathrm{I}_{\boldsymbol{a}}(0)-\mathrm{I}_{\boldsymbol{a}'}(0)\Big|+O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_4\right)}\Big)\\ &\qquad=O\Big(e^{\frac{r}{2\pi}\left(\mathrm{Vol}(\Delta_{\boldsymbol{\alpha}})-\delta_4\right)}\Big).\end{aligned}$$

□

Proof of Proposition 3.7. Let δ_3 and δ_4 respectively be as in Proposition 6.1 and Theorem 6.2, and let

$$\delta_2=\min\{\delta_3,\delta_4\}.$$

Then the result follows from Proposition 6.1 and Theorem 6.2 and the fact that the volume of any truncated hyperideal hyperbolic tetrahedron $\Delta_{\boldsymbol{\alpha}}$ is at most that of the regular ideal hyperbolic octahedron, which equals v_8. □

A Proof of the technical lemmas

Proof of Lemma 6.7. To see (1), by (6.10) with $N = n + 1$, we have

$$\left|\frac{f(x) - \sum_{k=0}^{n-1} a_k x^k}{x^n} - a_n\right| \leqslant C\sigma^{n+1}(n+1)!x;$$

and letting $x \to 0^+$, we have

$$a_n = \lim_{x\to 0^+} \frac{f(x) - \sum_{k=0}^{n-1} a_k x^k}{x^n}.$$

By (6.10) again with $N = n$, we have

$$\left|\frac{f(x) - \sum_{k=0}^{n-1} a_k x^k}{x^n}\right| \leqslant C\sigma^n n!;$$

and as a consequence,

$$|a_n| \leqslant C\sigma^n n!$$

for each $n \geqslant 0$. Therefore, the power series $\mathcal{B}\widehat{f}(x)$ converges for $x \in \mathbb{C}$ with $|x| < \sigma^{-1}$, defining a holomorphic function there.

To see (2), we use the identity that for any integer $n \geqslant 0$,

$$x^n = \frac{1}{x}\int_0^\infty \frac{e^{-\frac{t}{x}}}{n!} t^n dt, \tag{A.1}$$

which could be obtained using the Integration by Parts. Then for any $T \in (0, \sigma^{-1})$ and $N \geqslant 0$, we have

$$\begin{aligned}\mathcal{L}_T\mathcal{B}\widehat{f}(x) - \sum_{n=0}^{N-1} a_n x^n =& \frac{1}{x}\int_0^T e^{-\frac{t}{x}}\mathcal{B}\widehat{f}(t)dt - \frac{1}{x}\int_0^\infty e^{-\frac{t}{x}}\sum_{n=0}^{N-1}\frac{a_n}{n!}t^n dt\\ =& \frac{1}{x}\int_0^T e^{-\frac{t}{x}}\left(\mathcal{B}\widehat{f}(t) - \sum_{n=0}^{N-1}\frac{a_n}{n!}t^n\right)dt - \frac{1}{x}\int_T^\infty e^{-\frac{t}{x}}\sum_{n=0}^{N-1}\frac{a_n}{n!}t^n dt.\end{aligned}$$

To estimate the terms above, for $t < T$ we have

$$\left|\mathcal{B}\widehat{f}(t) - \sum_{n=0}^{N-1}\frac{a_n}{n!}t^n\right| \leqslant \sum_{n=N}^\infty \frac{|a_n|}{n!}|t|^n \leqslant \sum_{n=N}^\infty C\sigma^n|t|^n \leqslant \sum_{n=N}^\infty C(\sigma T)^n\left(\frac{t}{T}\right)^N \leqslant \frac{C}{1-\sigma T}\left(\frac{t}{T}\right)^N, \tag{A.2}$$

where the second inequality comes from (6.11) and the third inequality comes from that $\frac{t}{T} < 1$; and for $t > T$ we have

$$\left|\sum_{n=0}^{N-1}\frac{a_n}{n!}t^n\right| \leqslant \sum_{n=0}^{N-1} C(\sigma T)^n\left(\frac{t}{T}\right)^n \leqslant \frac{C}{1-\sigma T}\left(\frac{t}{T}\right)^N, \tag{A.3}$$

where the first inequality comes from (6.11) and the second inequality comes from that $\frac{t}{T} > 1$. Putting (A.2) and (A.3) together, and by (A.1), we have

$$\left|\mathcal{L}_T\mathcal{B}\widehat{f}(x) - \sum_{n=0}^{N-1} a_n x^n\right| \leqslant \frac{C}{1-\sigma T}\frac{1}{x}\int_0^\infty e^{-\frac{t}{x}}\left(\frac{t}{T}\right)^N dt = \frac{C}{1-\sigma T}N!\left(\frac{x}{T}\right)^N;$$

and by (6.10), we have

$$\begin{aligned}\left|f(x)-\mathcal{L}_T\mathcal{B}\widehat{f}(x)\right| \leqslant& \left|f(x)-\sum_{n=0}^{N-1}a_nx^n\right|+\left|\mathcal{L}_T\mathcal{B}\widehat{f}(x)-\sum_{n=0}^{N-1}a_nx^n\right| \\ \leqslant& C\sigma^N N!x^N+\frac{C}{1-\sigma T}N!\left(\frac{x}{T}\right)^N \leqslant C\left(1+\frac{1}{1-\sigma T}\right)N!\left(\frac{x}{T}\right)^N\end{aligned} \tag{A.4}$$

for all integers $N \geqslant 0$. Finally, for $x \in (0, \min\{x_0, T\})$, we let $N = \lfloor \frac{T}{x} \rfloor$ so that $\frac{T}{x} - 1 < N \leqslant \frac{T}{x}$. Then by the Stirling's Formula that

$$N! = \sqrt{2\pi N}\left(\frac{N}{e}\right)^N e^{\frac{\theta}{12N}}$$

for some $\theta \in (0,1)$, we have

$$N!\left(\frac{x}{T}\right)^N = \sqrt{2\pi N}e^{N\log\left(N\cdot\frac{x}{T}\right)-N+\frac{\theta}{12N}} \leqslant \sqrt{2\pi}e^{\frac{13}{12}}\sqrt{\frac{T}{x}}e^{-\frac{T}{x}}, \tag{A.5}$$

where the inequality comes from the choice of N so that $\sqrt{N} \leqslant \sqrt{\frac{T}{x}}$, $N \cdot \frac{x}{T} \leqslant 1$, $-N < 1 - \frac{T}{x}$ and $\frac{\theta}{12N} < \frac{1}{12}$. Then (2) follows from (A.4) and (A.5) with $D = \sqrt{2\pi}e^{\frac{13}{12}}C\big(1 + \frac{1}{1-\sigma T}\big)$. □

Proof of Lemma 6.8. For $n \geqslant 1$ and $k \geqslant 1$, we let

$$b_{n,k} = \sum_{\substack{m_1,\dots,m_n\geqslant 0\\ m_1+\cdots+m_n=k\\ m_1+2m_2+\cdots+nm_n=n}} \frac{k!}{m_1!\cdots m_n!}\prod_{j=1}^n a_j^{m_j}$$

so that $b_n = e^{a_0}\sum_{k=0}^n \frac{b_{n,k}}{k!}$. Then we have

$$b_{n,1} = a_n \quad \text{and} \quad b_{n,k+1} = \sum_{j=1}^{n-1} b_{j,k}a_{n-j}. \tag{A.6}$$

We will first prove that

$$|b_{n,k}| \leqslant C^k\sigma^n n! \tag{A.7}$$

by an induction in k. The base case $k = 1$ follows directly from A.6 and (6.10). Now suppose (A.7) holds for k and for all $n \geqslant 1$. Then

$$|b_{n,k+1}| \leqslant \sum_{j=1}^{n-1}|b_{j,k}||a_{n-j}| \leqslant C^{k+1}\sigma^n\sum_{j=1}^{n-1} j!(n-j)! \leqslant C^{k+1}\sigma^n n!,$$

where the last inequality comes from that $\sum_{j=1}^{n-1} j!(n-j)! \leqslant n!$.

Next we prove that

$$\left|\big(f(x)-a_0\big)^k - \sum_{n=0}^{N-1} b_{n,k}x^n\right| \leqslant C^k\sigma^N N!x^N\sum_{j=0}^{k-1}(\sigma x)^j \tag{A.8}$$

by an induction in k. The base case $k = 1$ follows directly from A.6 and (6.10). Now suppose (A.8) holds for k and for all $N \geqslant 1$. Then by (A.6), we have

$$\sum_{n=0}^{N-1} b_{n,k+1}x^n = \sum_{n=0}^{N-1}\sum_{j=1}^{n-1} b_{j,k}a_{n-j}x^n = \sum_{j=1}^{N-1} b_{j,k}x^j \sum_{l=1}^{N-j-1} a_l x^l;$$

and as a consequence, we have

$$\begin{aligned}&\big(f(x)-a_0\big)^{k+1} - \sum_{n=0}^{N-1} b_{n,k+1}x^n\\ =&\big(f(x)-a_0\big)\bigg(\big(f(x)-a_0\big)^k - \sum_{n=0}^{N-1} b_{n,k}x^n\bigg) + \sum_{j=1}^{N-1} b_{j,k}x^j\bigg(f(x)-a_0-\sum_{l=1}^{N-j-1} a_l x^l\bigg).\end{aligned}$$

By (6.10), we have

$$\bigg|f(x)-a_0-\sum_{l=1}^{N-j-1} a_l x^l\bigg| \leqslant C\sigma^{N-j}(N-j)!x^{N-j}$$

for each $j \leqslant N-1$. Together with the induction hypothesis (A.8) for k and (A.7), we have

$$\begin{aligned}&\bigg|\big(f(x)-a_0\big)^{k+1} - \sum_{n=0}^{N-1} b_{n,k+1}x^n\bigg|\\ \leqslant& C^{k+1}\sigma^N N!x^N\sum_{j=0}^{k-1}(\sigma x)^{j+1} + C^{k+1}\sigma^N x^N\sum_{j=1}^{N-1} j!(N-j)!\\ \leqslant& C^{k+1}\sigma^N N!x^N\sum_{j=0}^{k}(\sigma x)^j,\end{aligned}$$

where the last equality comes from that $\sum_{j=1}^{N-1} j!(N-j)! \leqslant N!$.

Finally, for $N = 0$, by (6.10), we have

$$\big|e^{f(x)}\big| \leqslant e^{|a_0|}|e^{|f(x)-a_0|} \leqslant e^C e^{C\sigma x_0}.$$

For $N \geqslant 1$, by (A.8), $|a_0| \leqslant C$, $x < x_0$, and the convention that $b_{n,k} = 0$ for $k > n$, we have

$$\begin{aligned}\bigg|e^{f(x)} - \sum_{n=0}^{N-1} b_n x^n\bigg| =&\bigg|e^{a_0}\sum_{k=1}^{\infty}\frac{1}{k!}\bigg(\big(f(x)-a_0\big)^k - \sum_{n=0}^{N-1} b_{n,k}x^n\bigg)\bigg|\\ \leqslant& e^C\sigma^N N!x^N\sum_{k=1}^{\infty}\frac{C^k}{k!}\sum_{j=0}^{k-1}(\sigma x_0)^j,\end{aligned}$$

where $\sum_{k=1}^{\infty}\frac{C^k}{k!}\sum_{j=0}^{k-1}(\sigma x_0)^j = \frac{e^C - e^{C\sigma x_0}}{1-\sigma x_0}$ when $\sigma x_0 \neq 1$, and $\sum_{k=1}^{\infty}\frac{C^k}{k!}\sum_{j=0}^{k-1}(\sigma x_0)^j = Ce^C$ when $\sigma x_0 = 1$, which converges in both cases. Then the result follows with

$$C_1 = e^C \max\bigg\{e^{C\sigma x_0}, \sum_{k=1}^{\infty}\frac{C^k}{k!}\sum_{j=0}^{k-1}(\sigma x_0)^j\bigg\},$$

which depends on C, σ and x_0, and $\sigma_1 = \sigma$. □

Proof of Lemma 6.9. We will construct intermediate terms $A_n(x)$, $B_n(x)$ and $C_n(x)$ that connect $I(x)$ with its desired Gevrey - 1 expansion $\sum_{n=0}^{\infty} b_n x^n$, and estimate the difference between the adjacent two of them.

As a preparation, we will define an injective function V of u such that $V(u)^2 = u^2 g(u)$, and describe its inverse function U and estimate its derivative. To do these, we need the following estimates. For any $u \in \mathbb{C}$ with $|u| < \rho$ and $s \in (0, \rho - |u|)$, the circle $|z - u| = s$ lies in the disk $|u| < \rho$; and by Cauchy's Derivative Formula, we have

$$\big|g'(u)\big| \leqslant \frac{1}{2\pi} \int_{|z-u|=s} \frac{|g(z)|}{|z-u|^2}\,|dz| \leqslant \frac{G}{s}.$$

Letting $s \to \rho - |u|$, we get

$$\big|g'(u)\big| \leqslant \frac{G}{\rho - |u|}.$$

Let $d = \frac{\rho c}{8G} \leqslant \frac{\rho}{8}$. If $|u| \leqslant d$, then

$$|g'(u)| \leqslant \frac{G}{\rho - d} \leqslant \frac{8G}{7\rho};$$

and integrating along the straight segment from 0 to u, we have

$$\big|g(u) - g(0)\big| \leqslant \frac{8Gd}{7\rho} = \frac{c}{7}.$$

As a consequence,

$$\mathrm{Re} g(u) \geqslant \frac{6c}{7} \quad \text{and} \quad \big|ug'(u)\big| \leqslant \frac{c}{7}$$

on the disk $|u| < d$. In particular, $\mathrm{Re} g(u) > 0$, and $\sqrt{g(u)}$ is well defined, holomorphic, and with the argument in $\{-\frac{\pi}{4}, \frac{\pi}{4}\}$ on the dist $|u| < d$. This implies that $\mathrm{Re}\big(\sqrt{g(u)}\overline{\sqrt{g(0)}}\big) > 0$; and as a consequence, we have

$$\Big|\sqrt{g(u)} + \sqrt{g(0)}\Big|^2 = \big|g(u)\big| + \big|g(0)\big| + 2\mathrm{Re}\Big(\sqrt{g(u)}\,\overline{\sqrt{g(0)}}\Big) \geqslant \big|g(0)\big|,$$

$$\Big|\sqrt{g(u)} - \sqrt{g(0)}\Big| = \frac{\big|g(u) - g(0)\big|}{\big|\sqrt{g(u)} + \sqrt{g(0)}\big|} \leqslant \frac{c}{7\big|\sqrt{g(0)}\big|} \leqslant \frac{1}{7}\big|\sqrt{g(0)}\big|,$$

and

$$\Big|\sqrt{g(u)}\Big| \geqslant \Big|\sqrt{g(0)}\Big| - \Big|\sqrt{g(u)} - \sqrt{g(0)}\Big| \geqslant \frac{6}{7}\big|\sqrt{g(0)}\big|.$$

Now define

$$V(u) \doteq u\sqrt{g(u)}$$

so that $u^2 g(u) = V(u)^2$.

We first prove that V is injective on the disk $|u| < d$. The estimates above imply

$$\Big|V'(u) - \sqrt{g(0)}\Big| \leqslant \Big|\sqrt{g(u)} - \sqrt{g(0)}\Big| + \frac{|ug'(u)|}{2|\sqrt{g(u)}|} \leqslant \frac{1}{7}\Big|\sqrt{g(0)}\Big| + \frac{1}{12}\frac{c}{\big|\sqrt{g(0)}\big|} < \frac{1}{4}\Big|\sqrt{g(0)}\Big|. \quad \text{(A.9)}$$

Therefore, for any two points $u_1 \neq u_2$ in the disk $|u| < d$, integrating along the straight line segment from u_1 to u_2 and using (A.9), we have

$$\Big|V(u_1) - V(u_2) - \sqrt{g(0)}(u_1 - u_2)\Big| \leqslant \int_{u_1}^{u_2} \Big|V'(u) - \sqrt{g(0)}\Big||du| \leqslant \frac{1}{4}\big|\sqrt{g(0)}\big||u_1 - u_2|; \quad \text{(A.10)}$$

and as a consequence,

$$\big|V(u_1)-V(u_2)\big| \geqslant \frac{3}{4}\Big|\sqrt{g(0)}\Big||u_1-u_2|, \tag{A.11}$$

implying the injectivity of V.

Next, we describe the inverse function U of V, and estimate its derivative. Let $l=\frac{\sqrt{c}\,d}{2}$. For any v_0 with $|v_0|<l$ and every $|u|=d$, we have

$$\Big|V(u)-v_0-\sqrt{g(0)}u\Big| \leqslant \Big|V(u)-\sqrt{g(0)}u\Big|+|v_0| \leqslant \frac{1}{4}\Big|\sqrt{g(0)}\Big|d+l < \Big|\sqrt{g(0)}\Big||u|,$$

where the second inequality comes from (A.10) with $u_1=u$ and $u_2=0$, and the last inequality comes from $c<|g(0)|$. Then by Rouché's Theorem, for each v_0 with $|v_0|<l$, there is exactly one zero u_0 of $V(u)-v_0$ in the disk $|u|<d$. Therefore, the map $U:\big\{v\in\mathbb{C}\;\big|\;|v|<l\big\}\to\big\{u\in\mathbb{C}\;\big|\;|u|<d\big\}$ sending v_0 to u_0 is well-defined, holomorphic, and satisfies $U(V(u))=u$. By (A.11), we have

$$\big|V'(u)\big| \geqslant \frac{3}{4}\Big|\sqrt{g(0)}\Big| \geqslant \frac{3\sqrt{c}}{4}$$

for each u with $|u|<d$, and as a consequence, we have

$$\big|U'(v)\big| \leqslant \frac{4}{3\sqrt{c}} < \frac{2}{\sqrt{c}} \tag{A.12}$$

for each v with $|v|<l$. We have the following steps I - IV.

Step I. For $n\geqslant 0$, we define

$$A_n(x)=\frac{1}{\sqrt{2\pi x}}\int_{-\rho_0}^{\rho_0} e^{-\frac{u^2g(u)}{x}}a_n(u)du.$$

Then for each $N\geqslant 1$, we have

$$\begin{aligned}\left|I(x)-\sum_{n=0}^{N-1}A_n(x)x^n\right| &= \left|\frac{1}{\sqrt{2\pi x}}\int_{-\rho_0}^{\rho_0} e^{-\frac{u^2g(u)}{x}}\left(f(u,x)-\sum_{n=0}^{N-1}a_n(u)x^n\right)du\right| \\ &\leqslant \frac{C\sigma^N N!x^N}{\sqrt{2\pi x}}\int_{\mathbb{R}} e^{-\frac{cu^2}{x}}\,du \leqslant \frac{C}{\sqrt{2c}}\,\sigma^N N!x^N.\end{aligned} \tag{A.13}$$

Step II. Let $\mu=\frac{1}{2}\min\big\{\rho_0,d,\frac{\ell}{4\sqrt{G}}\big\}$, and for $n\geqslant 0$, define

$$B_n(x)=\frac{1}{\sqrt{2\pi x}}\int_{-\mu}^{\mu} e^{-\frac{u^2g(u)}{x}}a_n(u)du.$$

By (6.11), we have

$$\sup_{|u|<\rho}|a_n(u)| \leqslant C\sigma^n n!. \tag{A.14}$$

Together with that $e^{-\frac{cu^2}{x}} \leqslant e^{-\frac{c\mu^2}{2x}}e^{-\frac{cu^2}{2x}}$ for all $u\in[-\rho_0,\rho_0]\setminus[-\mu,\mu]$, we have

$$\begin{aligned}\big|A_n(x)-B_n(x)\big| =& \left|\frac{1}{\sqrt{2\pi x}}\int_{[-\rho_0,\rho_0]\setminus[-\mu,\mu]} e^{-\frac{u^2g(u)}{x}}a_n(u)\,du\right| \\ \leqslant& \frac{C}{\sqrt{2\pi x}}\sigma^n n!e^{-\frac{c\mu^2}{2x}}\int_{\mathbb{R}} e^{-\frac{cu^2}{2x}}\,du \\ =& \frac{C}{\sqrt{c}}\sigma^n n!e^{-\frac{c\mu^2}{2x}} < \frac{C}{\sqrt{c}}\sigma^n n!(N-n)!\left(\frac{2x}{c\mu^2}\right)^{N-n},\end{aligned} \tag{A.15}$$

where the last inequality comes from that

$$e^v > \frac{v^{N-n}}{(N-n)!} \tag{A.16}$$

for all $v \geqslant 0$.

Step III. For $n \geqslant 0$, define the holomorphic function F_n on the disk $|v| < l$ by

$$F_n(v) \doteq a_n\big(U(v)\big)U'(v) = \frac{a_n(u)}{V'(u)} = \frac{a_n(u)}{\big(u\sqrt{g(u)}\big)'}$$

for u satisfying $v = V(u) = u\sqrt{g(u)}$. Then by (A.14) and (A.12), we have

$$\big|F_n(v)\big| \leqslant \frac{2C}{\sqrt{c}}\sigma^n n!. \tag{A.17}$$

Let $\gamma = \sqrt{c}\mu$; and for $n \geqslant 0$, define

$$C_n(x) = \frac{1}{\sqrt{2\pi x}}\int_{-\gamma}^{\gamma} e^{-\frac{v^2}{x}}F_n(v)dv.$$

We will show that

$$\big|B_n(x) - C_n(x)\big| \leqslant \frac{4CG}{c\sqrt{2\pi ec}}\,\sigma^n n!(N-n)!\left(\frac{2Gx}{c\gamma^2}\right)^{N-n}. \tag{A.18}$$

To this end, let $\Gamma = V([-\mu,\mu])$, and by the substitution $u = U(v)$, we have

$$B_n(x) = \frac{1}{\sqrt{2\pi x}}\int_{\Gamma} e^{-\frac{v^2}{x}}F_n(v)dv.$$

For $u \in [-\mu,\mu]$, by the definition of μ and the assumption of the Lemma, we have

$$\big|V(u)\big| = |u|\Big|\sqrt{g(u)}\Big| \leqslant \mu\sqrt{G} \quad \text{and} \quad \big|V(\pm\mu)\big| \geqslant \mu\sqrt{c} = \gamma. \tag{A.19}$$

Let $\theta = \frac{1}{2}\arccos\left(\frac{c}{G}\right) < \frac{\pi}{4}$. For $u \in [-\mu,\mu]$, since $\cos\big(\arg g(u)\big) = \frac{\mathrm{Re}g(u)}{|g(u)|} \geqslant \frac{c}{G}$, we have $\big|\arg\sqrt{g(u)}\big| \leqslant \theta$. Let Γ_+ and Γ_- respectively be as shown in Figure 6.

Then we have the following:

(1) Γ is homotopic to $\Gamma_- \cup [-\gamma,\gamma] \cup \Gamma_+$ relative to the end points $V(\pm\mu)$.

(2) The lengths

$$|\Gamma_+| + |\Gamma_-| \leqslant \big|V(\mu)\big| + \big|V(-\mu)\big| - 2\gamma(1-\theta) \leqslant 2\mu\sqrt{G}.$$

(3) Each $v \in \Gamma_\pm$ satisfies $|v| \geqslant \gamma$ and either $|\arg v| \leqslant \theta$ or $|\arg(-v)| \leqslant \theta$; and as a consequence, we have

$$\mathrm{Re}\big(v^2\big) \geqslant \gamma^2\cos(2\theta) \geqslant \frac{c\gamma^2}{G}$$

for each $v \in L$.

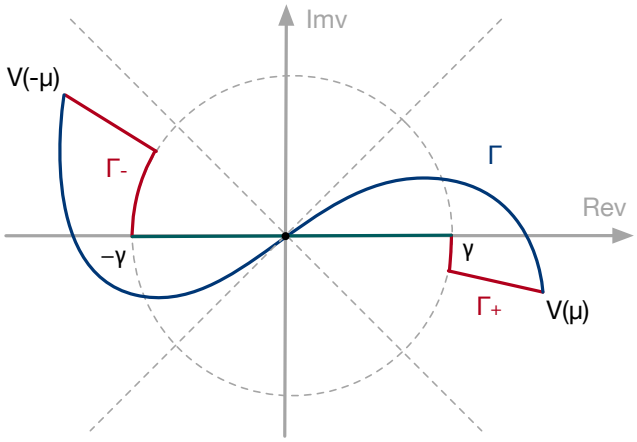


Figure 6: The curves Γ_+ and Γ_-.

Therefore, by (1), we have

$$B_n(x) - C_n(x) = \frac{1}{\sqrt{2\pi x}} \int_{\Gamma_+ \cup \Gamma_-} e^{-\frac{v^2}{x}} F_n(v) dv;$$

and by (2), (3), (A.17) and $\gamma = \mu\sqrt{c}$, we have

$$\left|B_n(x) - C_n(x)\right| \leqslant \frac{1}{\sqrt{2\pi x}} \int_{\Gamma_+ \cup \Gamma_-} e^{-\frac{\mathrm{Re} v^2}{x}} \left|F_n(v)\right| |dv| \leqslant \frac{4C\gamma\sqrt{G}}{c\sqrt{2\pi}} \sigma^n n! \frac{1}{\sqrt{x}} e^{-\frac{c\gamma^2}{Gx}}. \tag{A.20}$$

As $e^{\frac{c\gamma^2}{Gx}-1} \geqslant \frac{c\gamma^2}{Gx}$, we have $\frac{1}{\sqrt{x}} e^{-\frac{c\gamma^2}{2Gx}} \leqslant \frac{\sqrt{G}}{\gamma\sqrt{ec}}$, and as a consequence,

$$\frac{1}{\sqrt{x}} e^{-\frac{c\gamma^2}{Gx}} = \frac{1}{\sqrt{x}} e^{-\frac{c\gamma^2}{2Gx}} e^{-\frac{c\gamma^2}{2Gx}} < \frac{\sqrt{G}}{\gamma\sqrt{ec}} (N-n)! \left(\frac{2Gx}{c\gamma^2}\right)^{N-n}, \tag{A.21}$$

where the inequality comes from (A.16). Then (A.18) follows from (A.20) and (A.21).

Step IV. We will show that for each $n \geqslant 0$,

$$\left| C_n(x) - \frac{1}{\sqrt{2}} \sum_{k=0}^{N-n-1} \frac{F_n^{(2k)}(0)}{4^k k!} x^k \right| \leqslant \frac{20C}{3\sqrt{2c}} \sigma^n \gamma^{-2(N-n)} n! (N-n)! x^{N-n}. \tag{A.22}$$

To see this, as $2\gamma < l$, the function $F_n(v)$ is holomorphic in the disk $|v| < 2\gamma$, and we have

$$F_n(v) = \sum_{i=0}^{\infty} c_{n,i} v^i$$

with

$$c_{n,i} = \frac{F_n^{(i)}(0)}{i!};$$

and by the Cauchy Integral Theorem and (A.17), we have

$$|c_{n,i}| \leqslant \frac{1}{2\pi} \int_{|v|=2\gamma} \frac{|F_n(v)|}{|v|^{i+1}} |dv| \leqslant \frac{1}{2\pi} \frac{4\pi\gamma}{(2\gamma)^{i+1}} \frac{2C}{\sqrt{c}} \sigma^n n! = \frac{1}{(2\gamma)^i} \frac{2C}{\sqrt{c}} \sigma^n n!. \tag{A.23}$$

By the Gaussian Moment Formula that for every integer $k \geqslant 0$,

$$\frac{1}{\sqrt{2\pi x}} \int_{\mathbb{R}} e^{-\frac{v^2}{x}} v^{2k}\, dv = \frac{1}{4^k \sqrt{2}} \binom{2k}{k} k! x^k; \tag{A.24}$$

and by the fact that the integral of odd functions vanishes on $[-\gamma, \gamma]$, for any $m \geqslant 0$, we can write

$$\begin{aligned} & C_n(x) - \frac{1}{\sqrt{2}} \sum_{k=0}^{m-1} \frac{F_n^{(2k)}(0)}{4^k k!} x^k \\ =& \frac{1}{\sqrt{2\pi x}} \int_{-\gamma}^{\gamma} e^{-\frac{v^2}{x}} \Big(F_n(v) - \sum_{i=0}^{2m-1} c_{n,i} v^i \Big) dv - \frac{1}{\sqrt{2\pi x}} \sum_{j=0}^{m-1} c_{n,2j} \int_{\mathbb{R} \setminus [-\gamma,\gamma]} e^{-\frac{v^2}{x}} v^{2j} dv. \end{aligned} \tag{A.25}$$

On the one hand, for v with $|v| \leqslant \gamma$, by (A.23), we have

$$\left| F_n(v) - \sum_{i=0}^{2m-1} c_{n,i} v^i \right| \leqslant \frac{2C}{\sqrt{c}} \sigma^n n! \sum_{i=2m}^{\infty} \left(\frac{|v|}{2\gamma} \right)^i \leqslant \frac{4C}{\sqrt{c}} \sigma^n n! \left(\frac{|v|}{2\gamma} \right)^{2m}.$$

As a consequence, we obtain

$$\begin{aligned} \left| \frac{1}{\sqrt{2\pi x}} \int_{-\gamma}^{\gamma} e^{-\frac{v^2}{x}} \Big(F_n(v) - \sum_{i=0}^{2m-1} c_{n,i} v^i \Big) dv \right| &\leqslant \frac{(2\gamma)^{-2m}}{\sqrt{2\pi x}} \frac{4C}{\sqrt{c}} \sigma^n n! \int_{\mathbb{R}} e^{-\frac{v^2}{x}} v^{2m}\, dv \\ &\leqslant \frac{4^{-m+1} C}{\sqrt{2c}} \sigma^n n! \gamma^{-2m} m! x^m, \end{aligned} \tag{A.26}$$

where the last inequality comes from (A.24) for $k = m$ and $\binom{2m}{m} \leqslant 4^m$.

On the other hand, for v with $v > \gamma$ and for $j < m$, we have

$$|v|^{2j} = |v|^{2m} |v|^{-2(m-j)} \leqslant |v|^{2m} \gamma^{-2(m-j)};$$

and by (A.23), we have

$$|c_{n,2j}|\, |v|^{2j} \leqslant \frac{2C}{\sqrt{c}} \sigma^n n! \frac{1}{(2\gamma)^{2j}} \gamma^{-2(m-j)} |v|^{2m} = \frac{2C}{\sqrt{c}} \sigma^n n! 4^{-j} \gamma^{-2m} |v|^{2m}.$$

As a consequence,

$$\begin{aligned} \left| \frac{1}{\sqrt{2\pi x}} \sum_{j=0}^{m-1} c_{n,2j} \int_{\mathbb{R} \setminus [-\gamma,\gamma]} e^{-\frac{v^2}{x}} v^{2j} dv \right| &\leqslant \frac{1}{\sqrt{2\pi x}} \sum_{j=0}^{m-1} |c_{n,2j}| \int_{\mathbb{R} \setminus [-\gamma,\gamma]} e^{-\frac{v^2}{x}} |v|^{2j} dv \\ &\leqslant \frac{\gamma^{-2m}}{\sqrt{2\pi x}} \frac{2C}{\sqrt{c}} \sigma^n n! \Big(\sum_{j=0}^{m-1} 4^{-j} \Big) \int_{\mathbb{R} \setminus [-\gamma,\gamma]} e^{-\frac{v^2}{x}} |v|^{2m} dv \\ &\leqslant \frac{\gamma^{-2m}}{\sqrt{2\pi x}} \frac{2C}{\sqrt{c}} \sigma^n n! \Big(\sum_{j=0}^{m-1} 4^{-j} \Big) \int_{\mathbb{R}} e^{-\frac{v^2}{x}} v^{2m} dv \\ &\leqslant \frac{1}{\sqrt{2}} \frac{2C}{\sqrt{c}} \sigma^n n! \Big(\sum_{j=0}^{m-1} 4^{-j} \Big) \gamma^{-2m} m! x^m \\ &\leqslant \frac{8C}{3\sqrt{2c}} \sigma^n n! \gamma^{-2m} m! x^m, \end{aligned} \tag{A.27}$$

where the penultimate inequality comes from (A.24) again.

Then (A.22) follows from (A.25) with $m = N-n$, (A.26), (A.27) and the computation that $4^{-m+1} + \frac{8}{3} \leqslant \frac{20}{3}$ for $m \geqslant 0$.

Finally, for $n \geqslant 0$, define

$$c_n = \frac{1}{\sqrt{2}} \sum_{\substack{j,k\geqslant 0 \\ j+k=n}} \frac{F_j^{(2k)}(0)}{4^k k!}.$$

Then

$$\sum_{n=0}^{N-1} c_n x^n = \frac{1}{\sqrt{2}} \sum_{j=0}^{N-1} x^j \sum_{k=0}^{N-j-1} \frac{F_j^{(2k)}(0)}{4^k k!} x^k.$$

By(A.13), (A.15), (A.18), (A.22) and the Binomial Theorem, we have

$$\left| I(x) - \sum_{n=0}^{N-1} c_n x^n \right| \leqslant C\sqrt{\frac{2}{c}} \left(\frac{1}{2}\sigma^N + \frac{1}{\sqrt{2}} \left(\sigma + \frac{2}{c\mu^2} \right)^N + \frac{2G}{c\sqrt{\pi e}} \left(\sigma + \frac{2G}{c\gamma^2} \right)^N + \frac{10}{3} \left(\sigma + \frac{1}{\gamma^2} \right)^N \right) N! x^N.$$

As a consequence,

$$\left| I(x) - \sum_{n=0}^{N-1} c_n x^n \right| \leqslant C_2 \sigma_2^N N! x^N$$

with

$$C_2 = C\sqrt{\frac{2}{c}} \left(\frac{1}{2} + \frac{1}{\sqrt{2}} + \frac{2G}{c\sqrt{\pi e}} + \frac{10}{3} \right) \quad \text{and} \quad \sigma_2 = \max \left\{ \sigma, \sigma + \frac{2}{c\mu^2}, \sigma + \frac{2G}{c\gamma^2}, \sigma + \frac{1}{\gamma^2} \right\},$$

which is the desired real Gevrey- 1 estimate of $I(x)$. □

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